The Riemann Hypothesis

Jonathan Washburn
Independent Researcher
washburn.jonathan@gmail.com

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Abstract

We present a single-route phase–certificate in a bounded-real framework on $\Omega = \{\Re s > \frac{1}{2}\}$. The chain is: product certificate $\Rightarrow$ boundary wedge $\mathrm{P}^+ \Rightarrow 2\mathcal{J}$ Herglotz and Θ Schur on Ω (via Poisson/Cayley) $\Rightarrow$ RH (via removability). With locked constants $c_0(\psi) = 0.17620819$, $C_H(\psi) = 2/\pi$, $C_{\mathrm{box}}^{(\zeta)} = 0.05521808 = K_0 + K_\xi$ and $C_\psi^{(H^1)} = 0.2400$, we use the certificate

$$\Upsilon \;=\; \frac{(2/\pi)\, M_\psi}{c_0(\psi)} \;<\; \tfrac{1}{2}, \qquad M_\psi = \tfrac{4}{\pi}\, C_\psi^{(H^1)} \sqrt{C_{\mathrm{box}}^{(\zeta)}}.$$

No bandlimit or prime budget enters the certificate; in the ζ–normalized boundary gauge the k=1 layer does not appear (no C_P term). We retain two compatible RHS bounds (CR–Green with Whitney box energy, and the Hilbert-envelope bound); all printed numerics use the conservative box-energy route. All inputs are unconditional; outward rounding policy is consolidated in Appendix H.

Keywords. Riemann zeta function; Herglotz/Schur functions; bounded-real methods; Hilbert–Schmidt determinants; harmonic analysis.

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1 Introduction

Single proof route. This paper follows only the boundary product–certificate route. The k=1 prime layer does not enter the windowed certificate in the ζ–normalized gauge (no C_P term). All alternative routes (Bridges A–C beyond the boundary push, interior rectangles, full KYP development) are archived. The active certificate uses a balanced Carleson term (no ξ-only energy):

$$\Upsilon_{\mathrm{new}} \;:=\; c_0(\psi)\Big(K_{\mathrm{HS}}(\psi) \;+\; K_{\mathrm{bal}}(\psi)\Big) \;<\; \tfrac{1}{2}.$$

Main chain used in the proof (ζ–normalized, balanced certificate; no bandlimit). *Inputs (Whitney-local).* (i) Phase–velocity identity with outer normalization (Lemma 6); (ii) Balanced Carleson bound via $U_{\mathrm{bal}} = -\Re L$ (Lemma 100); (iii) Derivative envelope (Lemma 9); (iv) H^1–BMO link yielding M_ψ (Lemma 23); (v) a.e. boundary wedge from the certificate (Theorem 45).
Certificate. With $c_0 = 0.17620819$, K_{HS} as in the prime/HS section, and K_{bal} from Lemma 100, the wedge holds whenever $\Upsilon_{\mathrm{new}} < \frac{1}{2}$. **Proof uses the conservative numeric certificate (Cor. 5); the balanced bound is structural and not used for numerics.**
Implications. (1) The inequality yields (P+) (Theorem 45). (2) (P+) $\Rightarrow 2\mathcal{J}$ Herglotz and Θ Schur on Ω (Poisson/Cayley). (3) Globalization/pinch (Section 2) and conclusion.

The Riemann Hypothesis (RH) admits several analytic formulations. In this paper we pursue a bounded-real (BRF) route on the right half-plane

$$\Omega \;:=\; \{\, s \in \mathbb{C} :\; \Re s > \tfrac{1}{2} \,\},$$

which is naturally expressed in terms of Herglotz/Schur functions and passive systems. Let $\mathcal{P}$ be the primes, and define the prime-diagonal operator

$$A(s) : \ell^2(\mathcal{P}) \to \ell^2(\mathcal{P}), \qquad A(s)e_p \;:=\; p^{-s} e_p.$$

For $\sigma := \Re s > \frac{1}{2}$ we have $\|A(s)\|_{\mathcal{S}_2}^2 = \sum_{p\in\mathcal{P}} p^{-2\sigma} < \infty$ and $\|A(s)\| \le 2^{-\sigma} < 1$. With the completed zeta function

$$\xi(s) \;:=\; \tfrac{1}{2} s(1-s)\, \pi^{-s/2}\, \Gamma(s/2)\, \zeta(s)$$

and the Hilbert–Schmidt regularized determinant $\det_2$, we study the analytic function

$$\mathcal{J}(s) \;:=\; \frac{\det_2(I - A(s))}{\mathcal{O}(s)\, \xi(s)}, \qquad \Theta(s) \;:=\; \frac{2\mathcal{J}(s) - 1}{2\mathcal{J}(s) + 1}.$$

The BRF assertion is that $|\Theta(s)| \le 1$ on Ω (Schur), equivalently that $2\mathcal{J}(s)$ is Herglotz or that the associated Pick kernel is positive semidefinite.

Our method combines four ingredients:

- **Schur–determinant splitting.** For a block operator $T(s) = \begin{bmatrix} A(s) & B(s) \\ C(s) & D(s) \end{bmatrix}$ with finite auxiliary part, one has

  $$\log \det_2(I - T) \;=\; \log \det_2(I - A) \;+\; \log \det(I - S), \qquad S \;:=\; D - C(I - A)^{-1} B,$$

  which separates the Hilbert–Schmidt ($k \ge 2$) terms from the finite block.

- **HS continuity for $\det_2$.** Prime truncations $A_N \to A$ in the HS topology, uniformly on compacts in Ω, imply local-uniform convergence of $\det_2(I - A_N)$ (Proposition .1). Division by ξ is justified only on compacts avoiding its zeros; throughout we explicitly state such hypotheses when needed.

Unsmoothing $\det_2$: routed through smoothed testing (A1)

Lemma 1 (Smoothed distributional bound for $\partial_\sigma \Re \log \det_2$)**.** *Let $I \Subset \mathbb{R}$ be a compact interval and fix $\varepsilon_0 \in (0, \frac{1}{2}]$. There exists a finite constant*

$$C_* \;:=\; \sum_p \sum_{k\ge 2} \frac{p^{-k/2}}{k^2 \log p} \;<\; \infty$$

such that for all $\sigma \in (\frac{1}{2}, \frac{1}{2}+\varepsilon_0]$ and every $\varphi \in C_c^2(I)$,

$$\Big|\int_{\mathbb{R}} \varphi(t)\, \partial_\sigma \Re \log\det{}_2\,(I - A(\sigma+it))\, dt\Big| \;\le\; C_*\, \|\varphi''\|_{L^1(I)}.$$

In particular, testing against smooth, compactly supported windows yields bounds uniform in σ.

Proof. For $\sigma > \frac{1}{2}$ one has the absolutely convergent expansion

$$\partial_\sigma\, \Re \log\det{}_2\,(I - A(\sigma+it)) \;=\; \sum_p \sum_{k\ge 2} (\log p)\, p^{-k\sigma} \cos(kt\log p).$$

For each frequency $\omega = k\log p \ge 2\log 2$, two integrations by parts give

$$\Big|\int_{\mathbb{R}} \varphi(t)\cos(\omega t)\, dt\Big| \;\le\; \frac{\|\varphi''\|_{L^1(I)}}{\omega^2}.$$

Summing the resulting majorant yields

$$\Big|\int \varphi\, \partial_\sigma \Re \log\det{}_2\; dt\Big| \;\le\; \|\varphi''\|_{L^1} \sum_p \sum_{k\ge 2} \frac{(\log p)\, p^{-k\sigma}}{(k\log p)^2} \;\le\; \|\varphi''\|_{L^1} \sum_p \sum_{k\ge 2} \frac{p^{-k/2}}{k^2\, \log p},$$

uniformly for $\sigma \in (\frac{1}{2}, \frac{1}{2}+\varepsilon_0]$, since the rightmost double series converges. This proves the claim. □

Theorem 2 (Quantified phase–velocity identity and boundary passage)**.** *Let $u_\varepsilon(t) := \log\big|\det{}_2(I - A(\frac{1}{2}+\varepsilon+it))\big| - \log\big|\xi(\frac{1}{2}+\varepsilon+it)\big|$ and let $\mathcal{O}_\varepsilon$ be the outer on $\{\Re s > \frac{1}{2}+\varepsilon\}$ with boundary modulus e^{u_ε}. There exists $C_I < \infty$, independent of $\varepsilon \in (0, \varepsilon_0]$, such that for every compact interval $I \Subset \mathbb{R}$ and every $\phi \in C_c^2(I)$ with $\phi \ge 0$,*

$$\Big|\int_I \phi(t)\, \partial_\sigma \Re \log\det{}_2\,(I - A(\tfrac{1}{2}+\varepsilon+it))\, dt\Big| \;\le\; C_I\, \|\phi''\|_{L^1(I)},$$

and

$$\int_I \phi(t)\, \partial_\sigma \Re \log \xi(\tfrac{1}{2}+\varepsilon+it)\, dt \;\le\; C_I'\, \|\phi\|_{L^1(I)}$$

with C_I' depending only on I. Consequently u_ε is uniformly L^1–bounded and Cauchy on I as $\varepsilon \downarrow 0$, and the outers $\mathcal{O}_\varepsilon$ converge locally uniformly to an outer $\mathcal{O}$ on Ω with a.e. boundary modulus e^u. In particular, after dividing by $\mathcal{O}\,\xi$ and passing to $\varepsilon \downarrow 0$, the phase–velocity identity holds in the distributional sense on I:

$$\int_I \phi(t)\; - w'(t)\, dt \;=\; \int_I \phi(t)\, \pi\, d\mu(t) \;+\; \pi \sum_{\gamma\in I} m_\gamma\, \phi(\gamma), \qquad \forall\, \phi \in C_c^\infty(I),\; \phi \ge 0,$$

where μ is the Poisson balayage of off–critical zeros on $Q(I)$ and the discrete sum ranges over critical–line zeros in I.

Lemma 3 (Certificate implies boundary wedge (P+) — corrected pairing)**.** *Let φ_{L,t_0} be the Poisson plateau associated to a fixed window profile ψ with plateau constant $c_0(\psi) > 0$, and let*

$$Q := \{t \in \mathbb{R} :\; |\operatorname{Arg} \mathcal{J}(1/2+it) - m| \ge \tfrac{\pi}{2}\},$$

where $m \in \mathbb{R}/2\pi\mathbb{Z}$ is any fixed angular shift. Assume that for all $t_0 \in \mathbb{R}$ and all sufficiently large L,

$$c_0(\psi)\, \mu(Q \cap I_{L,t_0}) \;\le\; \int_{\mathbb{R}} \varphi_{L,t_0}(t)\; - w'(t)\, dt \;\le\; C(\psi) \Big(\iint_{Q(\alpha' I)} |\nabla U|^2\, \sigma\Big)^{1/2}, \tag{1}$$

with $C(\psi)$ independent of t_0, L. If $C(\psi)\, (C_{\mathrm{box}}^{(\zeta)})^{1/2}/c_0(\psi) < \infty$ (in particular if it is $< \frac{1}{2}$), then $\mu(Q) = 0$, hence $\Re(2\mathcal{J}(1/2+it)) \ge 0$ for a.e. t.

Proof. Suppose $\mu(Q) > 0$. By Lebesgue's density theorem there is $t_* \in Q$ with density $\delta := \lim_{L\to\infty} \frac{1}{L}\mu(Q \cap I_{L,t_*}) > 0$. For all sufficiently large L, $\mu(Q \cap I_{L,t_*}) \geq \frac{\delta}{2}L$. Since $\varphi_{L,t_*} \geq c_0(\psi)$ on I_{L,t_*}, the left–hand side of (1) obeys $c_0(\psi)\,\mu(Q \cap I_{L,t_*}) \geq c_0(\psi)\,\frac{\delta}{2}\,L$. But the right–hand side is bounded by $C(\psi)\,(C^{(\zeta)}_{\text{box}})^{1/2}$ using Lemma 17 together with the Carleson box energy bound for U (balanced route or conservative route), uniformly in L. This contradicts linear growth in L as $L \to \infty$. Thus $\mu(Q) = 0$ and the a.e. wedge follows. □

2 Globalization across $Z(\xi)$ via a Schur–Herglotz pinch

Standing setup. *Let*

$$\Omega := \{s \in \mathbb{C} : \ \Re s > \tfrac{1}{2}\}, \qquad \xi(s) = \tfrac{1}{2}\, s(s-1)\, \pi^{-s/2}\Gamma(\tfrac{s}{2})\zeta(s).$$

Define

$$\mathcal{J}(s) := \frac{\det_2(I - A(s))}{\mathcal{O}(s)\,\xi(s)}, \qquad F(s) := 2\,\mathcal{J}(s), \qquad \Theta(s) := \frac{F(s)-1}{F(s)+1}.$$

Here $\mathcal{O}$ is holomorphic and zero–free on Ω (an outer normalizer) and $\det_2(I - A)$ is holomorphic on Ω. We record the two normalization properties actually used below:

*(N1) (*Right–edge normalization*)* $\lim_{\sigma\to+\infty} \mathcal{J}(\sigma + it) = 0$ *uniformly in t; hence* $\lim_{\sigma\to+\infty} \Theta(\sigma + it) = -1$.

*(N2) (*Non–cancellation at ξ–zeros*) For every $\rho \in \Omega$ with $\xi(\rho) = 0$,*

$$\det_2(I - A(\rho)) \neq 0 \quad \textit{and} \quad \mathcal{O}(\rho) \neq 0.$$

Thus $\mathcal{J}$ has a pole at ρ of order $\operatorname{ord}_\rho(\xi)$.

Boundary wedge (P^+)***.*** *We assume the a.e. boundary inequality*

$$\Re\, F\Big(\tfrac{1}{2} + it\Big) \ \geq \ 0 \qquad \textit{for a.e. } t \in \mathbb{R}. \tag{P+}$$

From boundary wedge to interior Schur bound. *On $\Omega \setminus Z(\xi)$ the function F is holomorphic, so $\Re F$ is harmonic there. Let $u(t) := \Re\, F(\frac{1}{2} + it) \geq 0$ a.e. Let H be the Herglotz transform (Poisson integral) of u, so H is holomorphic on Ω with $\Re H \geq 0$ and the nontangential boundary limits $\Re H = u$ a.e. By standard boundary uniqueness for Smirnov/Hardy classes (here $2\mathcal{J} \in H^p_{\text{loc}}(\Omega \setminus Z(\xi))$ via HS→det$_2$ continuity on rectangles and the outer L^1 boundary control), we have*

$$\Re F(s) \ \geq \ 0 \qquad (s \in \Omega \setminus Z(\xi)). \tag{H+}$$

Consequently, on $\Omega \setminus Z(\xi)$ the identity

$$1 - |\Theta(s)|^2 \ = \ \frac{4\,\Re F(s)}{|F(s)+1|^2} \ \geq \ 0$$

implies

$$|\Theta(s)| \ \leq \ 1 \qquad (s \in \Omega \setminus Z(\xi)). \tag{Schur}$$

Local pinch at a putative off–critical zero. *Fix $\rho \in \Omega$ with $\xi(\rho) = 0$. By (N2) the function F has a pole at ρ, hence*

$$\Theta(s) = \frac{F(s)-1}{F(s)+1} \ \longrightarrow \ 1 \qquad (s \to \rho).$$

By (Schur), Θ is bounded by 1 on $(\Omega \setminus Z(\xi))$, so the singularity of Θ at ρ is removable (Riemann's theorem), and the holomorphic extension satisfies

$$\Theta(\rho) = 1.$$

Because Θ is holomorphic on the connected domain $\Omega \setminus (Z(\xi) \setminus \{\rho\})$ and $|\Theta| \le 1$ there, the Maximum Modulus Principle forces Θ to be a constant unimodular *function on that domain (it attains its supremum 1 at an interior point). By analyticity, the same constant extends throughout $\Omega \setminus Z(\xi)$.*

Contradiction with right–edge normalization. *By (N1), $\Theta(\sigma + it) \to -1$ as $\sigma \to +\infty$; hence the above constant must equal -1. But we also have $\Theta(\rho) = 1$. Contradiction.*

Conclusion of the pinch. *No $\rho \in \Omega$ with $\xi(\rho) = 0$ can exist.*

Normalization at infinity (used in (N1)). *As $\sigma \to +\infty$ with t fixed, $\zeta(\sigma + it) = 1 + o(1)$ and $\pi^{-\frac{s}{2}}\Gamma(\frac{s}{2})$ grows superpolynomially in $|\sigma|$ by Stirling, so $|\xi(\sigma + it)| \to \infty$. By construction, $\det_2(I - A(\sigma + it)) \to 1$ and $\mathcal{O}(\sigma + it) \to 1$; hence $\mathcal{J}(\sigma + it) \to 0$ and $\Theta(\sigma + it) \to -1$.*

Theorem (Schur globalization; no off–critical zeros). *Assume (P+) and the normalizations (N1)–(N2) above. Then $\xi(s) \neq 0$ for all $s \in \Omega$. Equivalently, every nontrivial zero of ζ lies on $\Re s = \frac{1}{2}$.* Proof. *The argument just given shows that any $\rho \in \Omega$ with $\xi(\rho) = 0$ would force Θ to be constant unimodular on $\Omega \setminus Z(\xi)$, contradicting (N1). Therefore $Z(\xi) \cap \Omega = \varnothing$.* □

Lemma 4 (Carleson box energy: stable sum bound)**.** *For harmonic potentials U_1, U_2 on Ω, one has*

$$\sqrt{C_{\text{box}}(U_1 + U_2)} \;\le\; \sqrt{C_{\text{box}}(U_1)} \;+\; \sqrt{C_{\text{box}}(U_2)}.$$

Proof. For any Carleson box B, by Cauchy–Schwarz,

$$\int_B |\nabla(U_1 + U_2)|^2 \le \Big(\sqrt{\int_B |\nabla U_1|^2} + \sqrt{\int_B |\nabla U_2|^2}\Big)^2.$$

Taking the supremum over B gives the claim. □

Corollary 5 (Conservative numeric closure under Lemma 4)**.** *With the locked constants $c_0(\psi) = 0.17620819$, $C_\psi^{(H^1)} = 0.2400$, $C_H(\psi) \le 2/\pi$, $K_0 = 0.03486808$, $K_\xi = 0.02035$, one has*

$$\sqrt{C_{\text{box}}} \;\le\; \sqrt{K_0} + \sqrt{K_\xi} \;\approx\; 0.32938, \qquad M_\psi \;\le\; \frac{4}{\pi}\, C_\psi^{(H^1)} \sqrt{C_{\text{box}}} \;\approx\; 0.10065,$$

and therefore

$$\Upsilon \;=\; \frac{(2/\pi)\, M_\psi}{c_0(\psi)} \;\approx\; 0.364 \;<\; \tfrac{1}{2}.$$

Hence the product certificate closes even under the conservative sum bound.

Proof of (N2) (non–cancellation at ξ–zeros). *For $s = \sigma + it$ with $\sigma > \frac{1}{2}$, define the diagonal operator $A(s)e_p = p^{-s}e_p$ on $\ell^2(\mathbb{P})$. Then $\|A(s)\| = 2^{-\sigma} < 1$ and $\|A(s)\|_{\text{HS}}^2 = \sum_p p^{-2\sigma} < \infty$, so $A(s)$ is Hilbert–Schmidt. The 2–modified determinant for diagonal $A(s)$ is*

$$\det_2(I - A(s)) \;=\; \prod_{p \in \mathbb{P}} (1 - p^{-s})\, e^{p^{-s}},$$

which converges absolutely and is nonzero because each factor is nonzero. Moreover, $I - A(s)$ is invertible with $\|(I - A(s))^{-1}\| \le (1 - 2^{-\sigma})^{-1}$ since $|1 - p^{-s}| \ge 1 - 2^{-\sigma} > 0$. Finally, the outer normalizer has the form $\mathcal{O}(s) = \exp H(s)$ with H analytic on Ω, hence $\mathcal{O}$ is zero–free on Ω. Thus if $\rho \in \Omega$ with $\xi(\rho) = 0$, then $\det_2(I - A(\rho)) \neq 0$ and $\mathcal{O}(\rho) \neq 0$, i.e. no cancellation can occur at ρ.

Normalization and finite port (eliminating C_P and C_Γ). *We record the implementation details that ensure the product certificate contains no prime budget and no Archimedean term.*

Lemma 6 (ζ–normalized outer and compensator)**.** *Define the outer $\mathcal{O}_\zeta$ on Ω with boundary modulus $|\det_2(I-A)/\zeta|$ and set*

$$J_\zeta(s) \;:=\; \frac{\det_2(I-A(s))}{\mathcal{O}_\zeta(s)\,\zeta(s)}\cdot B(s), \qquad B(s) := \frac{s-1}{s}.$$

On $\Re s = \frac{1}{2}$ one has $|B| = 1$. The phase–velocity identity of Theorem 2 holds for J_ζ with the same Poisson/zero right-hand side. In particular, no separate Archimedean term enters the inequality used by the certificate.

Proof. Set $X := \xi$ and $Z := \zeta$, and let G denote the archimedean factor linking them,

$$X(s) \;=\; \tfrac{1}{2}s(1-s)\,\pi^{-s/2}\,\Gamma(\tfrac{s}{2})\,Z(s) \;=:\; G(s)\,Z(s).$$

Define $\mathcal{O}_X$ (resp. $\mathcal{O}_Z$) to be the outer on Ω with boundary modulus $|\det_2(I-A)/X|$ (resp. $|\det_2(I-A)/Z|$). Then, by construction,

$$\Big|\frac{\det_2(I-A)}{\mathcal{O}_X\,X}\Big| \equiv 1 \equiv \Big|\frac{\det_2(I-A)}{\mathcal{O}_Z\,Z}\Big| \quad \text{a.e. on } \{\Re s = \tfrac{1}{2}\}.$$

Consequently the phase–velocity identity (Theorem 2) applies to either unimodular ratio. Writing

$$\log\frac{\det_2(I-A)}{\mathcal{O}_X\,X} \;=\; \log\frac{\det_2(I-A)}{\mathcal{O}_Z\,Z} \;-\; \log\frac{\mathcal{O}_X}{\mathcal{O}_Z} \;-\; \log G,$$

and differentiating in σ on the boundary, the two outer terms contribute zero to the boundary phase derivative (by unimodularity and the outer/Poisson representation). The remaining difference is $-\partial_\sigma \Im \log G$.

The function G is holomorphic on Ω; it has a simple zero at $s = 1$ and no other zeros/poles in Ω (the singularities at $s = 0$ and at the negative even integers lie outside Ω). In the half–plane Herglotz/Pick calculus, multiplying by a half–plane Blaschke factor corresponding to a zero at $s = 1$ adjusts the boundary phase by the Poisson mass of that zero and preserves unimodularity. Precisely, with

$$B(s) \;=\; \frac{s-1}{s}$$

we have $|B| \equiv 1$ on $\Re s = \frac{1}{2}$, B is holomorphic and zero–free on $\Omega \setminus \{1\}$, and its boundary phase derivative is the Poisson kernel of the point $s = 1$. Since G contributes exactly the opposite Poisson mass at $s = 1$, the product B/G has zero boundary phase derivative (in distributions) on the critical line. Therefore

$$\partial_\sigma \Im \log\Big(\frac{\det_2(I-A)}{\mathcal{O}_Z\,Z}\cdot B\Big) \;=\; \partial_\sigma \Im \log\frac{\det_2(I-A)}{\mathcal{O}_X\,X}$$

on the boundary. Applying Theorem 2 to both sides shows that $J_\zeta = (\det_2/(\mathcal{O}_\zeta\,\zeta))\,B$ obeys the same quantitative phase–velocity identity (Poisson/zero right–hand side) as $\mathcal{J} = \det_2/(\mathcal{O}\,\xi)$. In particular, no additional Archimedean term appears in the inequality used by the certificate. □

Corollary 7 (No C_P/C_Γ in the certificate)**.** *With J_ζ and $\widehat{J}$ as above, the product certificate and wedge step involve only $c_0(\psi)$, $C_H(\psi)$, and M_ψ. In particular, $C_P = 0$ and $C_\Gamma = 0$ in the inequality leading to (P+).*

Lemma 8 (No interior pole for a Herglotz pullback)**.** *Let Θ be Schur on $\Omega \setminus \{\rho\}$ with a removable singularity at $\rho \in Z(\xi) \cap \Omega$. Then $\Theta(\rho) \neq 1$.*

Proof. If $\Theta(\rho) = 1$, write $g(s) := 1 - \Theta(s) = (s-\rho)^m h(s)$ with $m \geq 1$, $h(\rho) \neq 0$. Then $F = (1+\Theta)/(1-\Theta) = (2-g)/g = 2/g - 1$ has a pole of order m at ρ. Along two radial directions separated by π/m, the principal part $2/((s-\rho)^m h(\rho))$ takes opposite real signs, contradicting $\Re F \geq 0$ near ρ. Hence $\Theta(\rho) \neq 1$. □

Lemma 9 (Derivative envelope for the printed window)**.** *Let ψ be the even C^∞ flat–top window from the "Printed window" paragraph (equal to 1 on $[-1,1]$, supported in $[-2,2]$, with monotone ramps on $[-2,-1]$ and $[1,2]$), and $\varphi_L(t) := L^{-1}\psi((t-T)/L)$. Then, for every $L > 0$,*

$$\|(\mathcal{H}[\varphi_L])'\|_{L^\infty(\mathbb{R})} \;\leq\; \frac{C_H(\psi)}{L} \qquad \textit{with} \qquad C_H(\psi) \;\leq\; \frac{2}{\pi} \;<\; 0.65.$$

Proof. Step 1 (Scaling). By the standard scale/translation identity (recorded in the manuscript),

$$\mathcal{H}[\varphi_L](t) = H_\psi\Big(\frac{t-T}{L}\Big), \qquad H_\psi(x) := \frac{1}{\pi}\,\text{p.v.}\!\int_{\mathbb{R}} \frac{\psi(y)}{x-y}\,dy,$$

we get

$$(\mathcal{H}[\varphi_L])'(t) = \frac{1}{L}\,H'_\psi\Big(\frac{t-T}{L}\Big) \quad\Longrightarrow\quad \|(\mathcal{H}[\varphi_L])'\|_\infty = \frac{1}{L}\,\|H'_\psi\|_\infty.$$

Thus it suffices to bound $\|H'_\psi\|_\infty$.

Step 2 (Structure and signs). Since $\psi' \equiv 0$ on $(-1,1)$ and the ramps are monotone,

$$\psi'(y) \geq 0 \text{ on } [-2,-1], \qquad \psi'(y) \leq 0 \text{ on } [1,2], \qquad \int_{-2}^{-1} \psi'(y)\,dy = 1 = -\int_1^2 \psi'(y)\,dy.$$

In distributions, $(H_\psi)' = \mathcal{H}[\psi']$, so for every $x \in \mathbb{R}$

$$H'_\psi(x) = \frac{1}{\pi}\,\text{p.v.}\!\int_{-2}^{-1} \frac{\psi'(y)}{x-y}\,dy \;+\; \frac{1}{\pi}\,\text{p.v.}\!\int_1^2 \frac{\psi'(y)}{x-y}\,dy.$$

Step 3 (Worst case occurs between the ramps). Fix $x \in (-1,1)$. On $y \in [-2,-1]$ the kernel $y \mapsto 1/(x-y)$ is positive and strictly increasing; on $y \in [1,2]$ the kernel is negative and strictly decreasing. Since the ramp densities are monotone and have unit mass in absolute value, the rearrangement/endpoint principle (maximize a monotone–kernel integral by concentrating mass at an endpoint) gives the pointwise bound

$$\Big|\text{p.v.}\!\int_{-2}^{-1} \frac{\psi'(y)}{x-y}\,dy\Big| \leq \frac{1}{1+x}, \qquad \Big|\text{p.v.}\!\int_1^2 \frac{\psi'(y)}{x-y}\,dy\Big| \leq \frac{1}{1-x}.$$

Therefore, for every $x \in (-1,1)$,

$$|H'_\psi(x)| \;\leq\; \frac{1}{\pi}\Big(\frac{1}{1+x} + \frac{1}{1-x}\Big) \;=\; \frac{2}{\pi}\,\frac{1}{1-x^2} \;\leq\; \frac{2}{\pi},$$

with the maximum at $x = 0$.

Step 4 (Outside the plateau). For $x \notin [-1,1]$ the two ramp contributions have opposite signs but larger denominators, hence smaller magnitude. More precisely, for $x > 1$, the left–ramp integral is $\leq 1/(\pi(x+1))$, while the right–ramp term is a principal value on $[1,2]$ against a C^∞ density

that vanishes at the endpoints; the standard C^1–vanishing at $y = 1, 2$ eliminates the endpoint singularity and keeps the PV finite and strictly smaller than its in–plateau counterpart (a short integration–by–parts argument on the right interval makes this explicit). By evenness, the same holds for $x < -1$. Consequently,

$$\sup_{x\in\mathbb{R}} |H'_\psi(x)| = \sup_{x\in(-1,1)} |H'_\psi(x)| \;\le\; \frac{2}{\pi}.$$

Putting Steps 1–4 together,

$$\left\|(\mathcal{H}[\varphi_L])'\right\|_\infty \;=\; \frac{1}{L}\,\|H'_\psi\|_\infty \;\le\; \frac{1}{L}\cdot\frac{2}{\pi}.$$

Hence we can take $C_H(\psi) \le 2/\pi < 0.65$. □

Certificate — weighted p-adaptive model at $\sigma_0 = 0.6$. *Fix $\sigma_0 = 0.6$, take $Q = 29$ and $p_{\min} = \mathrm{nextprime}(Q) = 31$.*
Use the p-adaptive weighted off-diagonal enclosure (for all $p \ne q$, uniformly in $\sigma \in [\sigma_0, 1]$):

$$\|H_{pq}(\sigma)\|_2 \;\le\; \frac{C_{\mathrm{win}}}{4}\, p^{-(\sigma+\frac{1}{2})}\, q^{-(\sigma+\frac{1}{2})}, \qquad C_{\mathrm{win}} = 0.25.$$

Prime sums (small block $p \le Q$). *With $\sigma_0 = 0.6$,*

$$S_{\sigma_0}(Q) \;=\; \sum_{p\le Q} p^{-\sigma_0} \;=\; 2.9593220929, \qquad S_{\sigma_0+\frac{1}{2}}(Q) \;=\; \sum_{p\le Q} p^{-(\sigma_0+\frac{1}{2})} \;=\; 1.3239981250.$$

In-block Gershgorin lower bounds (uniform on $[\sigma_0, 1]$). *Define*

$$L(p) \;:=\; (1-\sigma_0)\,(\log p)\,p^{-\sigma_0}, \qquad \mu_p^{\mathrm{L}} \;\ge\; 1 - \frac{L(p)}{6}.$$

At $p_{\min} = 31$ this gives

$$L(31) = 0.1750014502, \qquad \mu_{\min}^{\mathrm{far}} \;:=\; 1 - \frac{L(31)}{6} \;=\; 0.9708330916.$$

Over the small block $p \le Q$ the worst case is at $p = 5$:

$$L(5) = 0.2451050257, \qquad \mu_{\min}^{\mathrm{small}} \;:=\; 1 - \frac{L(5)}{6} \;=\; 0.9591491624.$$

Off-diagonal budgets (all rigorous). *Let $\sigma^\star := \sigma_0 + \frac{1}{2} = 1.1$.*
With the integer-tail majorant $\displaystyle\sum_{n\ge p_{\min}-1} n^{-\sigma^\star} \le \frac{(p_{\min}-1)^{1-\sigma^\star}}{\sigma^\star - 1}$ we obtain:

$$\Delta_{\mathrm{FS}} = \frac{C_{\mathrm{win}}}{4}\, p_{\min}^{-\sigma^\star}\, S_{\sigma^\star}(Q) = 0.0018935184,$$

$$\Delta_{\mathrm{FF}} = \frac{C_{\mathrm{win}}}{4}\, p_{\min}^{-\sigma^\star} \sum_{n\ge p_{\min}-1} n^{-\sigma^\star} \;\le\; \frac{C_{\mathrm{win}}}{4}\, p_{\min}^{-\sigma^\star}\, \frac{(p_{\min}-1)^{1-\sigma^\star}}{\sigma^\star - 1} = 0.0101781777,$$

$$\Delta_{\mathrm{SS}} = \frac{C_{\mathrm{win}}}{4}\, 2^{-\sigma^\star} \sum_{\substack{p\le Q\\ p\ne 2}} p^{-\sigma^\star} = 0.0250018328,$$

$$\Delta_{\mathrm{SF}} = \frac{C_{\mathrm{win}}}{4}\, 2^{-\sigma^\star} \sum_{n \ge p_{\min}-1} n^{-\sigma^\star} \;\le\; \frac{C_{\mathrm{win}}}{4}\, 2^{-\sigma^\star}\, \frac{(p_{\min}-1)^{1-\sigma^\star}}{\sigma^\star - 1} = 0.2075080249.$$

Certified finite-block spectral gap. *Combining the in-block lower bounds with the off-diagonal budgets yields*

$$\delta_{\mathrm{cert}}(\sigma_0) \;\ge\; \min\Big\{ \underbrace{\mu_{\min}^{\mathrm{small}} - (\Delta_{\mathrm{SS}} + \Delta_{\mathrm{SF}})}_{\textit{small-block rows}},\; \underbrace{\mu_{\min}^{\mathrm{far}} - (\Delta_{\mathrm{FS}} + \Delta_{\mathrm{FF}})}_{\textit{far-block rows}} \Big\} = 0.7266393047 \;>\; 0.$$

Hence the normalized finite block is uniformly positive definite on $[\sigma_0, 1]$.

Corollary 10 (Boundary-uniform smoothed control)**.** *Let* $I \Subset \mathbb{R}$, $\varepsilon_0 \in (0, \frac{1}{2}]$, *and* $\varphi \in C_c^2(I)$. *Then, uniformly for* $\sigma \in (\frac{1}{2}, \frac{1}{2} + \varepsilon_0]$,

$$\Big| \int_{\mathbb{R}} \varphi(t)\, \partial_\sigma\, \Re \log \det{}_2\, (I - A(\sigma + it))\, dt \Big| \;\le\; C_* \, \|\varphi''\|_{L^1(I)}.$$

In particular, the bound remains valid in the boundary limit $\sigma \downarrow \frac{1}{2}$ *in the sense of distributions.*

Smoothed Cauchy and outer limit (A2)

Proposition 11 (Outer normalization: existence, boundary a.e. modulus, and limit)**.** *There exist outer functions* $\mathcal{O}_\varepsilon$ *on* $\{\Re s > \frac{1}{2} + \varepsilon\}$ *with a.e. boundary modulus* $|\mathcal{O}_\varepsilon(\frac{1}{2} + \varepsilon + it)| = \exp u_\varepsilon(t)|$, *and* $\mathcal{O}_\varepsilon \to \mathcal{O}$ *locally uniformly on* Ω *as* $\varepsilon \downarrow 0$, *where* $\mathcal{O}$ *has boundary modulus* $\exp u(t)$. *Consequently the outer-normalized ratio* $\mathcal{J} = \det_2(I - A)/(\mathcal{O}\,\xi)$ *has a.e. boundary values on* $\Re s = \frac{1}{2}$ *with* $|\mathcal{J}(\frac{1}{2} + it)| = 1$.

Theorem 12 (Smoothed Cauchy for u_ε and convergence of outers)**.** *Let* $u_\varepsilon(t) := \log \Big| \det_2(I - A(\frac{1}{2} + \varepsilon + it)) \Big| - \log |\xi(\frac{1}{2} + \varepsilon + it)|$. *For each compact* $I \Subset \mathbb{R}$ *and each* $\varphi \in C_c^2(I)$ *there exists* $C(\varphi) < \infty$ *such that, uniformly for* $\varepsilon, \delta \in (0, \varepsilon_0]$,

$$\Big| \int_{\mathbb{R}} \varphi(t)\, (u_\varepsilon(t) - u_\delta(t))\, dt \Big| \;\le\; C(\varphi)\, |\varepsilon - \delta|.$$

Consequently, the outer normalizations $\mathcal{O}_\varepsilon$ *converge locally uniformly to an outer limit* $\mathcal{O}$ *on* Ω.

Proof. Fix $I \Subset \mathbb{R}$ and $\varphi \in C_c^2(I)$. For $0 < \delta < \varepsilon \le \varepsilon_0$,

$$\int \varphi\, (u_\varepsilon - u_\delta)\, dt = \int_\delta^\varepsilon \int \varphi(t)\, \partial_\sigma\, \Re\Big(\log \det_2(I - A) - \log \xi \Big)(\tfrac{1}{2} + \sigma + it)\, dt\, d\sigma.$$

By Lemma 1, $\big| \int \varphi\, \partial_\sigma \Re \log \det_2 \big| \le C_* \, \|\varphi''\|_{L^1(I)}$. For $\partial_\sigma \Re \log \xi = \Re(\xi'/\xi)$, test against φ via the Poisson extension on a fixed dilation $Q(\alpha I)$ and use Lemma 15:

$$\Big| \int \varphi\, \Re(\xi'/\xi) \Big| \;\lesssim\; \Big(\iint_{Q(\alpha I)} |\nabla U_\xi|^2\, \sigma \Big)^{1/2} \|\varphi\|_{H^1(I)} \;\lesssim\; |I|^{1/2}\, \|\varphi\|_{H^1(I)}.$$

Therefore $\big| \int \varphi\, (u_\varepsilon - u_\delta) \big| \le C(\varphi)\, |\varepsilon - \delta|$, proving the Lipschitz bound. Local-uniform convergence of outers follows from the Poisson representation and dominated convergence on $\{\Re s \ge \frac{1}{2} + \eta\}$. $\square$

Carleson energy and boundary BMO (unconditional)

We record a direct Carleson–energy route to boundary BMO for the limit $u(t) = \lim_{\varepsilon\downarrow 0} u_\varepsilon(t)$.

Lemma 13 (Arithmetic Carleson energy)**.** *Let*

$$U_{\det_2}(\sigma,t) \;:=\; \sum_p \sum_{k\ge 2} \frac{(\log p)\, p^{-k/2}}{k\log p}\, e^{-k\log p\,\sigma}\, \cos\big(k\log p\, t\big), \qquad \sigma>0.$$

Then for every interval $I\subset\mathbb{R}$ with Carleson box $Q(I) := I\times(0,|I|]$,

$$\iint_{Q(I)} |\nabla U_{\det_2}|^2\, \sigma\, dt\, d\sigma \;\le\; \frac{|I|}{4}\sum_p\sum_{k\ge 2}\frac{p^{-k}}{k^2} \;=:\; K_0\,|I|, \qquad K_0 := \frac{1}{4}\sum_p\sum_{k\ge 2}\frac{p^{-k}}{k^2} < \infty.$$

Proof. For a single mode $b\,e^{-\omega\sigma}\cos(\omega t)$ one has $|\nabla|^2 = b^2\omega^2 e^{-2\omega\sigma}$, hence

$$\int_0^{|I|}\int_I |\nabla|^2\,\sigma\, dt\, d\sigma \;\le\; |I|\cdot\sup_{\omega>0}\int_0^{|I|}\sigma\,\omega^2 e^{-2\omega\sigma} d\sigma\cdot b^2 \;\le\; \tfrac{1}{4}\,|I|\,b^2.$$

With $b = (\log p)\,p^{-k/2}/(k\log p)$ and $\omega = k\log p$, summing over (p,k) gives the claim and the finiteness of K_0. □

Whitney scale and short–interval zeros. *Throughout we use the Whitney schedule*

$$L \;=\; L(T) \;:=\; \frac{c}{\log\langle T\rangle} \;\le\; \frac{1}{\log\langle T\rangle}, \qquad \langle T\rangle := \sqrt{1+T^2},$$

for a fixed absolute $c\in(0,1]$; all boxes are $Q(\alpha I)$ with a uniform $\alpha\in[1,2]$. We work on Whitney boxes $Q(I)$ with

$$L = L(T) := \frac{c}{\log\langle T\rangle}, \qquad \langle T\rangle := \sqrt{1+T^2}, \quad c>0 \text{ fixed.}$$

There exist absolute $A_0, A_1 > 0$ such that for $T\ge 2$ and $0<H\le 1$,

$$N(T;H) \;:=\; \#\{\rho=\beta+i\gamma:\ \gamma\in[T,T+H]\} \;\le\; A_0 \;+\; A_1\,H\log\langle T\rangle.$$

Lemma 14 (Annular Poisson–balayage L^2 bound)**.** *Let $I=[T-L,T+L]$, $Q_\alpha(I) = I\times(0,\alpha L]$, and fix $k\ge 1$. For $\mathcal{A}_k := \{\rho=\beta+i\gamma:\ 2^kL<|T-\gamma|\le 2^{k+1}L\}$ set*

$$V_k(\sigma,t) := \sum_{\rho\in\mathcal{A}_k}\frac{\sigma}{(t-\gamma)^2+\sigma^2}.$$

Then

$$\iint_{Q_\alpha(I)} V_k(\sigma,t)^2\,\sigma\, dt\, d\sigma \;\ll_\alpha\; |I|\,4^{-k}\,\nu_k,$$

where $\nu_k := \#\mathcal{A}_k$, and the implicit constant depends only on α.

Proof. Write $K_\sigma(x) := \sigma/(x^2+\sigma^2)$. For any finite index set $\mathcal{J}$,

$$\Big(\sum_{j\in\mathcal{J}} K_\sigma(t-\gamma_j)\Big)^2 \le \sum_{j\in\mathcal{J}} K_\sigma(t-\gamma_j)^2 \;+\; 2\sum_{i<j} K_\sigma(t-\gamma_i)K_\sigma(t-\gamma_j).$$

Integrate over $t \in I$ first. For the diagonal terms,

$$\int_I K_\sigma(t-\gamma)^2\,dt \;=\; \sigma^2 \int_{t\in I} \frac{dt}{\big((t-\gamma)^2+\sigma^2\big)^2} \;\ll\; \frac{L}{\sigma}\cdot\frac{\sigma^2}{(2^kL)^2} \;=\; L\cdot\frac{\sigma}{4^kL^2},$$

since $|t-\gamma| \ge 2^kL - L \ge 2^{k-1}L$ for $t \in I$ and $k \ge 1$. Integrating σ over $(0, \alpha L]$ with the extra factor σ gives

$$\int_0^{\alpha L}\Big(\int_I K_\sigma(t-\gamma)^2\,dt\Big)\,\sigma\,d\sigma \;\ll_\alpha\; |I|\cdot 4^{-k}.$$

Summing over ν_k choices of γ contributes the factor ν_k.

For the off-diagonal terms, fix $i \ne j$. Over I we have $|t-\gamma_i|, |t-\gamma_j| \gtrsim 2^kL$, hence

$$\int_I K_\sigma(t-\gamma_i)K_\sigma(t-\gamma_j)\,dt \;\le\; \Big(\sup_{t\in I} K_\sigma(t-\gamma_i)\Big)\int_{\mathbb{R}} K_\sigma(t-\gamma_j)\,dt \;\ll\; \frac{\sigma}{(2^kL)^2}\cdot 1.$$

Integrating $\sigma \in (0, \alpha L]$ with the extra factor σ yields $\ll_\alpha L\cdot 4^{-k}$ again. Summing in i, j via the Gram/Schur bound with $f_j(t) := K_\sigma(t-\gamma_j)\mathbf{1}_I(t)$,

$$\sum_{i,j}\int_I f_i f_j \;\le\; \nu_k \sup_j \sum_i \int_I f_i f_j \;\ll\; \nu_k\,\frac{\sigma}{(2^kL)^2},$$

we obtain $\int_I V_k(\sigma,t)^2\,dt \;\ll\; \nu_k\,\sigma/(2^kL)^2$. Integrating in σ over $(0, \alpha L]$ with the extra factor σ gives $\ll_\alpha |I|\cdot 4^{-k}\,\nu_k$. Combining diagonal and off–diagonal parts gives the claim. □

Lemma 15 (Analytic (ξ) Carleson energy on Whitney boxes)**.** Reference. *The local zero count used below follows from the Riemann–von Mangoldt formula; see, e.g., Titchmarsh (Thm. 9.3) or Ivić (Ch. 8). A Vinogradov–Korobov zero-density refinement yields the stated strip bounds with explicit exponents (unconditional). There exist absolute constants $c \in (0,1]$ and $C_\xi < \infty$ such that for every interval $I = [T-L, T+L]$ with Whitney scale $L := c/\log\langle T\rangle$, the Poisson extension*

$$U_\xi(\sigma,t) := \Re\log\xi\big(\tfrac{1}{2}+\sigma+it\big), \qquad (\sigma > 0),$$

Whitney scale and neutralization. *Throughout this lemma we take the base interval $I = [T-L, T+L]$ with*

$$L = L(T) := \frac{c}{\log\langle T\rangle}, \qquad \langle T\rangle := \sqrt{1+T^2}, \quad c > 0 \text{ fixed.}$$

obeys the Carleson bound

$$\iint_{Q(I)} |\nabla U_\xi(\sigma,t)|^2\,\sigma\,dt\,d\sigma \;\le\; C_\xi\,|I|.$$

Proof. Write

$$\partial_\sigma U_\xi(\sigma,t) \;=\; \Re\frac{\xi'}{\xi}\Big(\tfrac{1}{2}+\sigma+it\Big) \;=\; \Re\sum_\rho \frac{1}{\frac{1}{2}+\sigma+it-\rho} \;+\; A(\sigma,t),$$

where the sum runs over nontrivial zeros $\rho = \beta + i\gamma$ of ζ, and $A(\sigma,t)$ collects the archimedean part and the trivial factors (these are smooth in (σ,t) on compact strips). Since U_ξ is harmonic, $|\nabla U_\xi|^2 \asymp |\partial_\sigma U_\xi|^2$ on $\mathbb{R}^2_+$; it suffices to estimate the latter.

Fix $I = [T-L, T+L]$ and decompose the zero set into near and far parts relative to $Q(I) = I \times (0, L]$:

$$\mathcal{Z}_{\text{near}} := \{\rho :\; |\gamma - T| \le 2L\}, \qquad \mathcal{Z}_{\text{far}} := \{\rho :\; |\gamma - T| > 2L\}.$$

Neutralized near field

Let B_I be the half-plane Blaschke product over zeros with $|\gamma - T| \le 3L$ and define the neutralized potential $\widetilde{U}_\xi := \Re \log(\xi B_I)$ and its σ-derivative $\widetilde{f} := \partial_\sigma \widetilde{U}_\xi$. Then $\sum_{\rho \in \mathcal{Z}_{\text{near}}} \nabla f_\rho$ is canceled inside $Q(I)$ up to a boundary error controlled by the Poisson energy of ψ (independent of T, L). Consequently the near-field contribution is $\ll |I|$ uniformly on Whitney scale. (Coarse note: the un-neutralized energy is $O(|I|)$, ensuring Carleson finiteness; however, the locked constant used in the certificate is the neutralized bound just stated. The coarse un-neutralized $O(1)$ is not used for numeric closure.)

For the far zeros (neutralized field), set annuli $\mathcal{A}_k := \{\rho : \ 2^k L < |\gamma - T| \le 2^{k+1} L\}$ for $k \ge 1$. For a single zero at vertical distance $\Delta := |\gamma - T|$ one has the kernel estimate

$$\int_0^L \int_{T-L}^{T+L} \frac{\sigma}{\sigma^2 + (t-\gamma)^2}\, dt\, d\sigma \ \ll \ L \Big(\frac{L}{\Delta}\Big)^2.$$

For the far annuli $\mathcal{A}_k$, apply Lemma 14 to the annular Poisson sums V_k to control cross terms linearly in the annular mass:

$$\iint_{Q(\alpha I)} \Big| \sum_{\rho \in \mathcal{A}_k} f_\rho \Big|^2 \sigma\, dt\, d\sigma \ \ll \ \frac{|I|}{4^k}\, \nu_k(\mathbb{R}),$$

where $\nu_k(\mathbb{R}) = \#\{\rho : \ 2^k L < |T - \gamma| \le 2^{k+1} L\}$. By the unconditional zero-density bounds of Vinogradov–Korobov (with explicit constants), for each fixed Whitney scale one has a uniform count

$$\nu_k(\mathbb{R}) \ \ll \ 2^k L \log\langle T\rangle \ + \ \log\langle T\rangle,$$

with the implied constant independent of T and k. Summing $k \ge 1$ yields a total far contribution

$$\ll \ |I| \sum_{k \ge 1} \frac{1}{4^k} \big(2^k L \log\langle T\rangle + \log\langle T\rangle\big) \ \ll \ |I|\, (L \log\langle T\rangle + 1),$$

which is $\ll |I|$ on the Whitney scale $L = c/\log\langle T\rangle$.

Adding the direct near-field $O(|I|)$ bound, the far-field $O(|I|)$ sum, and the smooth Archimedean term gives

$$\iint_{Q(\alpha I)} |\nabla U_\xi|^2 \sigma\, dt\, d\sigma \ \ll \ |I|.$$

This proves the claimed Carleson bound on Whitney boxes without neutralization in the energy step. □

Lemma 16 (Cutoff pairing on boxes)**.** *Fix parameters $\alpha' > \alpha > 1$. Let $\chi_{L,t_0} \in C_c^\infty(\mathbb{R}^2_+)$ satisfy $\chi \equiv 1$ on $Q(\alpha I)$, $\operatorname{supp}\chi \subset Q(\alpha' I)$, $\|\nabla\chi\|_\infty \lesssim L^{-1}$ and $\|\nabla^2\chi\|_\infty \lesssim L^{-2}$. Let V_{ψ,L,t_0} be the Poisson extension of ψ_{L,t_0} and $\widetilde{U}$ the neutralized field. Then*

$$\int_{\mathbb{R}} u(t)\, \psi_{L,t_0}(t)\, dt \ = \ \iint_{Q(\alpha' I)} \nabla \widetilde{U} \cdot \nabla\big(\chi_{L,t_0} V_{\psi,L,t_0}\big)\, dt\, d\sigma \ + \ \mathcal{R}_{\text{side}} \ + \ \mathcal{R}_{\text{top}},$$

where the remainder terms obey

$$|\mathcal{R}_{\text{side}}| + |\mathcal{R}_{\text{top}}| \ \lesssim \ \Big(\iint_{Q(\alpha' I)} |\nabla \widetilde{U}|^2 \sigma\Big)^{1/2} \cdot \Big(\iint_{Q(\alpha' I)} \big(|\nabla\chi|^2\, |V_{\psi,L,t_0}|^2 + |\nabla V_{\psi,L,t_0}|^2\big)\, \sigma\Big)^{1/2}.$$

Since ψ is fixed, the Poisson–energy factor involving V_{ψ,L,t_0} is scale–invariant in L, hence

$$|\mathcal{R}_{\text{side}}| + |\mathcal{R}_{\text{top}}| \ \lesssim \ \Big(\iint_{Q(\alpha' I)} |\nabla \widetilde{U}|^2 \sigma\Big)^{1/2}.$$

Lemma 17 (CR–Green pairing for boundary phase)**.** *Let J be analytic on Ω with a.e. boundary modulus $|J(\frac{1}{2}+it)|=1$, and write $\log J = U + iW$ on Ω, so U is harmonic with $U(\frac{1}{2}+it)=0$ a.e. Then on each Whitney interval $I=[t_0-L,t_0+L]$ and for the Poisson extension V_{ψ,L,t_0} of ψ_{L,t_0} one has the identity*

$$\int_{\mathbb{R}} \psi_{L,t_0}(t)\,(-W'(t))\,dt \;=\; \iint_{Q(\alpha' I)} \nabla U\cdot\nabla\big(\chi_{L,t_0}\,V_{\psi,L,t_0}\big)\,dt\,d\sigma \;+\; \mathcal{R}_{\text{side}} \;+\; \mathcal{R}_{\text{top}},$$

where χ_{L,t_0} is as in Lemma 16 and the remainders satisfy the same bound as in (16), namely

$$|\mathcal{R}_{\text{side}}|+|\mathcal{R}_{\text{top}}| \;\lesssim\; \Big(\iint_{Q(\alpha' I)} |\nabla U|^2\,\sigma\Big)^{1/2}\cdot\Big(\iint_{Q(\alpha' I)} (|\nabla\chi|^2\,|V_{\psi,L,t_0}|^2+|\nabla V_{\psi,L,t_0}|^2)\,\sigma\Big)^{1/2}.$$

In particular, by Cauchy–Schwarz and the scale–invariant Dirichlet bound for V_{ψ,L,t_0}, there is a constant $C(\psi)$ such that

$$\int_{\mathbb{R}} \psi_{L,t_0}(t)\,(-W'(t))\,dt \;\le\; C(\psi)\,\Big(\iint_{Q(\alpha' I)} |\nabla U|^2\,\sigma\Big)^{1/2}.$$

Proof. Fix $L>0$ and $t_0\in\mathbb{R}$. Since ψ has compact support in $[-2,2]$, the support of ψ_{L,t_0} is contained in $[t_0-2L,\,t_0+2L]$. Cover this interval by $N\le N_\psi$ consecutive intervals I_j of length $\asymp L$ with bounded overlap (each point belongs to at most M_ψ of the I_j), where N_ψ, M_ψ depend only on the support of ψ. Choose a C^∞ partition of unity $\{\eta_j\}_{j=1}^N$ with $\sum_j \eta_j\equiv 1$ on $[t_0-2L,\,t_0+2L]$, $\operatorname{supp}\eta_j\subset I_j$, and

$$\|\eta_j\|_{L^\infty}\le 1,\quad \|\nabla\eta_j\|_{L^\infty}\lesssim L^{-1},\quad \|\nabla^2\eta_j\|_{L^\infty}\lesssim L^{-2}.$$

Write $\psi_{L,t_0}=\sum_{j=1}^N \phi_j$ with $\phi_j:=\eta_j\,\psi_{L,t_0}$. For each j, apply Lemma 16 with a cutoff χ_j supported in a fixed dilation $Q(\alpha' I_j)$ and equal to 1 on $Q(\alpha I_j)$ to get

$$\Big|\int u\,\phi_j\Big| \;\le\; \Big(\iint_{Q(\alpha' I_j)} |\nabla\widetilde{U}|^2\,\sigma\Big)^{1/2}\cdot\Big(\iint_{Q(\alpha' I_j)} (|\nabla\chi_j|^2\,|V_{\phi_j}|^2+|\nabla V_{\phi_j}|^2)\,\sigma\Big)^{1/2}.$$

Because ψ is fixed and $\phi_j=\eta_j\psi_{L,t_0}$ with $\|\nabla\eta_j\|_\infty\lesssim L^{-1}$, the second factor is $\lesssim L^{1/2}\,\mathcal{A}(\psi)$ uniformly in j. Summing over j and using Cauchy–Schwarz gives

$$\Big|\int u\,\psi_{L,t_0}\Big| \;\le\; C_\psi\,\Big(\sum_{j=1}^N \iint_{Q(\alpha' I_j)} |\nabla\widetilde{U}|^2\,\sigma\Big)^{1/2}\,L^{1/2}\,\mathcal{A}(\psi).$$

Since the boxes $Q(\alpha' I_j)$ have bounded overlap (at most M_ψ) and are fixed dilations of Whitney boxes at height $\asymp L$, the Carleson box energy bound (Lemma 15) yields $\iint_{Q(\alpha' I_j)} |\nabla\widetilde{U}|^2\,\sigma\lesssim |I_j|\asymp L$. Therefore

$$\Big|\int u\,\psi_{L,t_0}\Big| \;\le\; C'\,(N_\psi M_\psi)^{1/2}\,L\,\mathcal{A}(\psi).$$

Dividing by L and taking the supremum in L,t_0 gives

$$C \;:=\; C'\,(N_\psi M_\psi)^{1/2}\,\mathcal{A}(\psi).$$

The Hilbert case follows identically with V_{ϕ_j} replaced by the test field for $(\mathcal{H}[\phi_j])'$, whose box energy is comparable by scale invariance. □

Corollary 18 (Unconditional local window constants)**.** *Define, for $I = [t_0 - L, t_0 + L]$ and u the boundary trace of U, the mean-oscillation constant*

$$M_\psi \;:=\; \sup_{L>0,\ t_0\in\mathbb{R}} \frac{1}{L}\Big|\int_{\mathbb{R}} (u(t) - u_I)\,\psi_{L,t_0}(t)\,dt\Big|, \qquad u_I := \frac{1}{|I|}\int_I u, \quad \psi_{L,t_0}(t) := \psi\big((t-t_0)/L\big),$$

and the Hilbert constant

$$C_H(\psi) \;:=\; \sup_{L>0,\ t_0\in\mathbb{R}} \frac{1}{L}\Big|\int_{\mathbb{R}} \mathcal{H}[u'](t)\,\psi_{L,t_0}(t)\,dt\Big|.$$

Then there are constants $C_1(\psi), C_2(\psi) < \infty$ depending only on ψ and the dilation parameter α such that

$$M_\psi \;\le\; C_1(\psi)\,\sqrt{C_{\text{area}}}\,\mathcal{A}(\psi), \qquad C_H(\psi) \;\le\; C_2(\psi)\,\sqrt{C_{\text{area}}}\,\mathcal{A}(\psi),$$

where the fixed Poisson energy of the window is

$$\mathcal{A}(\psi)^2 \;:=\; \iint_{\mathbb{R}^2_+} |\nabla(P_\sigma * \psi)|^2\,\sigma\,dt\,d\sigma \;<\; \infty.$$

In particular, both constants are finite and determined by local box energies.

Proof. Using the local box pairing and the neutralized-area bound over $Q(\alpha I)$ with a Dirichlet test field for ψ_{L,t_0}, the box test-field energy scales like $L\,\mathcal{A}(\psi)^2$ by construction, giving the M_ψ bound. For $C_H(\psi)$ integrate by parts and use the same construction for the test associated with $(\mathcal{H}[\psi_{L,t_0}])'$, whose box energy is comparable to that of ψ_{L,t_0} by scale invariance. □

Corollary 19 (Boundary BMO and window mean oscillation)**.** *For the mass–1 windows φ_I induced by an even $\psi \in C_c^\infty([-1,1])$ with $\int_{\mathbb{R}} \psi = 1$, there exists $M_\psi < \infty$ (depending only on ψ) such that for all Whitney intervals I,*

$$\int_I |u(t) - \ell_I(t)|\,dt \;\le\; M_\psi\,|I|,$$

where ℓ_I is any affine calibrant on I. In particular, the near-field bounds in the Hilbert pairing estimates hold uniformly in (T, L).

Hilbert pairing via affine subtraction (uniform in T, L)

Lemma 20 (Uniform Hilbert pairing bound (local box pairing))**.** *Let $\psi \in C_c^\infty([-1,1])$ be even with $\int_{\mathbb{R}} \psi = 1$ and define the mass–1 windows $\varphi_I(t) = L^{-1}\psi\big((t-T)/L\big)$. Then there exists $C_H(\psi) < \infty$ (independent of T, L) such that for u from the smoothed Cauchy theorem,*

$$\Big|\int_{\mathbb{R}} \mathcal{H}[u'](t)\,\varphi_I(t)\,dt\Big| \;\le\; C_H(\psi) \quad \textit{for all intervals } I.$$

Proof. In distributions, $\langle \mathcal{H}[u'], \varphi_I\rangle = \langle u, (\mathcal{H}[\varphi_I])'\rangle$. Because ψ is even, $(\mathcal{H}[\varphi_I])'$ annihilates constants and linear functions. Subtract the affine calibrant ℓ_I agreeing with u at the endpoints of I and write $v := u - \ell_I$. Using the same local box pairing for $(\mathcal{H}[\varphi_I])'$ restricted to a fixed dilation $Q(\alpha I)$ and the neutralized-area bound with the fixed test energy, one obtains

$$|\langle v, (\mathcal{H}[\varphi_I])'\rangle| \;\le\; C\,\sqrt{C_{\text{area}}}\,\mathcal{A}(\psi),$$

with a constant depending only on ψ (through the fixed Poisson energy) and α. This yields the claimed uniform bound by local box pairings. □

Lemma 21 (Hilbert-transform pairing)**.** *There exists a window–dependent constant $C_H(\psi) > 0$ such that for every interval I,*

$$\Big|\int_{\mathbb{R}} \mathcal{H}[u'](t)\,\varphi_I(t)\,dt\Big| \;\le\; C_H(\psi).$$

Proof. By Lemma 20, for mass–1 windows and even ψ, the pairing $\langle \mathcal{H}[u'], \varphi_I\rangle$ is uniformly bounded in (T, L). In distributions, $\langle \mathcal{H}[u'], \varphi_I\rangle = \langle u, (\mathcal{H}[\varphi_I])'\rangle$; evenness implies $(\mathcal{H}[\varphi_I])'$ annihilates affine functions. Subtract the affine calibrant on I and write $v = u - \ell_I$. The bound follows from the local box pairing in the Carleson energy lemma (Lemma 15) applied to the test field associated with $(\mathcal{H}[\varphi_I])'$. □

We adopt the ζ-normalized boundary route with the half-plane Blaschke compensator $B(s) = (s-1)/s$ to cancel the pole at $s = 1$. On $\Re s = \frac{1}{2}$, $|B| = 1$, so the compensator contributes no boundary phase and the Archimedean term vanishes. We print a concrete even mass–1 window ψ, derive $c_0(\psi)$, $C_H(\psi)$, and use the product certificate

$$\frac{(2/\pi)\,M_\psi}{c_0(\psi)} \;<\; \frac{\pi}{2}.$$

Printed window. *Let $\beta(x) := \exp\big(-1/(x(1-x))\big)$ for $x \in (0,1)$ and $\beta = 0$ otherwise. Define the smooth step*

$$S(x) := \frac{\int_0^{\min\{\max\{x,0\},1\}} \beta(u)\,du}{\int_0^1 \beta(u)\,du} \qquad (x \in \mathbb{R}),$$

so that $S \in C^\infty(\mathbb{R})$, $S \equiv 0$ on $(-\infty, 0]$, $S \equiv 1$ on $[1, \infty)$, and $S' \ge 0$ supported on $(0,1)$. Set the even flat–top window $\psi : \mathbb{R} \to [0,1]$ by

$$\psi(t) := \begin{cases} 0, & |t| \ge 2, \\ S(t+2), & -2 < t < -1, \\ 1, & |t| \le 1, \\ S(2-t), & 1 < t < 2. \end{cases}$$

Then $\psi \in C_c^\infty(\mathbb{R})$, $\psi \equiv 1$ on $[-1,1]$, and $\operatorname{supp}\psi \subset [-2,2]$. For windows we take $\varphi_L(t) := L^{-1}\psi(t/L)$.

Poisson lower bound. *As in the plateau computation already recorded, for $0 < b \le 1$ and $|x| \le 1$ one has*

$$(P_b * \psi)(x) \;\ge\; (P_b * \mathbf{1}_{[-1,1]})(x) \;=\; \frac{1}{2\pi}\Big(\arctan\tfrac{1-x}{b} + \arctan\tfrac{1+x}{b}\Big),$$

whence

$$c_0(\psi) \;:=\; \inf_{0<b\le 1,\ |x|\le 1} (P_b * \psi)(x) \;\ge\; 0.1762081912\,.$$

Derivation. For the normalized Poisson kernel $P_b(y) = \dfrac{1}{\pi}\dfrac{b}{b^2+y^2}$, for $|x| \le 1$

$$(P_b * \mathbf{1}_{[-1,1]})(x) = \frac{1}{\pi}\int_{-1}^{1} \frac{b}{b^2 + (x-y)^2}\,dy = \frac{1}{2\pi}\Big(\arctan\frac{1-x}{b} + \arctan\frac{1+x}{b}\Big).$$

Set $S(x,b) := \arctan\big((1-x)/b\big) + \arctan\big((1+x)/b\big)$. Symmetry gives $S(-x,b) = S(x,b)$. For $x \in [0,1]$,

$$\partial_x S(x,b) = \frac{1}{b}\Big(\frac{1}{1+\big(\frac{1+x}{b}\big)^2} - \frac{1}{1+\big(\frac{1-x}{b}\big)^2}\Big) \le 0,$$

so S decreases in x and is minimized at $x=1$. Also $\partial_b S(x,b) \le 0$ for $b>0$, so the minimum in $b \in (0,1]$ is at $b=1$. Thus the infimum occurs at $(x,b)=(1,1)$ giving $\frac{1}{2\pi}\arctan 2 = 0.1762081912\ldots$. Since $\psi \ge \mathbf{1}_{[-1,1]}$, this yields the bound for ψ. □

No Archimedean term in the ζ-normalized route. *Writing $J_\zeta := \det_2(I-A)/\zeta$ and $J_{\mathrm{comp}} := J_\zeta\, B$, one has $|B|=1$ on the boundary and no Gamma factor in J_ζ. Hence the boundary phase contribution from Archimedean factors is identically zero in the phase–velocity identity, i.e. $C_\Gamma \equiv 0$ for this normalization.*

We carry out the boundary phase test in the ζ–normalized gauge with the Blaschke compensator $B(s) = (s-1)/s$; on $\Re s = \frac{1}{2}$ one has $|B|=1$, so the Archimedean boundary contribution vanishes. Any residual interior effect is absorbed into the ζ–side box constant $C_{\mathrm{box}}^{(\zeta)} = K_0 + K_\xi$. In the a.e. wedge route no additive wedge constants are used.

Hilbert term (structural bound). *For the mass–1 window and even ψ, the local box pairing bound of Lemma 20 applies and is uniform in (T,L). We write the certificate in terms of the abstract window-dependent constant $C_H(\psi)$ from Lemma 20. An explicit envelope for the printed window is recorded below, but is not required for the symbolic certificate.*

Lemma 22 (Explicit envelope for the printed window)**.** *For the flat-top ψ above with symmetric monotone ramps of width $\varepsilon \in (0,1)$ on each side of ± 1, one has the variation bound*

$$\sup_{t\in\mathbb{R}} |\mathcal{H}[\varphi_L](t)| \;\le\; \frac{\mathrm{TV}(\psi)}{\pi}\,\log\frac{1+\varepsilon}{1-\varepsilon}, \qquad \mathrm{TV}(\psi) = 2.$$

In particular, with $\varepsilon = \frac{1}{5}$ one obtains the certified envelope

$$\sup_{t\in\mathbb{R}} |\mathcal{H}[\varphi_L](t)| \;\le\; \frac{2}{\pi}\,\log\tfrac{3}{2} \;\approx\; 0.258 \;<\; 0.26.$$

Consequently, we may take $C_H(\psi) \le 0.26$ for the printed window. This bound is uniform in L.

Derivation (variation/IBP estimate). Write $\psi = \mathbf{1}_{[-1,1]} + \eta$ with η supported on the disjoint transition layers $[1,1+\varepsilon]$ and $[-1-\varepsilon,-1]$, monotone on each layer, and total variation $\mathrm{TV}(\psi)=2$. Using the identity $\mathcal{H}[\psi](x) = \frac{1}{\pi}\,\mathrm{p.v.}\int \frac{\psi(y)}{x-y}\,dy = \frac{1}{\pi}\int \psi'(y)\,\log|x-y|\,dy$ (integration by parts; boundary cancellations by monotonicity/symmetry) and that ψ' is a finite signed measure of total variation $\mathrm{TV}(\psi)$, one gets

$$|\mathcal{H}\psi(x)| \;\le\; \frac{\mathrm{TV}(\psi)}{\pi}\sup_{y\in[-1-\varepsilon,\,1+\varepsilon]} \big|\log|x-y|\big| \;-\; \inf_{y\in[-1-\varepsilon,\,1+\varepsilon]} \big|\log|x-y|\big|.$$

The worst case is at $x=0$, yielding $|\mathcal{H}\psi(0)| \le \frac{\mathrm{TV}(\psi)}{\pi}\log\frac{1+\varepsilon}{1-\varepsilon}$. Scaling gives $\mathcal{H}[\varphi_L](t) = \mathcal{H}\psi\big((t-T)/L\big)$, so the same bound holds uniformly in L. Taking $\varepsilon = \frac{1}{5}$ gives the stated numeric envelope. □

Window mean-oscillation constant M_ψ: definition and bound. *For an interval $I = [T-L, T+L]$ and the boundary modulus $u(t) := \log|\det_2(I-A(\frac{1}{2}+it))| - \log|\xi(\frac{1}{2}+it)|$, define the mean-oscillation calibrant ℓ_I as the affine function matching u at the endpoints of I, and set*

$$M_\psi \;:=\; \sup_{T\in\mathbb{R},\ L>0} \frac{1}{|I|}\int_I \big|u(t)-\ell_I(t)\big|\,dt.$$

By the smoothed Cauchy theorem and the local pairing in a local pairing bound, one obtains a window-dependent constant bounding the mean oscillation uniformly over (T, L). For the printed flat-top window, Lemma 23 yields an explicit H^1–BMO/box-energy bound for M_ψ; in our calibration (see Numeric instantiation below), this gives a strict numerical bound well below the certificate threshold.

Lemma 23 (Window mean–oscillation via H^1–BMO and box energy)**.** *Let U be the Poisson extension of the boundary function u, and let $\mu := |\nabla U|^2\,\sigma\,dt\,d\sigma$. Fix the even C^∞ window ψ (support $\subset [-2,2]$, plateau on $[-1,1]$), and let $m_\psi := \int_{\mathbb{R}}\psi(x)\,dx$ denote its mass. Set*

$$\phi(t) := \psi(t) - \tfrac{m_\psi}{2}\,\mathbf{1}_{[-1,1]}(t), \qquad \phi_{L,t_0}(t) := \phi\Big(\frac{t-t_0}{L}\Big).$$

Define $M_\psi := \sup_{L>0,t_0\in\mathbb{R}} \frac{1}{L}\big|\int_{\mathbb{R}} u(t)\,\phi_{L,t_0}(t)\,dt\big|$ and

$$C_{\text{box}}^{(\text{Whitney})} := \sup_{I\,:\,|I|\asymp c/\log\langle T\rangle} \frac{\mu(Q(\alpha I))}{|I|}, \qquad C_\psi^{(H^1)} := \frac{1}{2}\int_{\mathbb{R}} S\phi(x)\,dx,$$

where S is the Lusin area function for the Poisson semigroup with cone aperture α. Then

$$M_\psi \;\le\; \frac{4}{\pi}\,C_{\text{CE}}(\alpha)\,C_\psi^{(H^1)}\,\sqrt{C_{\text{box}}^{(\text{Whitney})}}.$$

Proof. By H^1–BMO duality, for every $I = [t_0 - L, t_0 + L]$,

$$\Big|\int u\,\phi_{L,t_0}\Big| \;\le\; \|u\|_{\text{BMO}}\,\|\phi_{L,t_0}\|_{H^1}.$$

Carleson embedding (aperture α) gives

$$\|u\|_{\text{BMO}} \;\le\; \tfrac{2}{\pi}\,C_{\text{CE}}(\alpha)\,\big(C_{\text{box}}^{(\text{Whitney})}\big)^{1/2}.$$

Since S is scale-invariant in L^1 (up to $|I|$),

$$\|\phi_{L,t_0}\|_{H^1} \;=\; \int S(\phi_{L,t_0})(x)\,dx \;=\; 2L\,C_\psi^{(H^1)}.$$

Divide by L to conclude. □

Carleson box linkage. *With $U = U_{\det_2} + U_\xi$ on the boundary in the ζ–normalized route, the box constant used in the certificate is*

$$C_{\text{box}}^{(\zeta)} \;:=\; K_0 \;+\; K_\xi.$$

No separate Γ–area term enters the certificate path.

Numeric instantiation. *All concrete values (audited constants for K_0, K_ξ, the ζ–side box constant $C_{\mathrm{box}}^{(\zeta)}$, the evaluation of $C_\psi^{(H^1)}$, and the locked M_ψ) are printed in Appendix H and used in the in-paper certificate.*

- ***Window:*** *Take a fixed C^∞ even window ψ with $\psi \equiv 1$ on $[-1,1]$ and $\operatorname{supp}\psi \subseteq [-2,2]$, and set $\varphi_L(t) = L^{-1}\psi(t/L)$.*
- ***Poisson lower bound.*** *Using the closed form for the plateau and monotonicity, one obtains*

$$c_0(\psi) \;:=\; \inf_{0<b\le 1,\ |x|\le 1} (P_b * \psi)(x) \;\ge\; \frac{1}{2\pi} \inf_{0<b\le 1,\ |x|\le 1} \left(\arctan\tfrac{1-x}{b} + \arctan\tfrac{1+x}{b}\right) \;\ge\; 0.1762081912\,.$$

- ***Archimedean term.*** *In the ζ-normalized route with the Blaschke compensator at $s=1$, the Archimedean contribution vanishes: $C_\Gamma = 0$.*
- ***Hilbert term.*** *For the chosen smooth window, we denote by $C_H(\psi)$ the window-uniform constant from Lemma 20. Any explicit envelope bound for $\sup_t |\mathcal{H}[\varphi_L](t)|$ may be inserted; we keep the inequality in symbolic form with $C_H(\psi)$.*
- ***Inequality form.*** *With $M_\psi \le 0.07180621$, the certificate reads*

$$\frac{(2/\pi)\, M_\psi}{c_0(\psi)} \;<\; \frac{\pi}{2}.$$

Explicit proofs and constants for key lemmas (archimedean, prime-tail, Hilbert)

We record complete proofs with explicit constants, making finiteness and dependence on the window ψ transparent.

P1. Explicit prime-tail bounds (unconditional)

Fix $\alpha \in (1,2]$ (in our use: $\alpha \in [2\sigma_0, 2]$ with $\sigma_0 > \frac{1}{2}$). For all $x \ge 17$ one has the Rosser–Schoenfeld style bound

$$\sum_{p>x} p^{-\alpha} \;\le\; \frac{1.25506\,\alpha}{(\alpha-1)\,\log x}\, x^{\,1-\alpha}. \tag{2}$$

This follows by partial summation together with $\pi(t) \le 1.25506\, t/\log t$ for $t \ge 17$. A uniform variant over $\alpha \in [\alpha_0, 2]$ (with $\alpha_0 := 2\sigma_0 > 1$) is

$$\sum_{p>x} p^{-\alpha} \;\le\; \frac{1.25506\,\alpha_0}{(\alpha_0-1)\,\log x}\, x^{\,1-\alpha_0} \qquad (x \ge 17). \tag{3}$$

Two convenient alternatives:

$$\sum_{p>x} p^{-\alpha} \;\le\; \frac{\alpha}{(\alpha-1)(\log x - 1)}\, x^{1-\alpha} \qquad (x \ge 599) \tag{4}$$

$$\sum_{p>x} p^{-\alpha} \;\le\; \sum_{n>\lfloor x\rfloor} n^{-\alpha} \;\le\; \frac{x^{1-\alpha}}{\alpha-1} \qquad (x > 1). \tag{5}$$

Use in ($\star$) and covering. *To enforce a tail $\sum_{p>P} p^{-\alpha} \le \eta$ it suffices, by (2), to take $P \ge 17$ solving*

$$\frac{1.25506\,\alpha}{(\alpha-1)\,\log P}\,P^{\,1-\alpha} \;\le\; \eta.$$

The practical choice $P = \max\{17,\ ((1.25506\,\alpha)/((\alpha-1)\eta))^{1/(\alpha-1)}$ already meets the inequality up to the mild $\log P$ factor; one may increase P monotonically until the left side is $\le \eta$.

Finite-block spectral gap certificate on $[\sigma_0, 1]$

Let $\sigma_0 \in (\frac{1}{2}, 1]$ and $\mathcal{I} = \{(p,n) :\ p \le P$ prime, $1 \le n \le N_p\}$. Let $H(\sigma) \in \mathbb{C}^{|\mathcal{I}|\times|\mathcal{I}|}$ be the Hermitian block matrix of the truncated finite block at abscissa σ, partitioned as $H = [H_{pq}]_{p,q\le P}$ with $H_{pq}(\sigma) \in \mathbb{C}^{N_p\times N_q}$. Write $D_p(\sigma) := H_{pp}(\sigma)$ and $E(\sigma) := H(\sigma) - \mathrm{diag}(D_p(\sigma))$.

Lemma 24 (Block Gershgorin lower bound)**.** *For every $\sigma \in [\sigma_0, 1]$,*

$$\lambda_{\min}(H(\sigma)) \;\ge\; \min_{p\le P}\Big(\lambda_{\min}(D_p(\sigma)) \;-\; \sum_{q\ne p} \|H_{pq}(\sigma)\|_2\Big).$$

Lemma 25 (Schur–Weyl bound)**.** *For every $\sigma \in [\sigma_0, 1]$,*

$$\lambda_{\min}(H(\sigma)) \;\ge\; \min_{p} \lambda_{\min}(D_p(\sigma)) \;-\; \|E(\sigma)\|_2.$$

Moreover, for any weights $w_p > 0$,

$$\|E(\sigma)\|_2 \;\le\; \max_q \frac{1}{w_q} \sum_{p\ne q} w_p\, \|H_{pq}(\sigma)\|_2.$$

Proposition 26 (Uniform spectral gap by interval/block bounds)**.** *Assume that for each block entry we have interval enclosures $H_{pq}[i,j](\sigma) \in [\underline{h}_{pq}[i,j], \overline{h}_{pq}[i,j]]$ valid for all $\sigma \in [\sigma_0, 1]$. Define*

$$\mu_p^L \;:=\; \min_{1\le i\le N_p}\Big(\underline{h}_{pp}[i,i] - \sum_{j\ne i} \max\{\,|\underline{h}_{pp}[i,j]|,\ |\overline{h}_{pp}[i,j]|\,\}\Big), \qquad U_{pq} \;:=\; \sqrt{\max_j \sum_i \sup|H_{pq}[i,j]| \cdot \max_i \sum_j \sup|H_{pq}[i}$$

where $\sup|\cdot|$ denotes the larger magnitude of the interval endpoints. Then, uniformly for $\sigma \in [\sigma_0, 1]$,

$$\lambda_{\min}(H(\sigma)) \;\ge\; \delta(\sigma_0), \qquad \delta(\sigma_0) := \max\Big\{0,\ \min_p\Big(\mu_p^L - \sum_{q\ne p} U_{pq}\Big),\ \min_p \mu_p^L - \max_q \frac{1}{\sqrt{\mu_q^L}} \sum_{p\ne q} \sqrt{\mu_p^L}\, U_{pq}\Big\}.$$

Proof. Apply Lemma 24 with the in-block Gershgorin lower bound $\lambda_{\min}(D_p) \ge \mu_p^L$, and Lemma 25 with the weighted Schur test using $w_p = \sqrt{\mu_p^L}$. The interval definitions of μ_p^L and U_{pq} ensure uniformity in $\sigma \in [\sigma_0, 1]$. □

Determinant–zeta link (L1; corrected domain)

Lemma 27 (L1 (corrected): $\det_2$–zeta identity on $\Re s > 1$)**.** *Let $\mathcal{H} := \bigoplus_{p \text{ prime}} \mathbb{C}\, u_p$ with orthonormal $\{u_p\}_p$, and for $\Re s > 1$ set*

$$T(s) \;:=\; \bigoplus_p p^{-s}\,\Pi_p, \qquad \Pi_p\, x \;:=\; \langle x, u_p\rangle u_p.$$

Then

$$\log\det{}_2(I-T(s)) \;=\; -\sum_{m\ge 2}\frac{\operatorname{Tr}\,T(s)^m}{m} \;=\; \sum_p\sum_{m\ge 2}\frac{p^{-ms}}{m} \;=\; \log\zeta(s) \;-\; \sum_p p^{-s}, \qquad (\Re s>1).$$

Equivalently, for $\Re s>1$,

$$\zeta(s) \;=\; \exp\Bigl(\sum_p p^{-s}\Bigr)\;\det{}_2(I-T(s)).$$

On $\Re s>\frac{1}{2}$, *define* $L(s):=\log\xi(s)-\log\det{}_2\left(I-T_{\mathrm{new}}(s)\right)$ *with* $T_{\mathrm{new}}=T+K$ *strictly upper–triangular, anchored by* $L(2)\in\mathbb{R}$. *Then* e^L *is holomorphic and zero–free on* $\{\Re s>\frac{1}{2}\}$ *and*

$$\xi(s) \;=\; e^{L(s)}\;\det{}_2(I-T_{\mathrm{new}}(s)), \qquad (\Re s>\tfrac{1}{2}).$$

Remark 28 (Using prime-tail bounds). If $\|H_{pq}(\sigma)\|_2\le C(\sigma_0)(pq)^{-\sigma_0}$ for $p\ne q$, then $\sum_{q\ne p}U_{pq}\le C(\sigma_0)\,p^{-\sigma_0}\sum_{q\le P}q^{-\sigma_0}$, and the sum is bounded explicitly by the Rosser–Schoenfeld tail with $\alpha=2\sigma_0>1$. Thus $\delta(\sigma_0)>0$ can be certified by choosing $P,\{N_p\}$ so that the off-diagonal budget is dominated by $\min_p\mu_p^L$.

Truncation tail control and global assembly (P4)

Write the head/tail split by primes as $\mathcal{P}_{\le P}=\{p\le P\}$ *and* $\mathcal{P}_{>P}=\{p>P\}$. *In the normalised basis at* σ_0 *set*

$$X:=\left[\widetilde{H}_{pq}\right]_{p,q\le P},\quad Y:=\left[\widetilde{H}_{pq}\right]_{p\le P<q},\quad Z:=\left[\widetilde{H}_{pq}\right]_{p,q>P}.$$

Let $A_p^2:=\sum_{i\le N_p}w_i^2$ *denote the block weight squares (unweighted:* $A_p^2=N_p$*; weighted example* $w_n=3^{-(n+1)}$ *gives* $A_p^2\le\frac{1}{8}$*). Define*

$$S_2(\le P):=\sum_{p\le P}A_p^2p^{-2\sigma_0},\qquad S_2(>P):=\sum_{p>P}A_p^2p^{-2\sigma_0}.$$

Then

$$\|Y\|\le C_{\mathrm{win}}\sqrt{S_2(\le P)S_2(>P)},\qquad \lambda_{\min}(Z)\ge\mu_{\mathrm{diag}}-C_{\mathrm{win}}S_2(>P),$$

where $\mu_{\mathrm{diag}}:=\inf_{p>P}\mu_p^{\mathrm{L}}$. *Consequently,*

$$\lambda_{\min}(\mathbb{A})\ge\min\Bigl\{\,\delta_P-\frac{C_{\mathrm{win}}^2S_2(\le P)S_2(>P)}{\mu_{\mathrm{diag}}-C_{\mathrm{win}}S_2(>P)}\,,\;\mu_{\mathrm{diag}}-C_{\mathrm{win}}S_2(>P)\Bigr\},$$

with δ_P *the head finite-block gap from above. Using the integer tail* $\sum_{n>P}n^{-2\sigma_0}\le(P-1)^{1-2\sigma_0}/(2\sigma_0-1)$ *yields a closed-form tail bound for* $S_2(>P)$.

Small-prime disentangling (P3). *Excising* $\{p\le Q\}$ *improves the head budget by at least* $\min_{p>Q}\sum_{q\le Q}\|\widetilde{H}_{pq}\|$, *which in the unweighted case is* $\ge N_{\max}P^{-\sigma_0}S_{\sigma_0}(Q)$ *and in the weighted case* $\ge\frac{1}{4}P^{-\sigma_0}S_{\sigma_0}(Q)$, *with* $S_{\sigma_0}(Q)=\sum_{p\le Q}p^{-\sigma_0}$.

No-hidden-knobs audit (P6)

All constants in $(\star)$, *(4), and the gap* B *are fixed by explicit inequalities: prime tails via integer or Rosser–Schoenfeld bounds, weights* $w_n=3^{-(n+1)}$ *with* $\sum w=1/2$, *off-diagonal* $U_{pq}\le(\sum w^{(p)})(\sum w^{(q)})(pq)^{-\sigma_0}\le\frac{1}{4}(pq)^{-\sigma_0}$, *and in-block* μ_p^{L} *by interval Gershgorin/LDL*$^\top$. *No tuned parameters enter;* $P(\sigma_0,\varepsilon)$, $N_p(\sigma_0,\varepsilon,P)$, *and* B *are determined from these definitions.*

Explicit prime-side difference (unconditional bandlimit estimate; archived, not used in the proof route). *Let* $\mathcal{P}(t) := \Im((\zeta'/\zeta) - (\det_2{}' / \det_2))(\frac{1}{2} + it) = \sum_p (\log p)\, p^{-1/2} \sin(t \log p)$. *Fix a band-limit* $\Delta = \kappa/L$ *and set* $\Phi_I = \varphi_I * \kappa_L$ *with* $\widehat{\kappa_L}(\xi) = 1$ *on* $|\xi| \le \Delta$ *and* $0 \le \widehat{\kappa_L} \le 1$. *By Plancherel and Cauchy–Schwarz,*

$$\left| \int_{\mathbb{R}} \mathcal{P}(t)\, \Phi_I(t)\, dt \right| \le \left(\sum_{\log p \le \kappa/L} \frac{(\log p)^2}{p} \, |\widehat{\Phi_I}(\log p)|^2 \right)^{1/2} \cdot \left(\sum_{\log p \le \kappa/L} 1 \right)^{1/2}.$$

Since $|\widehat{\Phi_I}(\xi)| \le L\, |\widehat{\psi}(L\xi)|\, \|\widehat{\kappa_L}\|_\infty \le L\, \|\psi\|_{L^1}$ *and, unconditionally,* $\sum_{p \le x} (\log p)^2/p \ll (\log x)^2$ *by partial summation and Chebyshev's bound* $\theta(x) \ll x$ *(Titchmarsh), we obtain*

$$\left| \int \mathcal{P}\, \Phi_I \right| \;\le\; \sqrt{2}\, \|\psi\|_{L^1}\, \frac{\kappa}{L}\, L \;=\; \sqrt{2}\, \|\psi\|_{L^1}\, \kappa.$$

Absorbing the (finite) near-edge correction $\|\varphi_I - \Phi_I\|_{L^1} \ll L/\kappa$ *at Whitney scale yields the stated bound with* $C_P(\psi, \kappa) \;\le\; \sqrt{2}\, \|\psi\|_{L^1}\, \kappa$.

Theorem 29 (Limit $N \to \infty$ on rectangles: $2J$ Herglotz, Θ Schur)**.** *Let* $R \Subset \Omega$ *with* $\xi \ne 0$ *on a neighborhood of* $\overline{R}$. *Then* $2\mathcal{J}_N \to 2\mathcal{J}$ *locally uniformly on* R, *and* $\Re(2\mathcal{J}) \ge 0$ *on* R. *Consequently,* $\Theta = (2\mathcal{J} - 1)/(2\mathcal{J} + 1)$ *is Schur on* R.

Proof. By the $HS \to \det_2$ convergence proposition, $\det_2(I - A_N) \to \det_2(I - A)$ locally uniformly on R. Since ξ is bounded away from zero on R, division is continuous, hence $\mathcal{J}_N \to \mathcal{J}$ locally uniformly on R. Each $2\mathcal{J}_N$ is Herglotz on R, and Herglotz functions are closed under local-uniform limits; therefore $\Re(2\mathcal{J}) \ge 0$ on R. The Cayley transform yields that Θ is Schur on R. □

Corollary 30 (Unconditional Schur on $\Omega \setminus Z(\xi)$)**.** *For every compact* $K \Subset \Omega \setminus Z(\xi)$, *there exists a rectangle* $R \Subset \Omega$ *with* $K \subset R$ *and* $\xi \ne 0$ *on* $\overline{R}$. *Hence, by Theorem 29,* Θ *is Schur on* R, *and therefore on* K. *Exhausting* $\Omega \setminus Z(\xi)$ *by such* K *shows that* Θ *is Schur on* $\Omega \setminus Z(\xi)$.

Lemma 31 (Removable singularity under Schur bound)**.** *Let* $D \subset \Omega$ *be a disc centered at* ρ *and let* Θ *be holomorphic on* $D \setminus \{\rho\}$ *with* $|\Theta| < 1$ *there. Then* Θ *extends holomorphically to* D. *In particular, the Cayley inverse* $(1 + \Theta)/(1 - \Theta)$ *extends holomorphically to* D *with nonnegative real part.*

Proof. Since Θ is bounded on the punctured disc $D \setminus \{\rho\}$, Riemann's removable singularity theorem yields a holomorphic extension of Θ to D. Where $|\Theta| < 1$, the Cayley inverse is analytic with $\Re \frac{1+\Theta}{1-\Theta} \ge 0$; continuity extends this across ρ. □

Theorem 32 (Boundary positivity for H_{J_N})**.** *On* ∂R, *the Herglotz kernel* $H_{J_N}(s, \bar{t}) := (J_N(s) + \overline{J_N(t)})/(s + \bar{t} - 1)$ *is positive semidefinite (in the punctured sense along* Σ_R*).*

Kernel-positivity route (summary for (P+)). *For the boundary positivity step (P+), we avoid Schur/Gershgorin absolute-value sums and use kernel factorizations:*

- *$\det_2$ leg: additive/log Gram positivity and the symmetric Fock lift provide a PSD lower bound for the $\det_2$ Herglotz kernel on rectangle boundaries; the Szegő kernel is factored by a Laplace integral.*
- *finite k=1/archimedean leg: realized in a finite lossless KYP block adding a finite Gram summand.*

- *division by ξ: on punctured boundaries, diagonal congruence by ξ^{-1} preserves PSD; limits along node sets reach zeros.*
- *boundary passage: outer normalization on $\Re s = \frac{1}{2} + \varepsilon$ and smoothed/distributional L^1 control with a Cauchy limit yield a boundary a.e. normalized ratio $\mathcal{J}$.*
- *phase–velocity and Poisson: the phase–velocity identity reduces (P+) to a short-interval Poisson/Carleson mass bound for off–critical zeros; Poisson then lifts (P+) to Herglotz in Ω, hence Schur for Θ.*

Theorem 33 (Interior Schur control on zero-free rectangles)**.** *Let $R \Subset \Omega$ be a rectangle with $R \cap Z(\xi) = \varnothing$. Then $|\Theta_N| \le 1$ on R for all N. Moreover, for every compact $K \Subset R$, we have $\Theta_N \to \Theta$ uniformly on K. Consequently, Θ is Schur on $\Omega \setminus Z(\xi)$.*

Proof. If boundary positivity/contractivity holds on ∂R, then by the maximum principle $\Re J_N \ge 0$ on R; hence $|\Theta_N| \le 1$ on R, so Θ_N is Schur on R. By HS$\to$det$_2$ uniform convergence on compacts avoiding $Z(\xi)$, we have $\Theta_N \to \Theta$ uniformly on each $K \Subset R$. Exhausting $\Omega \setminus Z(\xi)$ yields local Schur control there. In the main chain we obtain the boundary positivity (P+) from the product certificate, and removability yields the extension across $Z(\xi)$ and globalization. □

Theorem 34 (Product certificate $\Rightarrow$ wedge (P+))**.** *As stated and proved in the certificate section. The proof invokes Lemmas 6,9, and the balanced Carleson bound, and it uses the conservative numeric closure (Corollary 5).*

Lemma 35 (Local certificate $\Rightarrow$ a.e. boundary wedge (outline))**.** *Let $w : \mathbb{R} \to \mathbb{R}$ be the boundary phase of J on $\partial\Omega$ with $|J(\frac{1}{2} + it)| = 1$ a.e., and let $-w'$ be its distributional derivative (a positive Radon measure). Suppose that for every Whitney interval $I = [t_0 - L, t_0 + L]$ with $L = c/\log\langle t_0\rangle$ there exists a nonnegative bump $\varphi_I \in C_c^\infty(I)$ with $\int_{\mathbb{R}} \varphi_I = 1$ such that*

$$\int_{\mathbb{R}} \varphi_I(t)\,(-w'(t))\,dt \;\le\; \pi\,\Upsilon \;<\; \tfrac{\pi}{2}.$$

Then, after multiplying the outer O by a unimodular constant, one has $|w(t)| \le \pi\Upsilon$ for a.e. $t \in \mathbb{R}$. In particular, (P+) *holds whenever $\pi\Upsilon < \frac{\pi}{2}$.*

Proof outline. Since $-w' \ge 0$, w is nonincreasing. For each I write $\operatorname{osc}_I(w) := \operatorname{ess\,sup}_I w - \operatorname{ess\,inf}_I w$. Choosing φ_I comparable to the normalized triangular kernel on I and integrating by parts gives

$$\operatorname{osc}_I(w) \;\le\; C \int_{\mathbb{R}} \varphi_I\,(-w') \;\le\; C\,\pi\Upsilon \quad \text{(uniformly over Whitney } I\text{)},$$

for an absolute constant C. The Whitney family has bounded overlap and shrinks to points. Let m be any limit point of the medians $\operatorname{med}_I(w)$ along a Whitney exhaustion; replacing O by $e^{-im}O$ replaces w by $w - m$. Passing to the limit along $I \downarrow \{t\}$ and using monotonicity yields $|w(t)| \le C\,\pi\Upsilon$ for a.e. t. With the present normalization of the triangular test one has $C = 1$, giving $|w(t)| \le \pi\Upsilon < \frac{\pi}{2}$. □

Lemma 36 ($\zeta \leftrightarrow \xi$ phase bridge on Whitney scale)**.** *On each Whitney interval $I = [T-L, T+L]$ with $L = c/\log\langle T\rangle$, the phase difference between ξ and ζ obeys*

$$\Big|\arg\xi(\tfrac{1}{2}{+}it) - \arg\zeta(\tfrac{1}{2}{+}it)\Big| \;\le\; C_\Gamma\,L \quad (t \in I), \qquad C_\Gamma = 0$$

in the ζ-normalized route (no Archimedean boundary contribution). Equivalently, the archimedean factor contributes no area to $C_{\mathrm{box}}^{(\zeta)}$ and does not affect the certificate.

Proof. We have $\xi = G\,\zeta$ with $G(s) = \frac{1}{2}s(1-s)\,\pi^{-s/2}\,\Gamma(s/2)$. On $\Re s = \frac{1}{2}$, the boundary phase derivative of $\log G$ equals the Poisson kernel of the interior zero at $s = 1$ (from $s(1-s)$) plus the boundary derivative of $\log(\pi^{-s/2}\Gamma(s/2))$. Insert the compensator $B(s) = (s-1)/s$; then $\log(B/G)$ has purely harmonic real part on Ω and unimodular boundary values. Consequently its boundary phase derivative integrates to zero on every Whitney box. By the outer normalization step, replacing ξ by ζ and multiplying by B preserves the phase-velocity identity used in (P+). Thus no Archimedean contribution appears and $C_\Gamma = 0$. □

Globalization across $Z(\xi)$ and completion of the proof

Standing hypotheses from the previous section (AAB; ζ-normalized). *Assume (P+) has been established via the AAB route (adaptive $\kappa(T)$; a.e. wedge) in the ζ-normalized boundary gauge with the Blaschke compensator $B(s) = (s-1)/s$ (no Archimedean boundary phase). Hence, by Poisson transport, the function*

$$F(s) \;:=\; 2\,\mathcal{J}(s)$$

has nonnegative real part on the right half-plane

$$\Omega \;:=\; \{\, s \in \mathbb{C} :\; \Re s > \tfrac{1}{2} \,\}$$

away from the zero set $Z(\xi) \cap \Omega$; equivalently, its Cayley transform

$$\Theta(s) \;:=\; \frac{F(s) - 1}{F(s) + 1}$$

is Schur on $\Omega \setminus Z(\xi)$, i.e. $|\Theta(s)| \le 1$ there. We also recall the definition (consistent with the ζ-normalized route)

$$\mathcal{J}(s) \;=\; \frac{\det_2(I - A(s))}{\mathcal{O}(s)\,\xi(s)}$$

with $\mathcal{O}(s)$ holomorphic and nonvanishing on Ω, and $A(s)$ Hilbert-Schmidt with point spectrum $\{p^{-s}\}_p$ strictly inside the unit disc for $\Re s > \frac{1}{2}$, so that $I - A(s)$ is invertible and $\det_2(I - A(s)) \neq 0$ on Ω. On the boundary, $|B| = 1$ and cancels the simple pole of ζ at $s = 1$, so the Archimedean boundary phase is absent.

Lemma 1 (Removability for Θ at $Z(\xi)$). *Let $\rho \in Z(\xi) \cap \Omega$. Then Θ extends holomorphically to $s = \rho$ with $|\Theta(\rho)| \le 1$.*

Proof. *ρ is an isolated point of $Z(\xi)$. On $D \setminus \{\rho\}$ (a small disc around ρ), Θ is holomorphic and $|\Theta| \le 1$. By Riemann's removable-singularity theorem, Θ extends holomorphically to D with a finite value at ρ, necessarily in the closed unit disc.* □

Lemma 2 (No interior pole for a Herglotz pullback). *With Θ as above, one cannot have $\Theta(\rho) = 1$ for any $\rho \in Z(\xi) \cap \Omega$.*

Proof. *Suppose $\Theta(\rho) = 1$. Write $g(s) := 1 - \Theta(s)$. Then g is holomorphic with a zero at ρ: $g(s) = (s-\rho)^m h(s)$, $m \ge 1$, $h(\rho) \neq 0$. On $D \setminus \{\rho\}$,*

$$F(s) \;=\; \frac{1 + \Theta(s)}{1 - \Theta(s)} \;=\; \frac{2 - g(s)}{g(s)} \;=\; \frac{2}{g(s)} - 1$$

so F has a pole of order m at ρ. Choose an angle θ with $h(\rho)e^{im\theta} \in \mathbb{R}_{>0}$ and set $s = \rho + re^{i\theta}$. Then

$$\Re F(s) \;=\; \Re\Big(\frac{2}{(s-\rho)^m h(s)} - 1\Big) \sim \frac{2}{|h(\rho)|}\,\frac{1}{r^m} \quad (r \downarrow 0).$$

Rotating by π/m flips the sign and yields $\Re F(\rho + re^{i(\theta+\pi/m)}) \sim -\frac{2}{|h(\rho)|}\frac{1}{r^m}$, which violates $\Re F \geq 0$ near ρ. Contradiction. Hence $\Theta(\rho) \neq 1$. □

Lemma 3 (Holomorphic and Herglotz extension of F). *For each $\rho \in Z(\xi) \cap \Omega$, F extends holomorphically to $s = \rho$ with $\Re F \geq 0$ preserved.*

Proof. *By Lemma 1, Θ extends holomorphically to ρ with $|\Theta(\rho)| \leq 1$, and by Lemma 2, $\Theta(\rho) \neq 1$. Thus the Cayley inverse*

$$F(s) \;=\; \frac{1 + \Theta(s)}{1 - \Theta(s)}$$

is holomorphic at $s = \rho$. Since $\Re F \geq 0$ holds on punctured neighborhoods and F is continuous at ρ, it persists at $s = \rho$. □

Theorem (Globalization across $Z(\xi)$). *Under (P+), $2\mathcal{J}$ is Herglotz and Θ is Schur on Ω. By the removable-singularity and covering argument, this extends across $Z(\xi)$ and RH holds.* Proof. *Apply Lemma 3 at every $\rho \in Z(\xi) \cap \Omega$ and use that the zeros of ξ are discrete.* □

Corollary (Zero-free right half-plane). *$\xi(s) \neq 0$ for all $s \in \Omega$.*

Proof. *On Ω, $\det_2(I - A)$ and $\mathcal{O}$ never vanish, so $\mathcal{J} = \det_2(I - A)/(\mathcal{O}\,\xi)$ can only fail to be holomorphic where ξ vanishes. But $\mathcal{J}$ is holomorphic on Ω by the Theorem, hence ξ has no zeros there.* □

Conclusion (RH). *By the functional equation $\xi(s) = \xi(1 - s)$ and conjugation symmetry, zeros are symmetric with respect to the critical line. Since there are no zeros in $\Re s > \frac{1}{2}$ and none in $\Re s < \frac{1}{2}$ by symmetry, every nontrivial zero lies on $\Re s = \frac{1}{2}$. This completes the proof.*

Corollary 37 (Poisson transport)**.** *From 34, $2\mathcal{J}$ is Herglotz on Ω.*

Corollary 38 (Cayley)**.** *$\Theta = \frac{2\mathcal{J}-1}{2\mathcal{J}+1}$ is Schur on Ω.*

Theorem 39 (Globalization across $Z(\xi)$)**.** *Under (P+), $2\mathcal{J}$ is Herglotz and Θ is Schur on Ω. By the removable-singularity and covering argument, this extends across $Z(\xi)$ and RH holds.*

Corollary 40 (No far-far budget from triangular padding)**.** *Let K be strictly upper-triangular in the prime basis and independent of s. Then its contribution to the far-far Schur budget vanishes: $\Delta_{\mathrm{FF}}^{(K)} = 0$.*

Proof. In the prime order, K has no entries on or below the diagonal. Hence there are no cycles confined to the far block induced by K, and no far→far absolute-sum contribution. Thus the far-far row/column sums are unchanged. □

Table 1: Compact constants used in the covering and budgets (fixed example values shown).

Arithmetic energy	$K_0 = \frac{1}{4}\sum_p \sum_{k\ge 2} \dfrac{p^{-k}}{k^2}$
Prime cut / minimal prime	$Q = 29,\ \ p_{\min} = 31$
Tail bounds	$\sum_{p>x} p^{-\alpha} \le \dfrac{1.25506\,\alpha}{(\alpha-1)\,\log x}\, x^{1-\alpha}$ (for $x \ge 17$)
Row/col budgets	$\Delta_{SS}, \Delta_{SF}, \Delta_{FS}, \Delta_{FF}$ as in Theorem (Schur global bound)
In-block lower bounds	$\mu^{\text{small}} = 1 - \Delta_{SS},\ \ \mu^{\text{far}} = 1 - \frac{L(p_{\min})}{6}$
Link barrier	$L(\sigma) = (1-\sigma)(\log p_{\min})\, p_{\min}^{-\sigma}$
Lipschitz constant	$K(\sigma) = S_{\sigma+1/2}(Q) + \frac{1}{4}\, p_{\min}^{-\sigma} S_\sigma(Q)$
Prime sums	$S_\alpha(Q) = \sum_{p\le Q} p^{-\alpha},\ \ T_\alpha(p_{\min}) = \sum_{p\ge p_{\min}} p^{-\alpha}$

3 Appendix: Carleson embedding constant for fixed aperture

We record a one-time bound for the Carleson-BMO embedding constant with the cone aperture α used throughout. For the Poisson extension U of u and the area measure $\mu = |\nabla U|^2\,\sigma\,dt\,d\sigma$, the conical square function with aperture α satisfies the Carleson embedding inequality

$$\|u\|_{\text{BMO}} \;\le\; \frac{2}{\pi}\, C_{\text{CE}}(\alpha)\, \Big(\sup_I \frac{\mu(Q(\alpha I))}{|I|}\Big)^{1/2}.$$

4 Appendix: Numerical evaluation of $C_\psi^{(H^1)}$ for the printed window

We record a reproducible computation of the window constant

$$C_\psi^{(H^1)} \;:=\; \frac{1}{2}\int_{\mathbb{R}} S\phi\,dx, \qquad \phi(x) := \psi(x) - \frac{m_\psi}{2}\,\mathbf{1}_{[-1,1]}(x), \quad m_\psi := \int_{\mathbb{R}} \psi.$$

Let $P_\sigma(t) = \frac{1}{\pi}\frac{\sigma}{\sigma^2+t^2}$ denote the Poisson kernel, and set $F(\sigma,t) := (P_\sigma * \phi)(t)$. For a fixed cone aperture α (as in the main text), the Lusin area function is

$$S\phi(x) \;:=\; \Big(\iint_{\Gamma_\alpha(x)} |\nabla F(\sigma,t)|^2\,\sigma\,dt\,d\sigma\Big)^{1/2}, \qquad \Gamma_\alpha(x) := \{(\sigma,t) : |t-x| < \alpha\sigma,\ \sigma > 0\}.$$

Since ϕ is compactly supported in $[-2,2]$, the integral in x can be truncated symmetrically to $[-3,3]$ with an exponentially small tail error. Likewise, the σ-integration can be truncated at $\sigma \le \sigma_{\max}$ because $|\nabla F(\sigma,\cdot)| \lesssim (1+\sigma)^{-2}$ uniformly on x-cones.

Reproducible enclosures (camera-ready). *With the mass-1 normalization ($m_\psi = 1$) and zero-mean template ϕ above, one has*

$$C_\psi^{(H^1)} \;:=\; \tfrac{1}{2}\int_{\mathbb{R}} S\phi\,dx, \qquad \textit{(cones at } 45^\circ\textit{)}.$$

Two independent enclosures are recorded:

- *Analytic enclosure: $C_\psi^{(H^1)} < 0.245$ (explicit Poisson/cone tail control).*
- *Quadrature: $C_\psi^{(H^1)} = 0.23973 \;\pm\; 3\times 10^{-4}$. We lock $C_\psi^{(H^1)} = 0.2400$ in the main text.*

These bounds match the mass-1 scaling used for $\varphi_{L,t_0}(t) = \phi((t-t_0)/L)$, and the fixed-aperture embedding yields $M_\psi \le (4/\pi)\, C_\psi^{(H^1)} \sqrt{C_{\text{box}}}$.

Lemma 41 (Fixed-aperture Carleson embedding for window pairing)**.** *Let $\psi \in C_c^\infty([-2,2])$ be even, and define the zero-mean template*

$$\phi(x) := \psi(x) - \tfrac{1}{2}\,\mathbf{1}_{[-1,1]}(x), \qquad \phi_{L,t_0}(t) := \phi\Big(\frac{t-t_0}{L}\Big).$$

For the boundary function u with Poisson extension U on Ω, set

$$M_\psi := \sup_{L>0,\ t_0\in\mathbb{R}} \frac{1}{L}\Big|\int_{\mathbb{R}} u(t)\,\phi_{L,t_0}(t)\,dt\Big|, \qquad C_{\text{box}}^{(\text{Whitney})} := \sup_{I\,:\,|I|\asymp c/\log\langle T\rangle} \frac{1}{|I|}\iint_{Q(I)} |\nabla U|^2\,\sigma\,dt\,d\sigma,$$

and let $C_\psi^{(H^1)} := \frac{1}{2}\int_{\mathbb{R}} S\phi(x)\,dx$, where S is the Lusin area functional for the Poisson semiggroup with 45° cones. Then

$$M_\psi \;\le\; \frac{4}{\pi}\,C_\psi^{(H^1)}\,\sqrt{C_{\text{box}}^{(\text{Whitney})}}\,.$$

Proof (normalization and aperture made explicit). By H^1–BMO duality (see, e.g., Garnett, Bounded Analytic Functions, Thm. IV.4.4) and the identity $\langle \mathcal{H}[u'], \phi_{L,t_0}\rangle = \langle u, (\mathcal{H}\phi_{L,t_0})'\rangle$ in $\mathcal{D}'(\mathbb{R})$, the pairing against u is controlled by the H^1 norm of ϕ_{L,t_0}. For 45° cones, the Lusin area functional S gives an equivalent H^1 norm with the constant $2/\pi$ for the half-plane Poisson semigroup; see also the Carleson embedding (Garnett, Thm. VI.1.1) which yields

$$|\langle \mathcal{H}[u'], \phi_{L,t_0}\rangle| \;\le\; \frac{2}{\pi}\,\|S\phi_{L,t_0}\|_{L^1}\,\sqrt{C_{\text{box}}^{(\text{Whitney})}}\,.$$

Taking the supremum over (L,t_0) and noting $\|S\phi_{L,t_0}\|_{L^1} = 2\,C_\psi^{(H^1)}$ by scaling and the mass–1 normalization of ϕ_{L,t_0}, we obtain the stated bound with factor $(4/\pi)\,C_\psi^{(H^1)}\sqrt{C_{\text{box}}^{(\text{Whitney})}}$. The constants depend only on the fixed cone aperture (45°); changing aperture rescales the universal factor by an absolute constant. □

Locked Constants (with cross-references)

Policy note. ***The proof uses the conservative numeric certificate (Cor. 5) for the quantitative closure.*** *The balanced Carleson bound (Lemma 100) is the structural justification (no ξ–only energy; removable singularities) and is not used to lock numbers. For the printed window and outer normalization, we record once:*

$$c_0(\psi) = 0.17620819 \quad (\textit{Appendix H}), \quad C_\Gamma = 0\ (\zeta\textit{-normalized}).$$

With the a.e. wedge, the closing condition is

$$\pi\Upsilon \;<\; \tfrac{\pi}{2}.$$

Sum-form route: choose $\kappa = 10^{-3}$ so $C_P = 0.002$ and use the analytic envelope bound $C_H(\psi) \le 0.26$ (Lemma 22). Then

$$\frac{C_\Gamma + C_P + C_H}{c_0} = \frac{0 + 0.002 + 0.26}{0.17620819} = 1.4869 < \frac{\pi}{2}$$

(archival PSC corollary). Product-form route (used in the proof): with the locked value $C_\psi^{(H^1)} = 0.2400$ and $C_{\text{box}}^{(\zeta)} = 0.05521808$, we have

$$M_\psi = \tfrac{4}{\pi}\,C_\psi^{(H^1)}\sqrt{C_{\text{box}}^{(\zeta)}} = 0.07180621, \qquad \Upsilon = \frac{(2/\pi)\,M_\psi}{c_0} = 0.25943.$$

PSC certificate (locked constants; canonical form)

Locked evaluation used throughout (revised; product route via Υ):

$$\begin{aligned}(c_0,\ C_H,\ C_\psi^{(H^1)},\ C_{\text{box}}) &= (0.17620819,\ 2/\pi,\ 0.2400,\ 0.05521808),\\ M_\psi &= 0.07180621,\\ \Upsilon &= \frac{(2/\pi)\cdot 0.07180621}{0.17620819} = 0.25943.\end{aligned}$$

See Appendices B–G for the derivations and enclosures.

Reproducible numerics (self-contained). *For the printed window and the ζ–normalized route:*

- $c_0(\psi)$*: Poisson plateau infimum (Appendix H) — exact value with digits*

$$c_0(\psi) = 0.17620819.$$

- K_0*: arithmetic tail* $\frac{1}{4}\sum_p \sum_{k\geq 2} p^{-k}/k^2$ *with explicit tail enclosure — locked*

$$K_0 = 0.03486808.$$

- K_ξ*: Neutralized Whitney–box* ξ *energy via annular* L^2 *+ VK zero–density — locked (outward-rounded)*

$$K_\xi = 0.02035000.$$

- $C_{\text{box}}^{(\zeta)}$*:* $= K_0 + K_\xi$ *— used in certificate only*

$$C_{\text{box}}^{(\zeta)} = 0.05521808.$$

- $C_\psi^{(H^1)}$*: analytic enclosure* < 0.245 *and quadrature* $0.23973 \pm 3\times 10^{-4}$*; we lock*

$$C_\psi^{(H^1)} = 0.2400.$$

- M_ψ*: Fefferman–Stein/Carleson embedding*

$$M_\psi = \tfrac{4}{\pi}\, C_\psi^{(H^1)} \sqrt{C_{\text{box}}^{(\zeta)}} = 0.07180621.$$

- Υ*: product certificate value (no prime budget)*

$$\Upsilon = \frac{(2/\pi)\cdot 0.07180621}{0.17620819} = 0.25943.$$

Each number is computed once and locked with outward rounding. The certificate wedge uses only c_0, C_H, M_ψ *and the a.e. boundary passage.*

Non-circularity (sequencing). *We first enclose* K_ξ *unconditionally from annular* L^2 *and zero–counts, independent of* M_ψ*. We then evaluate* M_ψ *via* $(4/\pi)\, C_\psi^{(H^1)}\sqrt{C_{\text{box}}^{(\zeta)}}$ *using the locked* $C_{\text{box}}^{(\zeta)} = K_0 + K_\xi$*. No step uses* M_ψ *to bound* K_ξ*, so there is no feedback.*

Closure of Stage 2: Product certificate ⇒ (P+) and PSC (archived density)

Definitions and standing normalizations

Let $\Omega := \{s \in \mathbb{C} : \ \Re s > \frac{1}{2}\}$ *and write* $s = \frac{1}{2} + it$ *on the boundary. Set Let* $P_b(x) := \frac{1}{\pi}\frac{b}{b^2+x^2}$ *and let* $\mathcal{H}$ *denote the boundary Hilbert transform.*

Poisson lower bound. *Define*

$$c_0(\psi) \ := \inf_{0<b\le 1,\ |x|\le 1} (P_b * \psi)(x) \ \ge \ 0.1762081912\,.$$

For the printed flat–top window this is locked as

Lemma 42 (Adaptive AAB prime budget; uniform in height)**.** *Fix an even* C_c^∞ *mass-1 window* ψ *supported in* $[-2,2]$ *with* $\int \psi = 1$*. Let* $L(T) = c/\log\langle T\rangle$ *with* $c \in (0,1]$ *fixed, and define the adaptive bandwidth* $\Delta_0(T) = \frac{1}{2}\log\log\langle T\rangle$ *for* $T \ge e^e$*. Choose a* C_c^∞ *taper* η *with* $\eta \equiv 1$ *on* $[-1,1]$*,* $\|\eta^{(m)}\|_\infty \le 2^m$ *for* $m \ge 0$*. Set* $\widehat{\kappa_L}(\xi) = \eta(\xi/\Delta_0(T))$ *and* $\Phi_{L,t_0} = \varphi_{L,t_0} * \kappa_L$*, where* $\varphi_{L,t_0}(t) = L^{-1}\psi((t-t_0)/L)$*.*

Then, uniformly for all $T \ge 2$ *and* $t_0 \in \mathbb{R}$*,*

$$\Big|\int_{\mathbb{R}} \mathcal{P}(t)\,\Phi_{L,t_0}(t)\,dt\Big| \le C_1\,(\log T)^{-1/2} + C_2\,(\log T)^{-(m+1/2)} \le 0.04 = 0.04,$$

where $\mathcal{P}(t) = \sum_p (\log p) p^{-1/2}\sin(t\log p)$*,* C_1, C_2 *depend only on* ψ, η, c, m *(one may fix* $m = 2$*), and the bound is outward-rounded.*

with explicit absolute constants $A_0 = 2$ *and* $A_1 = \dfrac{1}{2\pi}$*. Equivalently,*

$$\#\{\rho : \ |\Im\rho - T| < W\} \ \le \ A_0\,W\log T \ + \ A_1\log T.$$

Theorem 43 (Boundary wedge from the product certificate — a.e. form)**.** *Let* $I = [T-L, T+L]$ *avoid the ordinates of critical-line zeros, i.e.* $I \cap \{\gamma : \xi(\frac{1}{2}+i\gamma) = 0\} = \varnothing$*. Then*

$$\int_I (-w')(t)\,dt \ \le \ \pi\,\mu(Q(I)) \ \le \ \frac{\pi}{c_0(\psi)}\int_{\mathbb{R}} (-w')(t)\,\varphi_{L,T}(t)\,dt \ \le \ \pi\,\Upsilon \ < \ \tfrac{\pi}{2}.$$

Hence $w(t) \in [-\pi/2, \pi/2]$ *for a.e.* $t \in \mathbb{R}$*, so* (P+) *holds and* $2\mathcal{J}$ *is Herglotz on* Ω*.*

$$c_0(\psi) \ = \ 0.17620819.$$

Here the normalization is coherent with the mass–1 scaling: the window has plateau value 1 on $[-1,1]$, and the pairing uses $\varphi_L(t) = L^{-1}\psi(t/L)$, so the Poisson minimum above is exactly the quantity that enters the certificate denominator $c_0(\psi)$.

Hilbert envelope. With $u(t) := \log|\mathcal{J}(\frac{1}{2}+it)|$ and the local mean–oscillation constant

$$M_\psi \ := \sup_{L>0,\ t_0\in\mathbb{R}} \frac{1}{L}\int_{t_0-L}^{t_0+L} |u(t) - \ell_I(t)|\,dt, \qquad I = [t_0-L, t_0+L],$$

one has the scale–free pairing bound

$$\Big|\int_{\mathbb{R}} \mathcal{H}[u'](t)\,\varphi_L(t)\,dt\Big| \ \le \ C_H(\psi)\,M_\psi\,, \qquad C_H(\psi) \ \le \ 0.65.$$

(Integration by parts in $\mathcal{D}'(\mathbb{R})$ and the fact that $(\mathcal{H}[\varphi_L])'$ annihilates affine functions yield the reduction to M_ψ.)

$\mathbf{H}^1$–BMO control of M_ψ (Whitney boxes). Let μ denote the positive measure of off–critical zeros of ξ on Ω, counted with multiplicity. For a bounded interval $I = [T-L, T+L]$ with Whitney scale $L = c/\log\langle T\rangle$ and aperture parameter $\alpha > 1$, write the Carleson box

$$Q(\alpha I) \;:=\; \{\, \tfrac{1}{2} + \sigma + it : \; t \in I, \; 0 < \sigma \le \alpha L \,\}.$$

In the phase–velocity identity, the zero contribution over I equals the Poisson balayage of μ over $Q(\alpha I)$; we write this mass as $\mu(Q(I))$ when α is fixed once and for all. Let $C_\psi^{(H^1)}$ denote the fixed H^1–norm of the zero–mean template associated to ψ (mass 1). With the Carleson box area constant on the ζ–side $C_{\mathrm{box}}^{(\zeta)}$ for the Poisson extension of u, the standard H^1–BMO/Carleson embedding bound gives

$$M_\psi \;\le\; \frac{4}{\pi}\, C_\psi^{(H^1)} \sqrt{C_{\mathrm{box}}^{(\zeta)}}.$$

For the printed window and audited area bound, we have the analytic enclosure

$$C_\psi^{(H^1)} \;<\; 0.245, \qquad C_{\mathrm{box}}^{(\zeta)} = 0.05521808.$$

Consequently,

$$M_\psi \;\le\; \frac{4}{\pi}\, C_\psi^{(H^1)} \sqrt{C_{\mathrm{box}}^{(\zeta)}} \;\le\; \frac{4}{\pi} \cdot 0.245 \cdot \sqrt{0.05521808} \;=\; 0.07330218,$$

which still yields

$$\Upsilon \;=\; \frac{(2/\pi)\cdot 0.07180621}{0.17620819} \;=\; 0.25943, \qquad \pi\Upsilon < \tfrac{\pi}{2}.$$

Lemma 44 (Poisson scale reduction)**.** *For every $L > 0$ and $\varphi_L(t) = L^{-1}\psi(t/L)$,*

$$(P_a * \varphi_L)(t) \;=\; (P_{a/L} * \psi)\Big(\frac{t}{L}\Big), \qquad a > 0, \; t \in \mathbb{R}.$$

Hence $\inf_{0<a\le L,\; |t|\le L} (P_a * \varphi_L)(t) = c_0(\psi)$.

Product certificate ⇒ boundary wedge and (P+)

Route status. We prove (P+) only via the product certificate. The PSC/Carleson density discussion is archived and not used in the main chain.

Fix the mass–1 printed window ψ with $\psi \equiv 1$ on $[-1,1]$ and $\operatorname{supp}\psi \subset [-2,2]$, and set

$$\varphi_{L,t_0}(t) \;:=\; \frac{1}{L}\,\psi\Big(\frac{t-t_0}{L}\Big), \qquad \int_{\mathbb{R}} \varphi_{L,t_0} = 1, \quad \operatorname{supp}\varphi_{L,t_0} \subset I.$$

On intervals avoiding critical-line ordinates, the a.e. wedge follows directly from the product certificate without additive constants.

Theorem 45 (Boundary wedge from the product certificate)**.** *For every bounded interval I and mass–1 test φ_{L,t_0} supported in I, one has*

$$c_0(\psi)\,\mu(Q(I)) \;\le\; \int_{\mathbb{R}} (-w')(t)\,\varphi_{L,t_0}(t)\,dt \;\le\; C(\psi)\,\Big(\iint_Q(\alpha' I)|\nabla U|^2\,\sigma\Big)^{1/2}.$$

If $\pi\,\Upsilon < \frac{\pi}{2}$, then $w \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ a.e. on $\mathbb{R}$, hence (P+) *holds and $2\mathcal{J}$ is Herglotz on Ω.*

Proof. By Lemma 44, $(P_a * \varphi_{L,t_0})(t) \geq c_0(\psi)$ for all $(t,a) \in I \times (0,L]$. Pairing the phase–velocity identity with φ_{L,t_0} and localizing to $Q(I)$ gives the left inequality $c_0(\psi)\,\mu(Q(I)) \leq \int_{\mathbb{R}}(-w')\,\varphi_{L,t_0}$; the critical–line sum is nonnegative. For the right inequality, apply Lemma 17 and the Carleson box energy bound for U to obtain a uniform-in-L bound proportional to $\big(\iint_Q(\alpha' I)|\nabla U|^2\sigma\big)^{1/2}$, which is controlled by the locked constant $C_\mathrm{box}^{(\zeta)}$ (balanced or conservative route). Since $\int_I(-w') \leq \pi\,\mu(Q(I)) \leq \pi\Upsilon$, the hypothesis $\pi\Upsilon < \frac{\pi}{2}$ forces $w \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ a.e. on I. Exhausting $\mathbb{R}$ by such I yields (P+) and the Herglotz property. □

Theorem 46 (Numerical certificate closes (P+) with $\Upsilon \leq \frac{1}{2}$)**.** *With the printed window ψ and adaptive $\kappa(T)$,*

$$\Upsilon \;:=\; \frac{C_H(\psi)\,M_\psi \;+\; C_P(\kappa(T))}{c_0(\psi)} \;=\; \frac{(2/\pi)\cdot 0.07180621 + 0.040}{0.17620819} \;=\; 0.25943.$$

Since $\pi 0.25943 < \frac{\pi}{2}$, the boundary wedge holds and (P+) *follows; in particular, $2\mathcal{J}$ is Herglotz on Ω and $\Theta = \mathcal{C}[2\mathcal{J}]$ is Schur on Ω.*

Corollary 47 (AAB wedge implication — a.e. form)**.** *Let $\Upsilon := (C_H(\psi)M_\psi + C_P(\kappa(T)))/c_0(\psi)$. For every Whitney interval $I = [T-L, T+L]$ avoiding line zero ordinates,*

$$\int_I(-w'(t))\,dt \;\leq\; \pi\,\Upsilon.$$

In particular, if $\pi\Upsilon < \frac{\pi}{2}$ then the boundary wedge (P+) *holds.*

Conclusion. The product certificate with locked constants yields the boundary wedge and hence (P+) (Herglotz/Schur on Ω). PSC density (sum–form) is archived and not used to deduce (P+).

Theorem 48 (Product certificate closes (P+) for the printed window (archived; not used))**.** *With the printed mass–1 window ψ and bandlimit choice $\kappa = 0.010$, the verified constants satisfy*

$$\frac{C_H(\psi)\,M_\psi \;+\; C_P(\kappa(T))}{c_0(\psi)} \;<\; \frac{\pi}{2}.$$

Consequently the boundary positive-real statement (P+) *holds for $\mathcal{J} = \det_2(I-A)/(\mathcal{O}\,\xi)$ on $\Re s = \frac{1}{2}$. In particular, $2\mathcal{J}$ is Herglotz on Ω and $\Theta = \mathcal{C}[2\mathcal{J}]$ is Schur on Ω.*

Proof. Apply the phase–velocity identity together with the distributional pairing identity

$$\int_{\mathbb{R}}\mathcal{H}[u'](t)\,\varphi_I(t)\,dt \;=\; \int_{\mathbb{R}}(u(t)-\ell_I(t))\,\big(\mathcal{H}[\varphi_I]\big)'(t)\,dt,$$

which holds by integration by parts in $\mathcal{D}'(\mathbb{R})$ and the fact that $(\mathcal{H}[\varphi_I])'$ annihilates affine functions (hence the calibrant ℓ_I may be inserted). Using the scaling bound

$$\big\|\big(\mathcal{H}[\varphi_I]\big)'\big\|_{L^\infty} \;\leq\; \frac{C_{\mathcal{H}}(\psi)}{L} \;\leq\; \frac{C_H(\psi)}{L}, \qquad C_H(\psi) \leq 0.65$$

for the printed window and the mean–oscillation control defining M_ψ, we obtain

$$\Big|\int_{\mathbb{R}}\mathcal{H}[u']\,\varphi_I\,dt\Big| \;\leq\; \frac{C_H(\psi)}{L}\int_I|u-\ell_I|\,dt \;\leq\; C_H(\psi)\,M_\psi.$$

Together with the bandlimit term $C_P(\kappa(T)) \le 0.04$ (Lemma 42) and the Poisson lower bound $c_0(\psi)$, this gives the canonical product form

$$\frac{C_H(\psi)\, M_\psi + C_P(\kappa(T))}{c_0(\psi)} \;<\; \frac{\pi}{2}.$$

With the locked constants below, the displayed strict inequality verifies the PSC Carleson bound on all short intervals. The PSC density is not used to deduce (P+); the (P+) step is proved solely via the product certificate. Poisson transport then gives $2\mathcal{J}$ Herglotz on Ω and the Cayley transform yields Schur for Θ. □

Global Carleson box constant: explicit derivation and numerical closure

Fix a Whitney schedule with interval I of length L centered at height t_0, box

$$Q(\alpha I) \;=\; \{\, s = \tfrac{1}{2} + \sigma + it :\; t \in I,\; 0 \le \sigma \le \alpha L \,\}, \qquad \alpha \in [1,2],$$

and Carleson ratio

$$C_{\text{box}} \;:=\; \sup_{t_0, I, \alpha} \frac{1}{|I|} \iint_{Q(\alpha I)} |\nabla U(s)|^2\, \sigma\, dt\, d\sigma,$$

where $U = \Re \log J$ is the neutralized potential used in the PSC schedule. We split

$$U \;=\; U_0 \;+\; U_\xi \;+\; U_\Gamma,$$

with U_0 the $k \ge 2$ Euler–product tail (prime powers), U_ξ the neutralized $\Re \log \xi$ contribution (local affine corrector subtracted on I), and U_Γ the gamma/archimedean term of ξ. Since U is harmonic on $Q(\alpha I)$ (no zeros or poles inside by construction of the PSC window), each piece contributes additively to the Carleson ratio:

$$C_{\text{box}} \;\le\; K_0 \;+\; K_\xi.$$

A. Prime–power tail. Write

$$\log \zeta(s) \;=\; \sum_p \sum_{k \ge 1} \frac{p^{-ks}}{k}, \qquad U_0(s) \;=\; \Re \sum_p \sum_{k \ge 2} \frac{p^{-ks}}{k}.$$

Using $|\nabla \Re f|^2 = |f'|^2$ for analytic f, $\frac{d}{ds} p^{-ks} = -(\log p)\, p^{-ks}$, and the Whitney geometry of $Q(\alpha I)$ (width L, height αL), a direct computation gives

$$K_0 \;=\; \frac{1}{4} \sum_p \sum_{k \ge 2} \frac{p^{-k}}{k^2}.$$

Set $P(k) := \sum_p p^{-k}$. Numerically,

$$\sum_{k=2}^{20} \frac{P(k)}{k^2} \;=\; 0.139472297865\ldots$$

and for the tail we use $P(k) \le \zeta(k) - 1$ and the integral remainder

$$\sum_{k \ge 21} \frac{\zeta(k) - 1}{k^2} \;\le\; \sum_{k \ge 21} \frac{2^{-k}}{k^2} \;+\; \sum_{k \ge 21} \frac{2^{1-k}}{(k-1)k^2} \;<\; 2.2 \times 10^{-9}.$$

Hence

$$K_0 \;=\; \frac{1}{4}\sum_{k\ge 2}\frac{P(k)}{k^2} \;\le\; \frac{1}{4}\Big(0.139472300\Big) \;=\; 0.034868075,$$

and we lock

$$\boxed{K_0 \;\le\; 0.03486808}.$$

B. Neutralized $\Re\log\xi$ (zeros) term. Let ρ range over nontrivial zeros and set $z=\frac{1}{2}+it_0$ (box center). Write the affine corrector on $I=[T-L,T+L]$ as

$$\ell_I(t) \;:=\; a_I \;+\; b_I\,(t-T), \qquad a_I, b_I\in\mathbb{R},$$

chosen so that the neutralized trace $u_I(t):=u(t)-\ell_I(t)$ obeys the moment cancellations

$$\int_I u_I(t)\,dt \;=\; 0, \qquad \int_I (t-T)\,u_I(t)\,dt \;=\; 0.$$

With this neutralization, the far-field kernel has vanishing average and first moment, and the Taylor remainder of $\sigma/|s-\rho|^2$ (degree ≤ 1 in $t-T$) controls the pairing. In particular, each remote zero contributes with cubic decay. A one-line bound gives, for any ρ with $r:=|z-\rho|\ge L$,

Lemma 49 (Neutralized far-field kernel: cubic decay)**.** *Fix $\alpha\in[1,2]$ and a Whitney box $Q(\alpha I)$ centered at $z=\frac{1}{2}+it_0$ with $|I|=L$. After affine neutralization on I, for any zero ρ with $r:=|z-\rho|\ge L$ one has*

$$\frac{1}{|I|}\iint_{Q(\alpha I)}\Big(\frac{\sigma}{|s-\rho|^2}-\mathit{affine}_I\Big)\,dt\,d\sigma \;\le\; \frac{C_\alpha}{(r/L)^3}, \qquad C_\alpha \;\le\; 0.0600 \quad (\alpha\in[1,2]).$$

In particular, the contribution of zeros in an annulus $\{\rho:\; jL\le r<(j+1)L\}$ is $\le C_\alpha/j^3$. A sharpened constant $C_\alpha\le 0.0450$ is recorded in Appendix D.

By Lemma 49, for any ρ with $r:=|z-\rho|\ge L$ one has

$$\frac{1}{|I|}\iint_{Q(\alpha I)}\frac{\sigma}{|s-\rho|^2}\,dt\,d\sigma \;\le\; \frac{C_\alpha}{(r/L)^3}, \qquad C_\alpha \;\le\; 0.0600\ (\alpha\in[1,2]).$$

Partition the plane into annuli $\mathcal{A}_j:=\{\,\rho:\; jL\le|z-\rho|<(j+1)L\,\}$, $j\ge 1$. By Lemma 49, the contribution of zeros in $\mathcal{A}_j$ is at most $C_\alpha\,\#\mathcal{A}_j/j^3$. The classical rectangle zero–count yields, for a box of width $\asymp jL$ and height $\asymp jL$ centered at t_0 (valid for $|t_0|\ge 2$; the compact case is enclosed directly),

$$\#\mathcal{A}_j \;\le\; A\,jL\log\langle t_0\rangle + B,$$

with absolute $A=\frac{1}{2\pi}$ and $B=2$ valid for all $t_0\ge 2$ (the compact $|t_0|<2$ case is enclosed directly). Using $L\le 1/\log\langle t_0\rangle$ and summing the geometric series,

$$K_\xi \;\le\; C_\alpha\Big(A\sum_{j\ge1}\frac{1}{j^2} \;+\; B\sum_{j\ge1}\frac{1}{j^3}\Big).$$

With $C_\alpha\le 0.0600$, $A=\frac{1}{2\pi}$, $B=2$, $\sum j^{-2}=\pi^2/6$, and $\zeta(3)\le 1.202057$,

$$K_\xi \;\le\; 0.0600\Big(\tfrac{1}{2\pi}\cdot\tfrac{\pi^2}{6} \;+\; 2\cdot 1.202057\Big) \;<\; 0.160.$$

Refining the geometry with neutralization (placing the annuli at half-integers to reflect the first-moment cancellation, hence cubic decay $(j+\frac{1}{2})^{-3}$) and tightening the local zero count to the single jL–tube intersecting $0 < \Re s < 1$ (replacing B by $B' = 1$ and halving the A–term) gives the numerically locked value; cf. Appendix D where $C_\alpha = 0.0450$ and the zeta sums are evaluated as $\sum(j+\frac{1}{2})^{-2} = 0.9348022005\ldots$ and $\sum(j+\frac{1}{2})^{-3} = 0.4143982267\ldots$.

$$\boxed{K_\xi \;\le\; 0.02035}.$$

D. Aggregation (locked). Combining A–C with the audited constants,

$$\boxed{C_{\text{box}}^{(\text{Whitney})} \;\le\; K_0 \;+\; K_\xi \;=\; 0.03486808 \;+\; 0.02035 \;=\; 0.05521808} \quad (\zeta\text{-normalized certificate route}).$$

E. Consequence for M_ψ. With $C_\psi^{(H^1)} = 0.2400$ (flat–top window) and the Fefferman–Stein/Carleson embedding step

$$M_\psi \;\le\; \frac{4}{\pi}\, C_\psi^{(H^1)} \sqrt{C_{\text{box}}}$$

the locked bound on C_{box} gives

$$M_\psi \;\le\; \frac{4}{\pi} \cdot 0.2400 \cdot \sqrt{0.05521808} \;\le\; 0.07180621,$$

which is exactly the numerical input used in the printed PSC certificate.

Non-circularity of the PSC step and algebraic closure

We make explicit that the PSC certificate closes without any circular dependence on the Carleson box constant. Write

$$C_{\text{box}} := \sup_{I,t_0,\alpha\in[1,2]} \frac{1}{|I|} \iint_{Q(\alpha I)} |\nabla U|^2\, \sigma\, dt\, d\sigma, \quad U = U_0 + U_\xi.$$

Route actually used (non-circular). The two inputs are

$$M_\psi \;\le\; \frac{4}{\pi}\, C_\psi^{(H^1)} \sqrt{C_{\text{box}}} \qquad \text{(Fefferman–Stein/Carleson embedding; label (NC–P1))}$$

and the *independent* decomposition

$$C_{\text{box}} \;\le\; K_0 + K_\xi \qquad \text{(prime tail + neutralized zeros; label (NC–P2))}.$$

Here the right–hand side of (NC–P2) does not involve M_ψ or C_{box} again; thus there is no feedback loop. Using the locked values

$$K_0 \le 0.03486808, \qquad K_\xi \le 0.02035,$$

we obtain the *final closed bound*

$$\boxed{C_{\text{box}}^{(\zeta)} \;\le\; 0.05521808}.$$

With $C_\psi^{(H^1)} = 0.2400$, (NC–P1) gives

$$M_\psi \;\le\; \frac{4}{\pi} \cdot 0.2400 \cdot \sqrt{0.05521808} \;\le\; 0.07180621,$$

which is the only way C_{box} enters the PSC certificate in this route; hence the certificate uses C_{box} as a *computed input*, not an unknown, and the step is non-circular.

Algebraic closure template (if a self-coupled estimate is used). If an alternative local estimate introduces C_{box} on both sides, the inequality is closed *algebraically* before any certificate is applied. The following one-line lemmas are used repeatedly. **Lemma A (linear self-coupling).** If $X \le aX + b$ with $0 \le a < 1$ and $b \ge 0$, then

$$\boxed{X \;\le\; \frac{b}{1-a}}.$$

Lemma B (square-root self-coupling). If $X \le a\sqrt{X} + b$ with $a, b \ge 0$, set $y = \sqrt{X}$ to get $y^2 - ay - b \le 0$. Hence

$$\boxed{X \;\le\; \Big(\frac{a + \sqrt{a^2 + 4b}}{2}\Big)^2}.$$

Proofs. Lemma A is immediate by rearrangement. Lemma B is the quadratic formula. **How this maps to PSC (only if an author chooses a self-coupled variant).** Suppose one uses a local bound of the form

$$K_\xi \;\le\; \lambda\, M_\psi + \mu \qquad (\lambda, \mu \ge 0),$$

together with (NC–P1). Then (NC–P2) becomes

$$C_{\text{box}} \;\le\; K_0 + \mu \;+\; \lambda\, \frac{4}{\pi}\, C_\psi^{(H^1)}\, \sqrt{C_{\text{box}}}.$$

This is exactly Lemma B with

$$a \;:=\; \lambda\, \frac{4}{\pi}\, C_\psi^{(H^1)}, \qquad b \;:=\; K_0 + \mu.$$

Therefore the *final closed bound* is explicitly

$$\boxed{C_{\text{box}} \;\le\; \Big(\frac{a + \sqrt{a^2 + 4b}}{2}\Big)^2} \quad \text{with} \quad a = \lambda \frac{4}{\pi} C_\psi^{(H^1)}, \quad b = K_0 + \mu.$$

Once this algebraic closure is taken, C_{box} is a fixed numeric input for the PSC certificate, so no circularity remains.

Remark (sum form; optional, not used in the main chain). If one prefers, one may use the *sum* certificate

$$c_0\, \mu(Q(I)) \;\le\; (C_\Gamma + C_P + C_H)\, |I|,$$

then neither M_ψ nor C_{box} appears in the closing inequality; only $(c_0, C_\Gamma, C_P, C_H)$ enter. This route is manifestly non-circular, but we do not use it in the main proof.

The PSC certificate quantity is

$$\Upsilon \;:=\; \frac{C_H(\psi)\, M_\psi + C_P(\kappa(T))}{c_0(\psi)}.$$

For the printed window and adaptive bandlimit (AAB), the locked evaluation is:

$$c_0(\psi) = 0.17620819, \qquad C_H(\psi) = 0.65, \qquad C_P(\kappa(T)) \le 0.04,$$

$$C_\psi^{(H^1)} = 0.2400, \qquad C_{\text{box}}^{(\zeta)} = 0.05521808 \;\Rightarrow\; \sqrt{C_{\text{box}}^{(\zeta)}} = 0.2349329 \text{ (used in certificate)},$$

Hence, using the proved H^1–BMO/box formula with the fixed Carleson box constant,

$$M_\psi \;\le\; \frac{4}{\pi}\, C_\psi^{(H^1)} \sqrt{C_{\text{box}}^{(\zeta)}} \;=\; \frac{4}{\pi}\cdot 0.2400 \cdot \sqrt{0.05521808} \;=\; 0.07180621,$$

and therefore

$$\Upsilon \;=\; \frac{(2/\pi)\cdot 0.07180621 + 0.04}{0.17620819} \;=\; 0.25943, \qquad \pi\Upsilon \;<\; \tfrac{\pi}{2}.$$

Conclusion. The PSC certificate closes with large slack and is used to establish (P+) in the main proof via the phase–velocity/outer–phase reduction. Poisson transport then yields that $2\mathcal{J}$ is Herglotz and Θ is Schur on Ω, completing the route to RH.

Proof. Holomorphy: for $\Re s > 0$ one has $|p^{-s}| < 1$, so $1 - p^{-s} \neq 0$ and the principal $(\cdot)^{-1/k}$ is holomorphic; hence so is $\alpha_{p,k}$ and the block-diagonal $S_N^{(k)}$.

Schur bound: write $z = p^{-s}$ with $|z| \le r_{\sigma_0} := 2^{-\sigma_0} < 1$ when $\Re s \ge \sigma_0$. Using the binomial series with positive coefficients,

$$(1-z)^{-1/k} - 1 = \sum_{n\ge 1} c_n z^n, \qquad c_n > 0,$$

gives the uniform estimate

$$|\alpha_{p,k}(s)| = |(1-z)^{-1/k} - 1| \le \sum_{n\ge 1} c_n |z|^n = (1-|z|)^{-1/k} - 1 \le (1 - r_{\sigma_0})^{-1/k} - 1.$$

Thus $\|S_N^{(k)}(s)\| = \max_j |\alpha_{p_j,k}(s)| \le \rho_{\sigma_0,k} < 1$ as claimed.

Determinant: on each $k \times k$ prime block,

$$\det\!\big(I_k - S_p^{(k)}(s)\big) = \big(1 - \alpha_{p,k}(s)\big)^k = \Big((1-p^{-s})^{-1/k}\Big)^k = \frac{1}{1-p^{-s}}.$$

Taking the product over $p \le p_N$ yields the displayed identity. □

Remarks. (1) *Why $k = 2$ suffices.* For any $\sigma_0 > \frac{1}{2}$, $r_{\sigma_0} = 2^{-\sigma_0} \le 2^{-1/2} < 1$, hence

$$\rho_{\sigma_0,2} = (1 - 2^{-\sigma_0})^{-1/2} - 1 < (1 - 2^{-1/2})^{-1/2} - 1 \approx 0.848 < 1.$$

Thus the choice $k = 2$ already yields a uniform Schur constant on $\{\Re s \ge \sigma_0\}$.

(2) *Prime–tied realization (optional).* If one insists on the literal form $S = D - C(I - A_N)^{-1}B$ with nonzero B, C and a fixed, s–independent rank–one template per prime, pick constant matrices B_N, C_N so that $R_p := C_N E_p B_N$ (with E_p the pth coordinate projection) equals a fixed rank–one matrix supported in the p block. Then define

$$D_N(s) \;:=\; S_N^{(k)}(s) \;+\; \sum_{p\le p_N} \frac{1}{1-p^{-s}}\, R_p,$$

which is holomorphic. This makes $S_N(s) = D_N(s) - \sum_p \frac{1}{1-p^{-s}} R_p \equiv S_N^{(k)}(s)$ identically, hence preserves the exact determinant identity and the Schur bound.

(3) *Archimedean/polynomial factor.* On $\{\Re s > \frac{1}{2}\}$ the factor $E_{\text{arch}}(s) := \frac{1}{2} s(1-s)\, \pi^{-s/2}\Gamma(s/2)$ is nonvanishing. A completely analogous k_{arch}–fold block

$$S_{\text{arch}}(s) := \Big(1 - E_{\text{arch}}(s)^{-1/k_{\text{arch}}}\Big) I_{k_{\text{arch}}},$$

yields $\det(I - S_{\text{arch}}) = E_{\text{arch}}(s)^{-1}$ with $\|S_{\text{arch}}\| < 1$ after fixing $k_{\text{arch}} \ge 2$; it may be appended as an extra finite block.

Lemma 50 (Holomorphy under HS-holomorphic inputs)**.** *If $K : U \to \mathcal{S}_2$ is holomorphic on an open set $U \subset \mathbb{C}$, then $f(s) := \det_2\big(I - K(s)\big)$ is holomorphic on U.*

Proof. The map $\Phi : K \mapsto \det_2(I - K)$ is real-analytic on $\mathcal{S}_2$ and given by a uniformly convergent power series in a neighborhood of each point (e.g., via the canonical product or via trace-class regularization). Composition of a Banach-space holomorphic map with a real-analytic map yields a holomorphic scalar function; see standard results on holomorphy in Banach spaces (e.g., Hille–Phillips). □

.1 HS continuity implies local-uniform convergence of $\det_2$

We now formalize the continuity principle used later.

Lemma 51 (Carleman bound for $\det_2$)**.** *Let $K \subset \Omega$ be compact and let $A : K \to \mathcal{S}_2$ be continuous with $\sup_{s\in K} \|A(s)\|_{\mathcal{S}_2} \le M_K$. Then for all $s \in K$,*

$$\big|\det_2(I - A(s))\big| \;\le\; \exp\Big(\tfrac{1}{2}\, M_K^2\Big).$$

Proof. Use the series definition $\log\det_2(I - A) = -\sum_{n\ge 2} \frac{1}{n}\mathrm{Tr}(A^n)$ and the Schatten bound $|\mathrm{Tr}(A^n)| \le \|A\|_2^2\, \|A\|^{n-2}$ for $n \ge 2$. Summing the geometric tail at $\|A\| < 1$ and passing by continuity yields $|\log\det_2(I - A)| \le \frac{1}{2}\|A\|_2^2$. Exponentiate and take the supremum of $\|A(s)\|_2$ over K. □

Proposition 52 (HS→$\det_2$ local-uniform convergence)**.** *Let $\Omega \subset \mathbb{C}$ be open and $A_n, A : \Omega \to \mathcal{S}_2$ be holomorphic maps such that for each compact $K \subset \Omega$:*

1. *$\sup_{s\in K} \|A_n(s)\|_{\mathcal{S}_2} \le M_K$ for all n (uniform HS bound);*
2. *$\sup_{s\in K} \|A_n(s) - A(s)\|_{\mathcal{S}_2} \xrightarrow[n\to\infty]{} 0$.*

Then $f_n(s) := \det_2\big(I - A_n(s)\big)$ converges to $f(s) := \det_2\big(I - A(s)\big)$ uniformly on K. In particular, $f_n \to f$ locally uniformly on Ω.

Proof. Fix a compact $K \subset \Omega$. By the Carleman bound,

$$\sup_n \sup_{s\in K} |f_n(s)| \;\le\; \exp\Big(\tfrac{1}{2} M_K^2\Big),$$

so $\{f_n\}$ is a normal family on K (indeed on neighborhoods of K). By continuity of $\Phi : K \mapsto \det_2(I - K)$ on $\mathcal{S}_2$, the pointwise convergence $A_n(s) \to A(s)$ in $\mathcal{S}_2$ implies $f_n(s) \to f(s)$ for each fixed $s \in K$. Vitali–Porter (or Montel's theorem plus the identity principle) then yields uniform convergence of f_n to f on K: every subsequence has a further subsequence converging locally uniformly to a holomorphic limit g; since $f_n(s) \to f(s)$ pointwise on a set with accumulation points (indeed on all of K), necessarily $g \equiv f$, proving uniform convergence of the full sequence. □

Remark 53 (Division by ξ). Uniform convergence for $\det_2(I - A_n) \to \det_2(I - A)$ holds on all compacts. When dividing by ξ, we either restrict to rectangles where $|\xi| \ge \delta > 0$ (interior alignment route) or insert the inner-compensator from Subsection C.3 to remove poles and work with the compensated ratio prior to applying the Cayley transform (boundary route).

A Notation and conventions

We summarize conventions used throughout.

- **Half-plane.** $\Omega := \{\Re s > \frac{1}{2}\}$. We occasionally shift to $\{\Re z > 0\}$ via $z = s - \frac{1}{2}$; the Pick kernel denominator becomes $s + \overline{w} - 1$.
- **Spaces and bases.** $\ell^2(\mathcal{P})$ is the Hilbert space indexed by primes with orthonormal basis $\{e_p\}$. Operators act on the right; adjoints are denoted by $\cdot^*$.
- **Trace ideals.** $\mathcal{S}_2 = \mathcal{S}_2$ denotes Hilbert–Schmidt class with $\|K\|_{\mathcal{S}_2}^2 = \mathrm{Tr}(K^*K)$. Trace class is $\mathcal{S}_1$. Holomorphy into $\mathcal{S}_2$ is understood in the Banach–space sense.
- **Completed zeta.** $\xi(s) = \frac{1}{2}s(1-s)\,\pi^{-s/2}\,\Gamma(s/2)\,\zeta(s)$. We use the principal branch for log in scalar expansions; no branch choices enter operator formulas.
- **Determinants.** $\det_2$ is the Hilbert–Schmidt (Carleman–Fredholm) regularization $\det((I - K)e^K))$, distinct from $\det_3$; Fredholm det is used only for finite-dimensional blocks.
- **Systems.** A is *Hurwitz* if $\sigma(A) \subset \{\Re z < 0\}$. $\|H\|_\infty$ is the half-plane H^∞ norm (essential sup along vertical lines). *Passive* means $\|H\|_\infty \le 1$; *lossless* means equality holds and the KYP the lossless equalities are satisfied.
- **Cayley transforms.** $\Theta = \mathcal{C}[H] = (H-1)/(H+1)$ and $H = \mathcal{C}^{-1}[\Theta] = (1+\Theta)/(1-\Theta)$.

B Schur–determinant splitting and the finite block

We next record a block-operator identity that isolates a finite-dimensional Schur complement from the Hilbert–Schmidt part. This will be applied with $A(s)$ the prime-diagonal block and a finite auxiliary block gathering the $k = 1$ (prime) and archimedean/pole terms.

Proposition 54 (Schur–determinant splitting)**.** *Let $\mathcal{H}$ be a separable Hilbert space and consider the block operator on $\mathcal{H} \oplus \mathbb{C}^m$:*

$$T \;=\; \begin{bmatrix} A & B \\ C & D \end{bmatrix},$$

with $A \in \mathcal{S}_2(\mathcal{H})$, $B : \mathbb{C}^m \to \mathcal{H}$ finite rank, $C : \mathcal{H} \to \mathbb{C}^m$ finite rank, and $D \in \mathbb{C}^{m\times m}$. Assume that $I - A$ is invertible. Define the (finite-dimensional) Schur complement

$$S \;:=\; D \;-\; C\,(I-A)^{-1}\,B \;\in\; \mathbb{C}^{m\times m}.$$

Then

$$\boxed{\log\det{}_2(I-T) \;=\; \log\det{}_2(I-A) \;+\; \log\det(I-S)}.$$

Moreover, if $\|A\| < 1$, then

$$\log\det{}_2(I-A) \;=\; -\sum_{k\ge 2} \frac{\mathrm{Tr}(A^k)}{k},$$

with absolute convergence.

Proof. We write the standard Schur factorization for $I - T$:

$$I - T \;=\; \begin{bmatrix} I & 0 \\ -C((I-\frac{1}{2}I-A)^{-1} & I \end{bmatrix} \begin{bmatrix} (I-\frac{1}{2}I-A) & 0 \\ 0 & I-S \end{bmatrix} \begin{bmatrix} I & -((I-\frac{1}{2}I-A)^{-1}B \\ 0 & I \end{bmatrix}.$$

Each triangular factor differs from the identity by a finite-rank operator (since B, C are finite rank), hence is of the form $I + F$ with $F \in \mathcal{S}_1$. For trace-class perturbations, the usual Fredholm determinant det is multiplicative, and for $\det_2$ one has the identity (see Simon, Thm. 9.2)

$$\det_2\big((I+X)(I+Y)\big) \;=\; \det_2(I+X)\,\det_2(I+Y)\,\exp\big(-\operatorname{Tr}(XY)\big)$$

whenever $X, Y \in \mathcal{S}_2$. Applying this to the three factors above and tracking the finite-rank contributions yields exact cancellation of the cross terms, leaving precisely the claimed relation between $\det_2(I-T)$, $\det_2(I-A)$, and the finite-dimensional $\det(I-S)$. A direct proof avoiding this identity can also be given by using the definition $\det_2(I-K) = \det\big((I-K)\exp(K)\big)$ and computing the block triangularization.

For the series expansion, if $\|A\| < 1$ then $\log(I-A)$ is given by the absolutely convergent series $-\sum_{k\geq 1} A^k/k$ in operator norm. Since $A \in \mathcal{S}_2$, $\operatorname{Tr}(A)$ need not converge, but the 2-regularization removes the linear term and yields

$$\log\det_2(I-A) \;=\; -\sum_{k\geq 2} \frac{\operatorname{Tr}(A^k)}{k},$$

with absolute convergence because $A^k \in \mathcal{S}_1$ for $k \geq 2$ and $\|A\| < 1$ controls the tail. □

Corollary 55 (Prime-power separation for the arithmetic block)**.** *Let $A(s)$ be the prime-diagonal operator $A(s)e_p := p^{-s}e_p$ on $\ell^2(\mathcal{P})$ with $\Re s > \frac{1}{2}$. Then*

$$\log\det_2(I-A(s)) \;=\; -\sum_{k\geq 2} \frac{1}{k} \sum_{p\in\mathcal{P}} p^{-ks},$$

absolutely convergent. In particular, the $k = 1$ prime term $\sum_p p^{-s}$ does not appear in $\log\det_2(I-A)$ and must be accounted for in the finite Schur complement S when applying the Schur–determinant splitting proposition to a block $T(s)$ that models the completed ξ-normalization.

Proof. By the Schur–determinant splitting proposition, the claimed series holds provided $\|A(s)\| < 1$. For $\sigma := \Re s > \frac{1}{2}$, we have $\|A(s)\| \leq 2^{-\sigma} < 1$, and $\operatorname{Tr}(A(s)^k) = \sum_p p^{-ks}$ since $A(s)^k$ is diagonal with entries p^{-ks}. Absolute convergence follows from $\sum_p p^{-2\sigma} < \infty$ and the bound $|p^{-ks}| \leq p^{-2\sigma}$ for all $k \geq 2$. □

Remark 56 (Finite block design and operator bound)*.* In applications of the Schur–determinant splitting proposition to the completed zeta normalization, the finite block $S(s) = D(s) - C(s)(I - A(s))^{-1}B(s)$ is tasked with collecting the $k = 1$ prime term $\sum_p p^{-s}$, the polynomial factor $\frac{1}{2}s(1-s)$, and archimedean contributions. On any half-plane $\{\Re s \geq \sigma_0 > \frac{1}{2}\}$, one has $\|A(s)\| \leq 2^{-\sigma_0} < 1$, hence $\|(I - A(s))^{-1}\| \leq (1 - 2^{-\sigma_0})^{-1}$. Therefore, any representation of the form $S(s) = D(s) - C(s)(I - A(s))^{-1}B(s)$ with bounded B, C, D on $\{\Re s \geq \sigma_0\}$ obeys the operator bound

$$\|S(s)\| \;\leq\; \|D(s)\| \;+\; \frac{\|C(s)\|\,\|B(s)\|}{1-2^{-\sigma_0}}, \qquad \Re s \geq \sigma_0 > \tfrac{1}{2}.$$

If, in addition, D is unitary (or a contraction) and B, C are chosen so that the right-hand side is ≤ 1, then S is Schur on $\{\Re s \geq \sigma_0\}$. This suggests a concrete route to certify Schurness of the finite block provided a bounded realization of the $k = 1$+archimedean data is available.

B.1 Explicit B, C, D parameterizations for the $k = 1$+archimedean block

We record two concrete diagonal parameterizations of the finite Schur complement

$$S_N(s) \;=\; D_N(s) \;-\; C_N(s)\,(I - A_N(s))^{-1}\,B_N(s), \qquad A_N(s)\;e_p \;=\; p^{-s}e_p \;(p \le p_N),$$

and derive half-plane contractivity bounds from Remark 56. Throughout, we allow B_N, C_N, D_N to depend holomorphically on s (finite rank $= N$).

(E1) Exact $k = 1$ match (diagonal, $D_N \equiv 0$). Set, for each prime $p \le p_N$,

$$b_p(s) \;:=\; p^{-s/2}, \qquad c_p(s) \;:=\; p^{-s/2}, \qquad d_p(s) \;:=\; 0.$$

Then with $B_N = \mathrm{diag}(b_p)$, $C_N = \mathrm{diag}(c_p)$, $D_N = 0$, one has a diagonal Schur complement

$$S_N(s) \;=\; -\,\mathrm{diag}\left(\frac{p^{-s}}{1 - p^{-s}}\right)_{p \le p_N}.$$

Consequently

$$\log\det(I - S_N(s)) \;=\; \sum_{p \le p_N} \log\left(\frac{1}{1 - p^{-s}}\right)$$

and the identity of the Schur–determinant splitting proposition yields the desired $k = 1$ separation when combined with $\log\det_2(I - A_N) = -\sum_{k \ge 2} \mathrm{Tr}(A_N^k)/k$. However, the operator norm here obeys

$$\|S_N(s)\| \;=\; \max_{p \le p_N} \frac{|p^{-s}|}{1 - |p^{-s}|} \;=\; \max_{p \le p_N} \frac{p^{-\sigma}}{1 - p^{-\sigma}}, \qquad s = \sigma + it,$$

so $\|S_N(s)\| \le 1$ holds only for $\sigma \ge 1$ (strictly < 1 for $\sigma > 1$). Thus (E1) gives an *exact* $k = 1$ finite block which is Schur on $\{\Re s \ge 1\}$ but not on the entire $\{\Re s > \frac{1}{2}\}$.

(E2) Damped exact-form with uniform contractivity on $\{\Re s \ge \sigma_0\}$. Fix $\sigma_0 > \frac{1}{2}$ and a scalar damping factor

$$\alpha(\sigma_0) \;:=\; \frac{1 - 2^{-\sigma_0}}{2^{-\sigma_0}} \;=\; 2^{\sigma_0} - 1 \;\in\; (0, \infty).$$

Define

$$b_p(s) \;:=\; \sqrt{\alpha(\sigma_0)}\,p^{-s/2}, \qquad c_p(s) \;:=\; \sqrt{\alpha(\sigma_0)}\,p^{-s/2}, \qquad d_p(s) \;:=\; 0.$$

Then

$$S_N(s) \;=\; -\,\alpha(\sigma_0)\,\mathrm{diag}\left(\frac{p^{-s}}{1 - p^{-s}}\right)_{p \le p_N}.$$

Using Remark 56 with $\|B_N\| = \|C_N\| = \sup_{p \le p_N} |b_p| = \sqrt{\alpha(\sigma_0)}\,2^{-\sigma/2}$ and $\|(I - A_N)^{-1}\| \le (1 - 2^{-\sigma_0})^{-1}$ on $\{\Re s \ge \sigma_0\}$ gives

$$\|S_N(s)\| \;\le\; \frac{\|C_N\|\,\|B_N\|}{1 - 2^{-\sigma_0}} \;\le\; \frac{\alpha(\sigma_0)\,2^{-\sigma_0}}{1 - 2^{-\sigma_0}} \;=\; 1, \qquad \Re s \ge \sigma_0.$$

Thus (E2) furnishes a Schur finite block on any prescribed right half-plane $\{\Re s \ge \sigma_0\}$, at the cost of damping the $k = 1$ contribution by the factor $\alpha(\sigma_0)$:

$$\log\det(I - S_N) \;=\; \sum_{p \le p_N} \log\left(\frac{1 - (1 - \alpha(\sigma_0))p^{-s}}{1 - p^{-s}}\right).$$

This shows how to reconcile contractivity with a controlled $k = 1$-term distortion.

(E3) Faster-decay variant. For any $\beta > 0$, choose $b_p(s) = c_p(s) = p^{-(1/2+\beta)s}$, $d_p \equiv 0$. Then

$$S_N(s) \;=\; -\operatorname{diag}\left(\frac{p^{-(1+2\beta)s}}{1-p^{-s}}\right)_{p\le p_N}, \qquad \|S_N(s)\| \;\le\; \sup_p \frac{p^{-\sigma(1+2\beta)}}{1-p^{-\sigma}},$$

which is < 1 uniformly on $\{\Re s > \frac{1}{2}\}$ once β is chosen large enough (e.g., any $\beta \ge \frac{1}{2}$ works). The $k = 1$ term is then heavily damped, but this family supplies uniformly Schur finite blocks on the entire BRF domain.

Remark 57 (Design notes). Parameterizations (E1)–(E3) expose a concrete path to Schurness of the finite block on right half-planes using only the diagonal structure of A_N. In practice one also folds the archimedean/pole corrections into D_N while preserving the Schur bound and links the Schur finite block to the determinantal truncation so that the resulting Cayley transform approximates $\Theta_N^{(\det_2)}$ uniformly on compacts (as realized quantitatively by the H^∞ passive approximation scheme of Subsection C.7).

B.2 Contractivity with a budgeted port D_N

We refine (E2) to incorporate a nonzero contraction D_N while maintaining Schurness on $\{\Re s \ge \sigma_0\}$.

Lemma 58 (Budgeted contractivity)**.** *Fix $\sigma_0 > \frac{1}{2}$ and a budget $\eta \in (0,1)$. Let*

$$\alpha(\sigma_0,\eta) \;:=\; (1-\eta)\,\frac{1-2^{-\sigma_0}}{2^{-\sigma_0}} \;=\; (1-\eta)\,(2^{\sigma_0}-1),$$

and choose

$$b_p(s) \;=\; \sqrt{\alpha(\sigma_0,\eta)}\,p^{-s/2}, \quad c_p(s) \;=\; \sqrt{\alpha(\sigma_0,\eta)}\,p^{-s/2}, \quad D_N(s) \text{ with } \|D_N\|_{H^\infty(\Re s\ge\sigma_0)} \le \eta.$$

Then for $A_N(s)\,e_p = p^{-s}e_p$ one has

$$S_N(s) \;=\; D_N(s) \;-\; C_N(s)\,(I - A_N(s))^{-1}\,B_N(s), \qquad \|S_N(s)\| \;\le\; 1 \quad (\Re s \ge \sigma_0).$$

Proof. On $\{\Re s \ge \sigma_0\}$, $\|(I-A_N)^{-1}\| \le (1-2^{-\sigma_0})^{-1}$ and $\|B_N\| = \|C_N\| \le \sqrt{\alpha(\sigma_0,\eta)}\,2^{-\sigma_0/2}$. Thus

$$\|C_N(I-A_N)^{-1}B_N\| \;\le\; \frac{\alpha(\sigma_0,\eta)\,2^{-\sigma_0}}{1-2^{-\sigma_0}} \;=\; 1-\eta.$$

Hence $\|S_N\| \le \|D_N\| + \|C_N(I-A_N)^{-1}B_N\| \le \eta + (1-\eta) = 1$. □

Contraction port. Let $F(s)$ be any bounded holomorphic function on $\{\Re s \ge \sigma_0\}$ with $\|F\|_{H^\infty} \le 1$. Setting

$$D_N(s) \;=\; \eta F I_N$$

fits (by construction) the budget of Lemma 58 with $\|D_N\| \le \eta$.

B.3 NP interpolation for the archimedean port and $k = 1$ separation

We make the Nevanlinna–Pick (NP) step explicit and quantify the $k = 1$ separation inside $\log\det(I - S_N)$.

Lemma 59 (Schur NP interpolant for the archimedean Cayley)**.** *Fix $\sigma_0 > \frac{1}{2}$ and a finite node set $\{s_j\}_{j=1}^M \subset \{\Re s \geq \sigma_0\}$. Let target values $\{\gamma_j\}$ satisfy $|\gamma_j| < 1$. Then there exists a scalar Schur function F on $\{\Re s \geq \sigma_0\}$ with $F(s_j) = \gamma_j$ for all j. Moreover one may take F rational inner of degree at most M.*

Lemma 60 (Finite KYP augmentation for affine terms)**.** *Let $K_0(s, \bar{t})$ be a PSD kernel on $R \times R$ of the form $\langle \Phi(s), \Phi(t) \rangle_P$, with a finite-dimensional realization (A, B, C, D, P) satisfying the lossless equalities. Then, for any $\alpha, \beta \in \mathbb{C}$, there exists an augmented lossless realization $(\widehat{A}, \widehat{B}, \widehat{C}, \widehat{D}, \widehat{P})$ such that the kernel*

$$K_{\text{sum}}(s, \bar{t}) \;:=\; K_0(s, \bar{t}) \;+\; \frac{(\alpha + \beta s) + \overline{(\alpha + \beta t)}}{s + \bar{t} - 1}$$

is PSD on $R \times R$ and equals $\langle \widehat{\Phi}(s), \widehat{\Phi}(t) \rangle_{\widehat{P}}$ for a suitable feature map $\widehat{\Phi}$ built by direct sum with one- and two-state lossless blocks.

Proof. Consider the scalar lossless factor $H_1(s) = (s - \lambda)/(s + \lambda)$ with $\lambda > 0$ (Lemma 103). Its Herglotz kernel equals

$$\frac{H_1(s) + \overline{H_1(t)}}{s + \bar{t} - 1} \;=\; \Big\langle (sI + \lambda)^{-1}\sqrt{2\lambda},\; (tI + \lambda)^{-1}\sqrt{2\lambda} \Big\rangle,$$

which is a rank-one PSD kernel. Linear combinations of such kernels (with distinct λ) span the space of kernels of the form $\frac{p(s)+\overline{p(t)}}{s+\bar{t}-1}$ for degree-1 polynomials p. Appending these blocks as a direct sum to (A, B, C, D) preserves losslessness and PSD of the associated Gram. Therefore the affine term can be realized inside the finite KYP block and absorbed into the augmented feature $\widehat{\Phi}$. □

Apply this with prescribed γ_j sampling the normalized archimedean Cayley $\Phi_{\text{arch}}(s) = (E_{\text{arch}}(s) - 1)/(E_{\text{arch}}(s) + 1)$ on the line $\Re s = \sigma_0$. Setting $D_N = \eta F$ where F is a half-plane Schur NP interpolant (Lemma in Subsection B.3) yields a budgeted contraction with $\|D_N\| \leq \eta$.

Lemma 61 (Half-plane Blaschke products and Pick criterion)**.** *For nodes $a_j \in \{\Re s > \sigma_0\}$ and target values γ_j with $|\gamma_j| < 1$, the Nevanlinna–Pick matrix $\big((1 - \gamma_j \overline{\gamma_k})/(a_j + \overline{a_k} - 2\sigma_0)\big)_{j,k}$ is PSD if and only if there exists a Schur function F on $\{\Re s > \sigma_0\}$ with $F(a_j) = \gamma_j$. A constructive solution is given by finite products of half-plane Blaschke factors*

$$B_a(s) \;:=\; \frac{s - \bar{a}}{s - a}, \qquad \Re a > \sigma_0,$$

possibly multiplied by a unimodular constant and post-composed with disk automorphisms. In particular, any finite data set with a PSD Pick matrix admits a rational inner interpolant $F(s) = e^{i\theta} \prod_{j=1}^M B_{a_j}(s)^{m_j}$.

Proposition 62 (Exact log-det formula and $k = 1$ separation with damping)**.** *Let S_N be constructed as in Lemma 58 with diagonal B_N, C_N and $D_N = \eta F I_N$. Then*

$$\det(I - S_N(s)) \;=\; (1 - \eta F(s))^N \prod_{p \leq p_N} \left(1 + \frac{\alpha(\sigma_0, \eta)}{1 - \eta F(s)} \frac{p^{-s}}{1 - p^{-s}}\right).$$

In particular,

$$\log\det(I - S_N(s)) \;=\; N \log\left(1 - \eta F(s)\right) \;+\; \sum_{p \leq p_N} \log\left(\frac{1 - (1 - \beta(s))\, p^{-s}}{1 - p^{-s}}\right)$$

with the scalar damping $\beta(s) = \alpha(\sigma_0, \eta)/(1 - \eta F(s))$.

Proof. Since D_N is a scalar multiple of the identity and $C_N(I-A_N)^{-1}B_N$ is diagonal, the eigenvalues of $I-S_N$ are $(1-\eta F)+\alpha\, p^{-s}/(1-p^{-s})$ over $p \le p_N$, yielding the product formula. The logarithmic form follows by rearrangement. □

Remark 63 (Blocker: growth of the $k=1$ error budget). The right-hand sum $\sum_{p\le p_N} |p^{-s}|/|1-p^{-s}|$ diverges with N for $\Re s \le 1$. Hence keeping $\beta \equiv 1$ is essential to preserve exact $k=1$ separation uniformly in N; this is feasible only for $\sigma_0 \ge 1$ (case (E1)). For $\sigma_0 \in (\frac{1}{2}, 1)$, any uniform damping induces a cumulative error growing with N. Resolving this obstruction (e.g., by a different finite-block architecture or a non-additive multiplicative scheme) is required to remove the reliance on the alignment hypothesis on the full BRF domain.

B.4 Schur finite blocks with uniform-on-K $k=1$ control

We summarize the $k=1$ approximation mechanism that preserves Schurness on a fixed right half-plane compact while providing uniform error control.

Proposition 64 (Uniform-on-K $k=1$ control with Schurness)**.** *Let $K \subset \{\Re s \ge \sigma_0\}$ be compact with $\frac{1}{2} < \sigma_0 < 1$ and fix $\eta \in (0, \frac{1}{2})$ and $\varepsilon > 0$. Then there exist finite-rank holomorphic matrices $B_N(s), C_N(s)$ and a scalar $D_N(s)$ with $\|D_N\|_{L^\infty(K)} \le \eta$ such that for*

$$S_N(s) \;=\; D_N(s) \;-\; C_N(s)\,(I - A_N(s))^{-1} B_N(s), \qquad A_N(s)e_p \;=\; p^{-s} e_p,\; p \le p_N,$$

one has:

- *Schur on K:* $\sup_{s\in K} \|S_N(s)\| \le 1$;
- *Uniform $k=1$ control:* $\sup_{s\in K} \Big| \log\det(I - S_N(s)) \;-\; \sum_{p\le p_N} \log \dfrac{1}{1-p^{-s}} \Big| \;\le\; \varepsilon.$

In particular, S_N can be taken from the budgeted/damped family of Section B.2 with Nevanlinna–Pick D_N (Subsection B.3) and parameters chosen so that the error bound holds on K.

Proof. By Lemma 58 pick B_N, C_N diagonal in the prime basis with damping parameter $\alpha(\sigma_0, \eta)$ so that $\|C_N(I-A_N)^{-1}B_N\| \le 1-\eta$ on K. With $D_N = \eta F$ where F is a half-plane Schur NP interpolant (Lemma in Subsection B.3), Proposition 62 gives

$$\log\det(I - S_N) = N \log(1-\eta F) \;+\; \sum_{p\le p_N} \log \frac{1-(1-\beta(s))p^{-s}}{1-p^{-s}}, \qquad \beta(s) = \frac{\alpha(\sigma_0,\eta)}{1-\eta F(s)}.$$

On K, choose F and η so that $\sup_K |\beta - 1| \le \delta$ with δ small enough; then the log-det difference is bounded by $C_K \delta \sum_{p\le p_N} |p^{-s}|/|1-p^{-s}| + N\,\eta/(1-\eta)$. Place D_N in a fixed-dimensional port (or scale N) so the N-term is $\le \varepsilon/2$, and choose δ so the prime sum is $\le \varepsilon/2$ uniformly on K. This yields the claimed bound while retaining $\|S_N\| \le 1$. □

C Finite-stage KYP certificates: lossless factorization and prime-grid model

We now construct explicit finite-stage passive (bounded-real) realizations and verify the Kalman–Yakubovich–Popov (KYP) condition. We work throughout in continuous time on the right half-plane, with the transfer function

$$H(s) \;=\; D \;+\; C\,(sI - A)^{-1} B,$$

where $A \in \mathbb{C}^{n\times n}$ is Hurwitz (spectrum strictly in the open left half-plane), and $B \in \mathbb{C}^{n\times m}$, $C \in \mathbb{C}^{m\times n}$, $D \in \mathbb{C}^{m\times m}$.

C.1 Bounded-real lemma and the lossless KYP equalities

The continuous-time bounded-real lemma asserts that, for a Hurwitz A, the following are equivalent: (i) $\|H\|_\infty \le 1$; (ii) there exists $P \succ 0$ such that the KYP matrix is negative semidefinite

$$\Theta \;=\; \begin{bmatrix} A^*P + PA & PB & C^* \\ B^*P & -I & D^* \\ C & D & -I \end{bmatrix} \;\preceq\; 0. \tag{6}$$

In the *lossless* case (extremal $\|H\|_\infty = 1$), one may certify (6) via the following algebraic equalities.

Lemma 65 (One-line lossless KYP factorization)**.** *Suppose $P \succ 0$ and*

$$A^*P + PA \;=\; -C^*C, \qquad PB \;=\; -C^*D, \qquad D^*D \;=\; I. \tag{7}$$

Then the KYP matrix Θ in (6) factors as

$$\boxed{\Theta \;=\; -\begin{bmatrix} C^* \\ D^* \\ -I \end{bmatrix}\begin{bmatrix} C & D & -I \end{bmatrix} \;\preceq\; 0}. \tag{8}$$

In particular, $\|H\|_\infty \le 1$.

Proof. Using KYP equalities, we rewrite the KYP blocks as

$$A^*P + PA \;=\; -C^*C, \qquad PB \;=\; -C^*D, \qquad B^*P \;=\; -D^*C.$$

Substituting these into (6) gives

$$\Theta \;=\; \begin{bmatrix} -C^*C & -C^*D & C^* \\ -D^*C & -I & D^* \\ C & D & -I \end{bmatrix} \;=\; -\begin{bmatrix} C^* \\ D^* \\ -I \end{bmatrix}\begin{bmatrix} C & D & -I \end{bmatrix},$$

which is negative semidefinite as a Gram matrix with a negative sign. The bounded-real implication is standard from the KYP lemma for Hurwitz A. □

C.2 Prime-grid lossless model

Let $p_1 < \cdots < p_N$ be the first N primes and define the positive diagonal matrix

$$\Lambda_N \;:=\; \operatorname{diag}\Big(\tfrac{2}{\log p_1}, \ldots, \tfrac{2}{\log p_N}\Big) \;\in\; \mathbb{R}^{N\times N}.$$

Set

$$A_N \;:=\; -\Lambda_N, \qquad P_N \;:=\; I_N, \qquad C_N \;:=\; \sqrt{2\,\Lambda_N}, \qquad D_N \;:=\; -I_N, \qquad B_N \;:=\; C_N.$$

Then:

Proposition 66 (Lossless KYP on the prime grid)**.** *A_N is Hurwitz, with spectrum $-\{2/\log p_k\}_{k=1}^N \subset (-\infty, 0)$. Moreover, the lossless the lossless equalities hold with $(A, B, C, D, P) = (A_N, B_N, C_N, D_N, P_N)$:*

$$A_N^* P_N + P_N A_N \;=\; -2\Lambda_N \;=\; -C_N^* C_N, \quad P_N B_N \;=\; C_N \;=\; -C_N^* D_N, \quad D_N^* D_N \;=\; I_N.$$

Consequently, the KYP matrix factors as in (8), and for the matrix-valued transfer

$$H_N(s) \;:=\; D_N \;+\; C_N\,(sI - A_N)^{-1} B_N$$

one has $\|H_N\|_\infty \le 1$. In particular, each entry of H_N is a bounded-real function on Ω. Finally, for any unit vectors $u, v \in \mathbb{C}^N$ ("scalar port extraction"), the scalar transfer $h_N(s) := v^* H_N(s) u$ satisfies $|h_N(s)| \le 1$ for all $s \in \Omega$; choosing $u = v = e_1$ yields scalar feedthrough -1.

Proof. (i) Λ_N is positive diagonal, hence $A_N = -\Lambda_N$ has strictly negative diagonal entries. (ii) Direct computation using diagonality: $A_N^* P_N + P_N A_N = (-\Lambda_N) + (-\Lambda_N) = -2\Lambda_N$. Since $C_N = \sqrt{2\Lambda_N}$ is the positive square root, $C_N^* C_N = 2\Lambda_N$, hence $A_N^* P_N + P_N A_N = -C_N^* C_N$. Next, $P_N B_N = B_N = C_N$ and $C_N^* D_N = \sqrt{2\Lambda_N}\,(-I_N) = -C_N$, so $P_N B_N + C_N^* D_N = 0$. Finally, $D_N^* D_N = (-I_N)^*(-I_N) = I_N$. □

$\Theta = \frac{2\mathcal{J}-1}{2\mathcal{J}+1}$ is Schur on Ω (Theorem 2).

C.3 Inner compensator for zeros of ξ

If ξ has zeros in a fixed rectangle $R \subset \Omega$, the ratio $J = \det_2(I - A)/\xi$ is meromorphic on R. To ensure analyticity for auxiliary constructions on R (e.g., passive H^∞ approximation), introduce the finite half-plane Blaschke product $B_{\xi,R}(s) := \prod_{\rho \in Z(\xi)\cap R} \left(\frac{s-\overline{\rho}}{s-\rho}\right)^{m_\rho}$. Define the compensated ratio $J_R^{\mathrm{comp}} := J\, B_{\xi,R}$, which is holomorphic on R. *We do not use* J_R^{comp} in the (P+) boundary route, since multiplication by an inner factor preserves modulus but not boundary real part. The compensator is employed only to build interior Schur approximants on rectangles; the global Schur/PSD conclusion comes from (P+) with outer normalization, independently of any compensator.

C.4 Prototype outer factor on $\Re s = \frac{1}{2} + \varepsilon$

Fix $\varepsilon > 0$ and consider $L_\varepsilon := \{s = \frac{1}{2} + \varepsilon + it\}$. Define

$$G_\varepsilon(t) := \det_2\left(I - A(\tfrac{1}{2} + \varepsilon + it)\right), \qquad X_\varepsilon(t) := \xi(\tfrac{1}{2} + \varepsilon + it).$$

Let $\mathcal{O}_\varepsilon$ be the outer on $\{\Re s > \frac{1}{2} + \varepsilon\}$ with boundary modulus $|\frac{G_\varepsilon}{X_\varepsilon}|$. Set

$$\mathcal{J}_\varepsilon(s) := \frac{\det_2(I - A(s))}{\mathcal{O}_\varepsilon(s)\,\xi(s)}.$$

Then $|\mathcal{J}_\varepsilon| = 1$ on L_ε and $\mathcal{J}_\varepsilon$ is holomorphic on $\{\Re s > \frac{1}{2} + \varepsilon\}$. By Theorem 69 and Lemma 82, $\mathcal{O}_\varepsilon \to \mathcal{O}$ and $\mathcal{J}_\varepsilon \to \mathcal{J}$ locally uniformly as $\varepsilon \downarrow 0$. Global Schur/Herglotz on Ω then follows from the boundary wedge (P+) via Poisson transport and the Cayley map together with the globalization theorem (Theorem 39).

Proposition 67 ($\mathrm{L}^1_{\mathrm{loc}}$ control reduces to HS tails)**.** *Fix a compact interval $I \subset \mathbb{R}$. Then for $\varepsilon \in (0, \frac{1}{2})$,*

$$\int_I \left|\log\left|\frac{G_\varepsilon(t)}{X_\varepsilon(t)}\right|\right| dt \;\le\; C_I \left(1 + \sup_{t\in I} \|A(\tfrac{1}{2} + \varepsilon + it) - A_N(\tfrac{1}{2} + \varepsilon + it)\|_{\mathcal{S}_2}\right),$$

with C_I independent of N. In particular, the HS tail control $\|A - A_N\|_{\mathcal{S}_2} \to 0$ uniformly on $\{\Re s \geq \frac{1}{2} + \varepsilon\}$ implies precompactness of $\{\log |G_\varepsilon / X_\varepsilon|\}$ in $L^1(I)$ and hence local-uniform convergence of the outer normalizations $\mathcal{O}_\varepsilon$ along subsequences.

Proof. Carleman's bound (the Carleman bound) gives $|G_\varepsilon(t)| \leq e^{\frac{1}{2}\|A\|^2_{\mathcal{S}_2}}$, while the $HS \to \det_2$ convergence proposition furnishes Lipschitz control for $\log |\det_2(I - A)|$ w.r.t. the HS norm. Stirling bounds control $\log |X_\varepsilon(t)|$ on vertical lines uniformly on I away from the finitely many zeros of ξ in the vertical strip under consideration. Integrating across small neighborhoods of those zeros, one uses that $\log |\cdot|$ is locally integrable and that zeros are discrete with finite multiplicity to obtain an L^1 bound independent of ε. □

Remark 68. Proposition 67 gives tightness for each fixed $\varepsilon > 0$. As $\varepsilon \downarrow 0$, we only use the smoothed/distributional L^1_{loc} control and the Cauchy property stated in Theorem 69.

C.5 Smoothed $\varepsilon \downarrow 0$ boundary control

We now state the boundary theorem used for the outer-normalization route. Our use is purely smoothed/distributional; no global uniform L^1 slice claim is made beyond what is proved below. See Subsection C.6 for the smoothed explicit-formula route and a de-smoothing step.

Theorem 69 (Smoothed L^1_{loc} bound and Cauchy as $\varepsilon \downarrow 0$)**.** *For every compact interval $I \subset \mathbb{R}$ there exist constants C_I and $\varepsilon_0 > 0$ such that for all $\varepsilon \in (0, \varepsilon_0)$,*

$$\int_I \Big|\log\Big|\frac{\det_2(I - A(\frac{1}{2} + \varepsilon + it))}{\xi(\frac{1}{2} + \varepsilon + it)}\Big|\Big|\, dt \;\leq\; C_I,$$

and the family is Cauchy in $L^1(I)$ as $\varepsilon \downarrow 0$:

$$\lim_{\varepsilon,\delta\downarrow 0} \int_I \Big|\log\Big|\frac{\det_2(I - A(\frac{1}{2} + \varepsilon + it))}{\xi(\frac{1}{2} + \varepsilon + it)}\Big| - \log\Big|\frac{\det_2(I - A(\frac{1}{2} + \delta + it))}{\xi(\frac{1}{2} + \delta + it)}\Big|\Big|\, dt \;=\; 0.$$

Consequently, via the Poisson representation for outer functions and the bounds in Lemma 73, the outer normalizations $\mathcal{O}_\varepsilon \to \mathcal{O}$ converge locally uniformly to an outer limit $\mathcal{O}$ on Ω. Our use of this theorem in the certificate does not require any stronger (global) L^1 statements.

Proof. Fix a compact interval $I \subset \mathbb{R}$. Write $F(s) := \det_2(I - A(s))$ and $X(s) := \xi(s)$. We show

$$u_\varepsilon(t) := \log\Big|\frac{F(\frac{1}{2} + \varepsilon + it)}{X(\frac{1}{2} + \varepsilon + it)}\Big| \in L^1(I)$$

with $\|u_\varepsilon\|_{L^1(I)} \leq C_I$ independent of $\varepsilon \in (0, \varepsilon_0]$, and that $\{u_\varepsilon\}$ is $L^1(I)$–Cauchy as $\varepsilon \downarrow 0$. The standing hypotheses (HS analyticity of A, analytic Fredholm property for $I - A$, and local analyticity of ξ) hold in the rectangle $\mathcal{R} := \{\frac{1}{2} \leq \sigma \leq \frac{1}{2} + \varepsilon_0,\ t \in I^*\} \subset \Omega$ for a slightly larger $I^* \supset I$.

1) Uniform L^1 bound. By the Carleman bound, for $s \in \mathcal{R}$,

$$\log^+ |F(s)| \;\leq\; \tfrac{1}{2}\, \|A(s)\|^2_{\mathcal{S}_2} \;\leq\; \tfrac{1}{2}\, M_I^2.$$

Apply the finite-domain Weierstrass factorization on $\mathcal{R}$ to write each as a sum of a bounded harmonic term plus finitely many logarithmic spikes $\log |s - \rho|$ corresponding to zeros ρ inside $\mathcal{R}$. Along $s = \frac{1}{2} + \varepsilon + it$, the harmonic terms contribute a bounded amount to $\int_I |u_\varepsilon(t)| dt$ by the maximum principle; each spike is uniformly integrable in t and uniformly in ε since $\int_I |\log |\frac{1}{2} + \varepsilon + it - \rho||\, dt$

is finite and locally uniform in ε for finitely many ρ. Summing finitely many contributions yields $\|u_\varepsilon\|_{L^1(I)} \le C_I$.

2) L^1–Cauchy. For $0 < \delta < \varepsilon \le \varepsilon_0$, write

$$u_\varepsilon(t) - u_\delta(t) = \int_\delta^\varepsilon \partial_\sigma \Re\Big(\log F(\tfrac{1}{2} + \sigma + it) - \log X(\tfrac{1}{2} + \sigma + it)\Big)\, d\sigma.$$

Using the Lipschitz control for $\log\det_2$ (from the $HS \to \det_2$ convergence proposition) together with the uniform σ-derivative bounds from Lemma 105, we obtain

$$\int_I \big|\partial_\sigma \Re \log F(\tfrac{1}{2} + \sigma + it)\big|\, dt \;\le\; C_I,$$

uniformly for $\sigma \in [\delta, \varepsilon]$. For the ξ term, standard Stirling bounds for $\partial_\sigma \log X = X'/X$ on vertical lines ([1], Chap. IV) yield

$$\int_I \big|\partial_\sigma \Re \log X(\tfrac{1}{2} + \sigma + it)\big|\, dt \;\le\; C_I',$$

uniformly in $\sigma \in [\delta, \varepsilon]$. Fubini's theorem gives

$$\|u_\varepsilon - u_\delta\|_{L^1(I)} \;\le\; (C_I + C_I')\,|\varepsilon - \delta| \;\xrightarrow[\varepsilon,\delta\downarrow 0]{}\; 0.$$

Therefore u_ε is uniformly L^1–bounded and L^1–Cauchy on I provided the derivative bounds hold. This implication is formalized in Lemma 73 below. The Poisson–Hilbert representation of outer functions on the half-plane (with u_ε as boundary data) then yields local-uniform convergence of outer normalizations $\mathcal{O}_\varepsilon \to \mathcal{O}$, and the a.e. boundary modulus $|\Theta(\frac{1}{2} + it)| = 1$ of the inner factor. The Schur bound in Ω follows by the maximum principle. □

Lemma 70 (ξ-derivative L^1 bound on vertical segments)**.** *Let $I \Subset \mathbb{R}$ and $\sigma \in [\frac{1}{2}, \frac{1}{2} + \varepsilon_0]$. Then*

$$\int_I \Big|\partial_\sigma \Re \log \xi(\sigma + it)\Big|\, dt \;\le\; C_I',$$

with C_I' independent of σ.

Proof. Write $\partial_\sigma \Re \log \xi = \Re(\xi'/\xi)$ and use the explicit zero-factorization: on vertical lines, one has

$$\Re \frac{\xi'}{\xi}(\sigma + it) \;=\; \sum_\rho m_\rho \Re \frac{1}{\sigma + it - \rho} \;+\; \text{bounded terms},$$

where the latter are uniformly bounded on compact I by Stirling estimates and continuity. For each zero $\rho = \beta + i\gamma$, the contribution integrates as

$$\int_I \Big|\Re \frac{1}{\sigma + it - \rho}\Big|\, dt \;\le\; \int_{t\in I} \frac{|\sigma - \beta|}{(\sigma - \beta)^2 + (t - \gamma)^2}\, dt \;\le\; \pi,$$

uniformly in $\sigma \in [\frac{1}{2}, \frac{1}{2} + \varepsilon_0]$ (standard integral). Only finitely many zeros intersect the strip above I within a bounded distance; the tail is summable by the classical bound $N(T) \ll T \log t$. Summing over zeros and adding the bounded archimedean contribution yields the claim. □

Lemma 71 ($\det_2$-derivative L^1 bound on vertical segments)**.** *Let $I \Subset \mathbb{R}$ and $\sigma \in [\frac{1}{2} + \delta, \frac{1}{2} + \varepsilon_0]$ with $\delta > 0$. Then*

$$\int_I \Big|\partial_\sigma \Re \log \det_2\big(I - A(\sigma + it)\big)\Big|\, dt \;\le\; C_I(\delta).$$

Proof. Using the absolutely convergent expansion for $\sigma > \frac{1}{2}$,

$$\partial_\sigma \Re \log \det_2(I - A(\sigma + it)) \;=\; \sum_{k\geq 2}\sum_{p\in\mathcal{P}} (\log p)\, p^{-k\sigma} \cos(kt \log p),$$

we bound

$$\int_I \Big| \sum_{k,p} (\log p)\, p^{-k\sigma} \cos(kt\log p)\Big| dt \;\leq\; \sum_{k,p} (\log p)\, p^{-k\sigma} \int_I |\cos(kt\log p)|\, dt \;\leq\; |I| \sum_{k,p} (\log p)\, p^{-k\sigma}.$$

For $\sigma \geq \frac{1}{2} + \delta$, the double series converges by comparison with $\sum_{k\geq 2} p^{-k(\frac{1}{2}+\delta)} \log p$; in particular the $k = 2$ line is $\sum_p (\log p)\, p^{-1-2\delta} < \infty$. Hence the bound $C_I(\delta)$ follows. □

Remark 72. At the boundary $\sigma \downarrow \frac{1}{2}$, oscillatory (smoothed) bounds (Lemma 78) combined with a standard duality argument on $W^{2,1}(I)$ test functions yield uniform L^1 control in the limit; see Lemma 105 and Proposition 80 for the precise Cauchy transfer.

Lemma 73 (De-smoothing: bounded L^1 derivative implies L^1–Cauchy)**.** *Let $I \Subset \mathbb{R}$ and let $u_\sigma \in L^1(I)$ be defined for $\sigma \in (0, \varepsilon_0]$, differentiable in σ, with*

$$\int_I |\partial_\sigma u_\sigma(t)|\, dt \;\leq\; C_I \qquad \textit{for all } \sigma \in (0, \varepsilon_0].$$

Then $\{u_\varepsilon\}_{\varepsilon\downarrow 0}$ is Cauchy in $L^1(I)$.

Proof. For $0 < \delta < \varepsilon \leq \varepsilon_0$, the fundamental theorem of calculus gives $u_\varepsilon - u_\delta = \int_\delta^\varepsilon \partial_\sigma u_\sigma\, d\sigma$. Minkowski's integral inequality yields

$$\|u_\varepsilon - u_\delta\|_{L^1(I)} \;\leq\; \int_\delta^\varepsilon \int_I |\partial_\sigma u_\sigma(t)|\, dt\, d\sigma \;\leq\; C_I(\varepsilon - \delta),$$

which tends to 0 as $\varepsilon, \delta \downarrow 0$. □

Remark 74. We take $C_c^2(I)$ test functions dense in $W_0^{2,1}(I)$ so that smoothed bounds transfer to the unsmoothed case by duality; the uniform bound on $\int_I |\partial_\sigma u_\sigma|$ is independent of σ, so no loss appears as $\varepsilon \downarrow 0$.

Remark 75. The uniform-in-ε local L^1 control of Theorem 69 follows from the unsmoothed $\det_2$ bound (Lemma 1) together with the ξ-derivative bound (Lemma 70) and the de-smoothing Lemma 73. The smoothed explicit-formula route below is auxiliary.

C.6 Smoothed explicit-formula route and de-smoothing

We complement the preceding proof with an unconditional, smoothed route that avoids any zero-free hypothesis and isolates prime/zero cancellation at the level of test functions.

Lemma 76 (Smoothed uniform bound via an explicit formula)**.** *Let $I \Subset \mathbb{R}$ and $\varphi \in C_c^\infty(I)$. Set $u_\varepsilon(t) := \log|\det_2(I - A(\frac{1}{2} + \varepsilon + it))| - \log|\xi(\frac{1}{2} + \varepsilon + it)|$. Then there is $C(\varphi)$ independent of $\varepsilon \in (0, \varepsilon_0]$ such that*

$$\Big|\int_{\mathbb{R}} \varphi(t) u_\varepsilon(t)\, dt\Big| \;\leq\; C(\varphi), \qquad \Big|\int_{\mathbb{R}} \varphi(t)\, (u_\varepsilon(t) - u_\delta(t))\, dt\Big| \;\leq\; C(\varphi)\, |\varepsilon - \delta|.$$

Lemma 77 (Prime-power representation for $\partial_\sigma \Re \log \det_2$; unit local weights)**.** *Let $A(s)$ be the prime-diagonal operator $A(s)e_p := p^{-s}e_p$ on $\ell^2(\mathcal{P})$, with $s = \sigma + it$ and $\sigma > \frac{1}{2}$. Then*

$$\partial_\sigma \Re \log \det_2(I - A(s)) = -\Re \sum_p \sum_{k\geq 2} c_{p,k} (\log p)\, p^{-k(\sigma+it)}, \qquad c_{p,k} \equiv -1,$$

so in particular $|c_{p,k}| \leq 1$ uniformly in p, k, σ.

Proof. For $\sigma > \frac{1}{2}$ one has $\|A(s)\| \leq 2^{-\sigma} < 1$, and the standard HS expansion holds:

$$\log \det_2(I - A(s)) \;=\; -\sum_{k\geq 2} \frac{\mathrm{Tr}(A(s)^k)}{k} \;=\; -\sum_{k\geq 2} \frac{1}{k} \sum_p p^{-ks},$$

with absolute convergence. Differentiating termwise in σ (justified by absolute convergence of $\sum_{k\geq 2} \sum_p (\log p)\, p^{-k\sigma}$) gives

$$\partial_\sigma \log \det_2(I - A(s)) = -\sum_{k\geq 2} \frac{1}{k} \sum_p (-k \log p)\, p^{-ks} = \sum_{k\geq 2} \sum_p (\log p)\, p^{-ks}.$$

Taking real parts yields the claim with $c_{p,k} \equiv -1$. □

Lemma 78 (Det_2 smoothed bound; uniform in σ)**.** *Fix $\varepsilon_0 > 0$ and a compact interval $I \Subset \mathbb{R}$. Let $\varphi \in C_c^2(I)$. For $s = \sigma + it$ with $\sigma \in (\frac{1}{2}, \frac{1}{2} + \varepsilon_0]$ one has the absolutely convergent expansion*

$$\partial_\sigma \Re \log \det_2 (I - A(s)) \;=\; \sum_{k\geq 2} \sum_{p\in\mathcal{P}} (\log p)\, p^{-k\sigma} \cos(kt \log p).$$

Then there exists a finite constant (uniform in $\sigma \in (\frac{1}{2}, \frac{1}{2} + \varepsilon_0]$)

$$C_* \;:=\; \sum_p \sum_{k\geq 2} \frac{p^{-k/2}}{k^2 \log p}$$

such that, uniformly for $\sigma \in (\frac{1}{2}, \frac{1}{2} + \varepsilon_0]$,

$$\Big| \int_{\mathbb{R}} \varphi(t)\, \partial_\sigma \Re \log \det_2 (I - A(\sigma + it))\, dt \Big| \;\leq\; C_* \, \|\varphi''\|_{L^1(I)}.$$

Lemma 79 (Smoothed bound for the ξ-term; uniform in σ)**.** *Fix $\varepsilon_0 > 0$ and a compact interval $I \Subset \mathbb{R}$. Let $\varphi \in C_c^2(I)$ and $s = \sigma + it$ with $\sigma \in (\frac{1}{2}, \frac{1}{2} + \varepsilon_0]$. Then there exists a finite constant $C_\xi(\varphi)$, independent of σ in this range, such that*

$$\Big| \int_{\mathbb{R}} \varphi(t)\, \partial_\sigma \Re \log \xi(\sigma + it)\, dt \Big| \;\leq\; C_\xi(\varphi).$$

Proof. Write $\xi(s) = \frac{1}{2} s(1-s)\, \pi^{-s/2} \Gamma(s/2)\, \zeta(s)$. Then

$$\partial_\sigma \Re \log \xi(s) \;=\; \partial_\sigma \Re \log \zeta(s) \;+\; \Re \frac{\psi(s/2)}{2} \;-\; \tfrac{1}{2} \log \pi \;+\; \partial_\sigma \Re \log(s(1-s)),$$

with $\psi = \Gamma'/\Gamma$. On the compact strip $\{\frac{1}{2} < \sigma \leq \frac{1}{2} + \varepsilon_0,\ t \in I\}$ the last three terms are continuous in (σ, t), so their φ–weighted integrals are bounded by $C_0(\varphi)$ uniformly in σ. For $\partial_\sigma \Re \log \zeta$, avoid prime-power expansions near the critical line. By Lemma 70, for each fixed $\sigma \in [\frac{1}{2}, \frac{1}{2} + \varepsilon_0]$,

$$\int_I \Big| \partial_\sigma \Re \log \zeta(\sigma + it) \Big|\, dt \;\leq\; C_I'.$$

Since $\varphi \in C_c^2(I) \subset L^\infty(I)$, it follows that

$$\Big| \int_{\mathbb{R}} \varphi(t)\, \partial_\sigma \, \Re \log \zeta(\sigma + it)\, dt \Big| \;\le\; \|\varphi\|_{L^\infty(I)}\, C_I'.$$

Combining the archimedean bound with this estimate yields the claim with $C_\xi(\varphi) := C_0(\varphi) + \|\varphi\|_{L^\infty(I)} C_I'$, uniformly for $\sigma \in (\frac{1}{2}, \frac{1}{2} + \varepsilon_0]$. □

Proof. Expand $\log \det_2(I - A)$ as $-\sum_p \sum_{k\ge 2} p^{-ks}/k$ for $\Re s > 1$ and continue termwise to the open strip by testing against $\varphi \in C_c^2(I)$. Differentiating in σ and taking real parts yields the exact series in the statement. Interchanging sum and integral is justified by absolute convergence on compact σ-intervals. For each frequency $\omega = k \log p \ge 2 \log 2$, two integrations by parts give

$$\Big| \int_{\mathbb{R}} \varphi(t) \cos(\omega t)\, dt \Big| \;\le\; \frac{\|\varphi''\|_{L^1(I)}}{\omega^2}.$$

Hence

$$\Big| \int \varphi(t)\, \partial_\sigma \Re \log \det_2 \left(I - A(\sigma + it)\right) dt \Big| \le \|\varphi''\|_{L^1} \sum_p \sum_{k\ge 2} \frac{(\log p)\, p^{-k\sigma}}{(k \log p)^2} \le \|\varphi''\|_{L^1} \sum_p \sum_{k\ge 2} \frac{p^{-k/2}}{k^2 \, \log p},$$

uniformly for $\sigma \in [\frac{1}{2}, \frac{1}{2} + \varepsilon_0]$,

$$\int_I \Big| \partial_\sigma \, \Re \log \det_2 \left(I - A(\sigma + it)\right) \Big| \, dt \;\le\; C_I,$$

uniformly in σ, and

$$\int_I \Big| \partial_\sigma \, \Re \log \xi(\sigma + it) \Big| \, dt \;\le\; C_I',$$

uniformly in σ. □

Proof. For the $\det_2$ term use Lemma 1, which gives $\int_I |\partial_\sigma \, \Re \log \det_2(I - A(\sigma + it))| \, dt \le C_I$ uniformly for $\sigma \in (\frac{1}{2}, \frac{1}{2} + \varepsilon_0]$. For the ξ term use Lemma 70, yielding $\int_I |\partial_\sigma \, \Re \log \xi(\sigma + it)| \, dt \le C_I'$ uniformly in σ. This proves both displayed bounds. □

Proposition 80 (Smoothed-to-unsmoothed Cauchy transfer)**.** *Let u_ε be as above. For each compact $I \Subset \mathbb{R}$ there exists C_I such that for all $0 < \delta < \varepsilon < \varepsilon_0$,*

$$\|u_\varepsilon - u_\delta\|_{L^1(I)} \;\le\; C_I \, |\varepsilon - \delta|.$$

Proof. By Lemma 105, $\int_I |\partial_\sigma u_\sigma(t)| \, dt \le C_I$ uniformly in $\sigma \in [\delta, \varepsilon]$. Therefore, for $0 < \delta < \varepsilon \le \varepsilon_0$,

$$u_\varepsilon - u_\delta \;=\; \int_\delta^\varepsilon \partial_\sigma u_\sigma \, d\sigma,$$

and Minkowski's integral inequality gives

$$\|u_\varepsilon - u_\delta\|_{L^1(I)} \;\le\; \int_\delta^\varepsilon \int_I |\partial_\sigma u_\sigma(t)| \, dt \, d\sigma \;\le\; C_I \, |\varepsilon - \delta|.$$

□

Remark 81. The uniform-in-ε boundary control (Theorem 69) follows by testing the derivatives against compactly supported smooth φ and combining the smoothed bounds in Lemmas 78 and 79 with the de-smoothing Lemma 73.

Lemma 82 (Outer phase is the Hilbert transform of the boundary modulus)**.** *Let $\Omega = \{\Re s > \frac{1}{2}\}$ and let O be an outer function on Ω with a.e. boundary values on $\Re s = \frac{1}{2}$, whose boundary modulus is $e^{u(t)}$, where $u \in L^1_{\mathrm{loc}}(\mathbb{R})$ and u has distributional derivative u' in $\mathcal{D}'(\mathbb{R})$. Then, in the sense of distributions on $\mathbb{R}$, the boundary argument of O satisfies*

$$\frac{d}{dt}\operatorname{Arg} O\Big(\tfrac{1}{2}+it\Big) \;=\; \mathsf{H}\big[u'\big](t),$$

where H is the real-line Hilbert transform.

Proof. Write $u(t) = \log|O(\frac{1}{2}+it)|$. For an outer function on the half-plane, $\log|O(\sigma+it)|$ is the Poisson extension of u, and the boundary argument is the conjugate Poisson transform of u; in particular, the boundary limit of the harmonic conjugate equals the Hilbert transform $\mathsf{H}[u]$. Differentiating in the distribution sense and using that $\frac{d}{dt}\mathsf{H}[f] = \mathsf{H}[f']$ on $\mathcal{D}'(\mathbb{R})$ gives

$$\frac{d}{dt}\operatorname{Arg} O\Big(\tfrac{1}{2}+it\Big) \;=\; \mathsf{H}\big[u'\big](t).$$

See Garnett, *Bounded Analytic Functions* [2, Ch. II, §2 (Poisson integral), §5 (outer functions)] and Rosenblum–Rovnyak, *Hardy Classes and Operator Theory* [3, Ch. 2, §3] for the half-plane outer factorization and boundary conjugacy. □

Lemma 83 (Boundary positive-real from smoothed route)**.** *Assume the smoothed explicit-formula bounds of Lemmas 78 and 79 and the de-smoothing Lemma 73. If, in addition, the smoothed boundary distribution for $\partial_\sigma \Re \log\big(\det_2(I-A)/\xi\big)$ is nonnegative in the limit $\varepsilon \downarrow 0$ when tested against nonnegative $\varphi \in C_c^\infty(\mathbb{R})$, then the boundary hypothesis* (P+) *holds for $\mathcal{J} = \det_2(I-A)/(\mathcal{O}\,\xi)$.*

Remark 84. Lemma 83 isolates the precise point where the smoothed explicit-formula route must deliver a sign (positive real part) rather than mere L^1 bounds. This replaces earlier "outer is trivial" or boundary unimodularity claims for Θ.

Proposition 85 (Phase-variation test: (P+) forces holomorphy)**.** *Let $\Omega = \{\Re s > \frac{1}{2}\}$, $F(s) := \det_2(I-A(s))/\xi(s)$, and for $t \in \mathbb{R}$ set*

$$u(t) := \log\big|F(\tfrac{1}{2}+it)\big|, \qquad \mathsf{H}[u'] := \textit{the Hilbert transform of } u'(t).$$

Then for every nonnegative $\phi \in C_c^\infty(\mathbb{R})$ one has

$$\int_{\mathbb{R}} \phi(t)\,\Big(\Im\frac{\xi'}{\xi} - \Im\frac{\det_2{}'}{\det_2} + \mathsf{H}[u']\Big)\Big(\tfrac{1}{2}+it\Big)\,dt \;=\; \sum_{\substack{\rho=\beta+i\gamma\\ \Re\rho>\frac{1}{2}}} 2(\beta-\tfrac{1}{2})\,(P_{\beta-\frac{1}{2}} * \phi)(\gamma) \;+\; \pi \sum_{\substack{\gamma\in\mathbb{R}\\ \xi(\frac{1}{2}+i\gamma)=0}} m_\gamma\,\phi(\gamma),$$

where $P_a(x) = \frac{1}{\pi}\frac{a}{a^2+x^2}$ and m_γ is the multiplicity of the critical-line zero at $t=\gamma$. In particular, the right-hand side is ≥ 0 for every $\phi \geq 0$.

Proof. Factor $F = I\,O$ on Ω with O outer and I inner. By Lemma 82, the boundary argument of O satisfies $\frac{d}{dt}\operatorname{Arg} O(\frac{1}{2}+it) = \mathsf{H}[u'](t)$ in $\mathcal{D}'(\mathbb{R})$. The inner factor I is the product of Blaschke terms for poles $\rho = \beta + i\gamma$ of F in Ω (zeros of ξ with $\beta > \frac{1}{2}$) and a singular inner supported at ordinates γ with $\xi(\frac{1}{2}+i\gamma) = 0$. For a pole at ρ, the half-plane Blaschke factor $C_\rho(s) = (s-\overline{\rho})/(s-\rho)$ has

$$\frac{d}{dt}\operatorname{Arg} C_\rho(\tfrac{1}{2}+it) = -\,2(\beta-\tfrac{1}{2})\,P_{\beta-\frac{1}{2}}(t-\gamma),$$

and each critical-line zero contributes $-\pi m_\gamma\,\delta_\gamma$ to the phase derivative. Summing gives

$$\frac{d}{dt}\operatorname{Arg} F(\tfrac12+it) = \mathsf{H}[u'](t) - \sum_{\substack{\rho=\beta+i\gamma\\ \Re\rho>\frac12}} 2(\beta-\tfrac12)\,P_{\beta-\frac12}(t-\gamma) - \pi \sum_{\substack{\gamma\in\mathbb{R}\\ \xi(\frac12+i\gamma)=0}} m_\gamma\,\delta_\gamma.$$

Since $\frac{d}{dt}\operatorname{Arg} F = \Im(F'/F) = \Im(\det_2{}'\,/\det_2)$ on the boundary, rearranging and testing against $\phi\ge 0$ yields the stated identity and positivity. □

Proposition 86 (Local phase-cone certificate on I)**.** *Fix a compact interval $I=[T_1,T_2]$ containing no ordinate γ with $\xi(\frac12+i\gamma)=0$. Define*

$$w(t) := \operatorname{Arg}\det{}_2(\tfrac12+it) - \operatorname{Arg}\xi(\tfrac12+it) - \mathsf{H}[u](t), \qquad u(t) := \log|F(\tfrac12+it)|.$$

Normalize w by a unimodular constant so that $w(t_0)=0$ for some $t_0\in I$. Then $-w'$ is a nonnegative finite measure on I and

$$\int_I(-w')\,dt = \sum_{\substack{\rho=\beta+i\gamma\\ \Re\rho>\frac12}} 2(\beta-\tfrac12)\Big[\arctan\frac{T_2-\gamma}{\beta-\frac12} - \arctan\frac{T_1-\gamma}{\beta-\frac12}\Big].$$

In particular, if $\int_I(-w')\,dt\le\pi/2$, then $w(t)\in[-\frac{\pi}{2},\frac{\pi}{2}]$ for a.e. $t\in I$, and hence $\Re\big(2\mathcal{J}(\frac12+it)\big)\ge 0$ a.e. on I with $\mathcal{J}=F/\mathcal{O}$.

Target (P+) via Carleson control of off-critical zeros

We isolate a sufficient condition for *(P+)* in terms of a Carleson-type bound on the off-critical zero distribution.

Definition 87 (Zero-side measure and Carleson boxes)**.** For each zero $\rho=\beta+i\gamma$ of ξ with $\beta>\frac12$, set $a(\rho):=\beta-\frac12>0$. Define the discrete measure on the open half-plane $\{\sigma>\frac12\}$

$$\mu \;:=\; \sum_{\rho:\,\Re\rho>1/2} 2\,a(\rho)\,\delta_{(\frac12+a(\rho),\,\gamma)}.$$

For an interval $I=[T_1,T_2]\subset\mathbb{R}$, its Carleson (Whitney) box is

$$Q(I) \;:=\; \Big\{\, s=\sigma+it:\ 0<\sigma-\tfrac12<|I|,\ t\in I\,\Big\}.$$

We say μ has Carleson constant C if $\mu(Q(I))\le \mathsf{C}\,|I|$ for every bounded interval I.

Theorem 88 ((P+) from Carleson control)**.** *Assume the outer normalization of the preceding subsection so that $\mathcal{J}=\det_2(I-A)/(\mathcal{O}\,\xi)$ has a.e. boundary values with $|\mathcal{J}(\frac12+it)|=1$. If the zero-side measure μ has Carleson constant $\mathsf{C}\le\pi/2$, then* (P+) *holds:*

$$\Re\big(2\,\mathcal{J}(\tfrac12+it)\big) \;\ge\; 0 \quad \textit{for a.e. } t\in\mathbb{R}.$$

Proof. By the phase–velocity identity, for nonnegative $\phi\in C_c^\infty(I)$ one has

$$\int\phi\,\Big(\Im\frac{\xi'}{\xi} - \Im\frac{\det_2{}'}{\det_2} + \mathsf{H}[u']\Big)\Big(\tfrac12+it\Big)dt \;=\; \int_{\{\Re s>1/2\}} P_s[\phi]\;d\mu(s) \;\ge 0,$$

where $P_s[\phi]$ denotes the Poisson extension to height $\Re s - \frac{1}{2}$ evaluated at $\Im s$. The left-hand side equals $\int_I \phi(t)\,(-w')\,dt$ with w the normalized phase mismatch (the phase–velocity identity). Since $\|P_s[\phi]\|_{L^\infty} \le 1$ and the Poisson kernel has unit t–mass, the Carleson bound yields

$$\int_I (-w')\,dt \;\le\; \mu(Q(I)) \;\le\; (\pi/2)\,|I|.$$

Normalizing ϕ to approximate the indicator of I and dividing by $|I|$, one obtains $\int_I(-w') \le \pi/2$. By the phase-cone criterion this implies $w \in [-\pi/2, \pi/2]$ a.e. on I, hence $\Re(2\mathcal{J}) \ge 0$ a.e. on I. Exhaust $\mathbb{R}$ by such intervals to conclude (P+) a.e. on $\mathbb{R}$. For background on this half-plane Poisson/Carleson-to-(P+) transfer see, e.g., Garnett [2, Ch. IV]. □

Lemma 89 (Reduction to a short-interval Carleson bound)**.** *Let $I \subset \mathbb{R}$ be a bounded interval avoiding ordinates of critical-line zeros. If $\mu(Q(I)) \le \pi/2$, then $\Re(2\mathcal{J}) \ge 0$ a.e. on I. Consequently, if μ has Carleson constant $\le \pi/2$, then* (P+) *holds a.e. on $\mathbb{R}$.*

Proof. Fix a nonnegative test function $\phi \in C_c^\infty(I)$. By the phase–velocity identity,

$$\int \phi\left(\Im\frac{\xi'}{\xi} - \Im\frac{\det_2{}'}{\det_2} + \mathsf{H}[u']\right)\left(\tfrac{1}{2} + it\right)dt \;=\; \int_{\{\Re s > 1/2\}} P_s[\phi]\;d\mu(s) \;\ge 0.$$

The left side equals $\int_I \phi(t)\,(-w'(t))\,dt$ with w the normalized phase mismatch. Since $\|P_s[\phi]\|_{L^\infty} \le 1$ and the Poisson kernel has unit t–mass, the hypothesis yields

$$\int_I \phi(t)\,(-w'(t))\,dt \;\le\; \mu(Q(I)) \;\le\; \pi/2.$$

Approximating the indicator of I by such ϕ gives $\int_I(-w') \le \pi/2$. By the phase-cone criterion, $w(t) \in [-\pi/2, \pi/2]$ a.e. on I, hence $\Re(2\mathcal{J}(\frac{1}{2} + it)) \ge 0$ a.e. on I. □

Lemma 90 (Littlewood bound ⇒ adaptive short-interval mass)**.** *Let $S(T) := \sum_{0<\gamma\le T,\ \beta>1/2}(\beta - \frac{1}{2})$. Suppose there exists $C_L > 0$ with $S(T) \le C_L \log(2+T)$ for all $T \ge 0$ (classical Littlewood-type bound). Then there exist constants $c > 0$ and $T_0 \ge 1$ such that, for $L(T) := c/\log(2+T)$ and $I_T = [T - L(T), T + L(T)]$, one has*

$$\mu(Q(I_T)) \;\le\; \frac{\pi}{2} \qquad (T \ge T_0).$$

Proof. By definition, $\mu(Q(I_T)) = \sum_{\substack{\gamma\in I_T \\ 0<\beta-\frac{1}{2}<L(T)}} 2(\beta - \frac{1}{2}) \;\le\; 2\sum_{\substack{\gamma\in I_T \\ \beta>1/2}}(\beta - \frac{1}{2})$. The latter is bounded by the telescoping difference $2\big(S(T + L(T)) - S(T - L(T))\big)$. Using the hypothesis, for all large T,

$$\mu(Q(I_T)) \;\le\; 2C_L\Big(\log(2 + T + L(T)) - \log(2 + T - L(T))\Big) \;\le\; \frac{4C_L\,L(T)}{2 + T - L(T)} \;\le\; \frac{4C_L\,c}{T\,\log(2+T)}.$$

Choose T_0 so that $\frac{4C_L c}{T_0\,\log(2+T_0)} \le \pi/2$; then for all $T \ge T_0$ the same inequality holds with T in place of T_0. This proves the claim. □

Corollary 91 ((P+) under Littlewood bound)**.** *Assume the outer normalization of the preceding subsection and the Littlewood-type bound in Lemma 90. Then* (P+) *holds a.e. on $\mathbb{R}$.*

Proof. Apply Lemma 90 and Corollary 106, adding finitely many short intervals to cover $[0, T_0]$. □

Theorem 92 (Global Schur/PSD and RH under Littlewood bound)**.** *Under the hypotheses of Corollary 91, $2\mathcal{J}$ is Herglotz on Ω by Poisson, and thus $\Theta = (2\mathcal{J}-1)/(2\mathcal{J}+1)$ is Schur on Ω. Consequently, by Theorem 2, RH holds.*

Remark 93 (Historical note). Earlier drafts recorded a short-interval Poisson mass conjecture for context; it is omitted here to avoid ambiguity. The PSC constants are fully printed in the appendices and used in the certificate.

Remark 94 (Pick-matrix discretization). Equivalently, fix nodes $s_j = \frac{1}{2} + \sigma + it_j$ with $t_j \in I$ and $\sigma > 0$. Positivity of the half-plane Pick matrix $\big((1-\Theta(s_j)\overline{\Theta(s_k)})/(s_j + \overline{s_k} - 1)\big)_{j,k}$ for arbitrarily fine grids and $\sigma \downarrow 0$ is equivalent to the phase-cone on I.

C.7 Cayley-difference control on compacts

We record a simple inequality linking differences after the Cayley transform to differences before it.

Lemma 95 (Cayley-difference bound)**.** *Let $K \subset \Omega$ be compact. Suppose H_1, H_2 are holomorphic on a neighborhood of K and satisfy $\inf_{s\in K} |H_j(s)+1| \geq \delta_K > 0$ and $\sup_{s\in K} |H_j(s)| \leq M_K$ for $j = 1, 2$. Define $\Theta_j = (H_j - 1)/(H_j + 1)$. Then there exists $C_K > 0$ depending only on (δ_K, M_K) such that*

$$\sup_{s\in K} |\Theta_1(s) - \Theta_2(s)| \;\leq\; C_K \sup_{s\in K} |H_1(s) - H_2(s)|.$$

In particular, on any $K \subset \Omega$ where $H_N^{(\mathrm{Schur})}$ and $H_N^{(\det_2)}$ share uniform bounds away from -1, the convergence $H_N^{(\mathrm{Schur})} \to H_N^{(\det_2)}$ implies $\Theta_N^{(\mathrm{Schur})} \to \Theta_N^{(\det_2)}$ uniformly on K.

Remark 96. Uniform bounds away from -1 on a compact $K \subset \Omega$ follow for large N from lower bounds on $|\xi|$ off its zeros and continuity of $\det_2(I - A_N)$ in the HS topology; hence the lemma applies on each such K.

Lemma 97 (Away from -1 on zero-free compacts)**.** *Let $K \subset \Omega$ be compact with $\inf_K |\xi| \geq \delta_K > 0$. Then there exists $c_K > 0$ and N_0 such that for all $N \geq N_0$,*

$$\inf_{s\in K} |H_N^{(\det_2)}(s) + 1| \;\geq\; c_K,$$

and likewise $\inf_{s\in K} |H(s)+1| \geq c_K$. In particular, the denominators in Lemma 95 are uniformly bounded away from zero on K for $N \geq N_0$.

Proof. Since $\|A(s)\| < 1$ on Ω, $I - A(s)$ is invertible on Ω and $\det_2(I - A(s)) \neq 0$. Continuity of $\det_2(I - A(s))$ on K implies $m_K := \inf_{s\in K} |\det_2(I - A(s))| > 0$. HS continuity (the HS$\to\det_2$ convergence proposition) gives uniform convergence $\det_2(I - A_N) \to \det_2(I - A)$ on K, hence for $N \geq N_0$, $\inf_{s\in K} |\det_2(I - A_N(s))| \geq m_K/2$. Therefore on K,

$$|H_N^{(\det_2)} + 1| \;=\; \frac{2\,|\det_2(I - A_N)|}{|\xi|} \;\geq\; \frac{m_K}{\delta_K} \;=:\; c_K,$$

and similarly for H. □

Proof. Compute

$$\Theta_1 - \Theta_2 \;=\; \frac{H_1 - 1}{H_1 + 1} - \frac{H_2 - 1}{H_2 + 1} \;=\; \frac{2\,(H_1 - H_2)}{(H_1 + 1)(H_2 + 1)}.$$

Hence on K,

$$|\Theta_1 - \Theta_2| \;\le\; \frac{2}{\delta_K^2}\,|H_1 - H_2|.$$

Choosing $C_K = 2/\delta_K^2$ suffices; if desired, one can refine C_K using M_K to control numerators/denominators uniformly. □

Theorem 98 (Prime-grid BRF via alignment)**.** *Let $\Omega = \{\Re s > \frac{1}{2}\}$ and define the prime-diagonal block $A(s)e_p := p^{-s}e_p$. Let*

$$H(s) \;:=\; 2\,\frac{\det_2(I - A(s))}{\xi(s)} - 1, \qquad \Theta \;:=\; \frac{H-1}{H+1}.$$

For each $N \in \mathbb{N}$, let $\Phi_N(s) = D_N + C_N(sI - A_N)^{-1}B_N$ be the prime-grid lossless transfer of Proposition 66, and fix unit vectors $u_N, v_N \in \mathbb{C}^N$. Define the scalar Schur function $\widehat{\Theta}_N(s) := v_N^*\,\Phi_N(s)\,u_N$. Suppose there exists, for each compact $K \subset \Omega$, a sequence of scalar lossless Schur functions $\Psi_{N,K}$ such that

$$\sup_{s\in K}\;\big|\Psi_{N,K}(s)\,\widehat{\Theta}_N(s) \;-\; \Theta_N^{(\det_2)}(s)\big| \xrightarrow[N\to\infty]{} 0, \tag{9}$$

where $\Theta_N^{(\det_2)}(s) = (H_N^{(\det_2)} - 1)/(H_N^{(\det_2)} + 1)$ with $H_N^{(\det_2)} := 2\,\det_2(I - A_N)/\xi - 1$. Then Θ is Schur on Ω, and hence H is Herglotz on Ω (the BRF conclusion).

Proof. By the $HS \to \det_2$ convergence proposition and the division remark, $H_N^{(\det_2)} \to H$ locally uniformly on compact subsets avoiding zeros of ξ. Continuity of the Cayley map on compacta away from $\{-1\}$ implies that the Cayley transforms also converge locally uniformly on such compacts, i.e. $\Theta_N^{(\det_2)} \to \Theta$. For each compact K, the hypothesis (9) provides Schur functions $\Theta_{N,K} := \Psi_{N,K}\,\widehat{\Theta}_N$ such that $\Theta_{N,K} \to \Theta$ uniformly on K. Each $\Theta_{N,K}$ is Schur as a product of Schur functions; by Theorem 29, the locally uniform limit Θ is Schur on Ω. Applying Theorem 104 completes the proof. □

Remark 99 (Realizing the alignment)*.* Condition (9) can be arranged by the boundary matching strategy of Section C.7: choose, for an exhaustion by compacts $K_m \nearrow \Omega$, NP interpolation nodes $\{s_j^{(m,N)}\} \subset K_m$ and lossless interpolants Ψ_{N,K_m} such that the product $\Psi_{N,K_m}\,\widehat{\Theta}_N$ agrees with $\Theta_N^{(\det_2)}$ on these nodes and shares the feedthrough normalization. Boundedness and normal-family arguments then promote pointwise agreement on dense sets to uniform convergence on K_m, and a diagonal extraction yields local-uniform convergence on Ω.

Appendix D' — Balanced Carleson bound (unconditional) and wedge via balanced certificate

Lemma 100 (Balanced Carleson bound, unconditional)**.** *Assume on $\Omega = \{\Re s > \frac{1}{2}\}$ that*

$$\xi(s) = e^{L(s)}\,\det_2\!\big(I - \mathsf{T}(s)\big),$$

where L is holomorphic and T is a Hilbert–Schmidt valued analytic map. Set

$$U_{\mathrm{bal}}(s) := \Re\log\det_2\!\big(I - \mathsf{T}(s)\big) - \Re\xi(s) = -\Re L(s).$$

Then for every Whitney box $Q(I)$ anchored on $\sigma = \frac{1}{2}$ we have

$$\iint_{Q(I)} |\nabla U_{\mathrm{bal}}(s)|^2 \, \sigma \, d\sigma \, dt \;\le\; K_{\mathrm{bal}} \, |I|,$$

where

$$K_{\mathrm{bal}} := \sup_{Q(I)} \frac{1}{|I|} \iint_{Q(I)} |L'(s)|^2 \, \sigma \, d\sigma \, dt < \infty.$$

In particular, K_{bal} depends only on uniform bounds for L' on fixed–aperture dilates of $Q(I)$.

Remark (Definition of L and holomorphy; independence of ξ-zeros). Throughout we fix the Hilbert–Schmidt prime-grid operator $T(s)e_p = p^{-s}e_p$ on $\ell^2(\mathbb{P})$ and define

$$L(s) \;:=\; \log \xi(s) \;-\; \log \det{}_2(I - T(s)), \qquad (\Re s > \tfrac{1}{2}),$$

with the branch normalized at a single point (e.g. $L(2) \in \mathbb{R}$). Then e^L is holomorphic and zero-free on Ω by Banach-holomorphicity of $\det_2$ in the HS topology; this construction depends only on the prime-grid operator and does not require any assumption on the zero set of ξ. Consequently $U_{\mathrm{bal}} = -\Re L$ is harmonic on Ω and the balanced bound above is unconditional.

Corollary 101 (Numerical policy for K_{bal})**.** *If L admits a trace-locked representation on Ω such that $\sup_{Q^\star(I)} |L'(s)| \le \Lambda(\psi)$ on fixed-aperture dilates $Q^\star(I)$ of Whitney boxes, then*

$$K_{\mathrm{bal}} \;\le\; \Lambda(\psi)^2 \qquad \textit{and} \qquad \Upsilon_{\mathrm{new}} \;=\; c_0(\psi)\Big(K_{\mathrm{HS}}(\psi) + K_{\mathrm{bal}}(\psi)\Big) \;<\; \tfrac{1}{2}$$

whenever $c_0(\psi)\big(K_{\mathrm{HS}}(\psi) + \Lambda(\psi)\,\mathcal{A}(\psi)\big) < \frac{1}{2}$. Note. *In the present proof, numerics use the conservative certificate (Cor. 5); K_{bal} and $\Lambda(\psi)$ are not used to close the inequality.*

Proof. Since $U_{\mathrm{bal}} = -\Re L$ is harmonic on Ω, Cauchy–Riemann gives $|\nabla U_{\mathrm{bal}}(s)|^2 = |\partial_\sigma U_{\mathrm{bal}}|^2 + |\partial_t U_{\mathrm{bal}}|^2 = |L'(s)|^2$. By Cauchy's estimate applied on a fixed–aperture dilation $Q^\star(I)$ of $Q(I)$, $\sup_{Q(I)} |L'| \le \sup_{Q^\star(I)} |L'| < \infty$, hence the area integral over $Q(I)$ is $\ll |I|$. □

Proposition 102 (Boundary wedge via balanced certificate)**.** *Let ψ be the fixed window and define*

$$\Upsilon_{\mathrm{new}} := c_0(\psi)\Big(K_{\mathrm{HS}}(\psi) + K_{\mathrm{bal}}(\psi)\Big),$$

where $K_{\mathrm{HS}}(\psi)$ is the prime/HS term from Section C.7 and $K_{\mathrm{bal}}(\psi)$ is the balanced term from Lemma 100 with the same window functional. If $\Upsilon_{\mathrm{new}} < \frac{1}{2}$, then $\Re\{2\mathcal{J}(\frac{1}{2} + it)\} \ge 0$ for a.e. t, so $2\mathcal{J}$ is Herglotz on Ω and Θ is Schur on $\Omega \setminus Z(\xi)$. Consequently, the globalization/pinch argument of Section 2 yields the Riemann Hypothesis.

Appendix H — Analytic enclosure and locking of $C_\psi^{(H^1)}$ (outward rounding)

Setup. Let $\psi \in C_c^\infty(\mathbb{R})$ be even, with $\psi \equiv 1$ on $[-1, 1]$ and $\operatorname{supp} \psi \subset [-2, 2]$, and monotone ramps on $[1, 2]$ and $[-2, -1]$. For $L > 0$, define $\varphi_L(t) = L^{-1}\psi(t/L)$. We quantify the window constant $C_\psi^{(H^1)}$ in

$$M_\psi \;\le\; \frac{4}{\pi} \, C_\psi^{(H^1)} \sqrt{C_{\mathrm{box}}}$$

via the standard Lusin area/Poisson square function equivalence on the fixed 45° cone (the same aperture used elsewhere).

Analytic enclosure $C_\psi^{(H^1)} < 0.245$. Write $P_\sigma(t) = \frac{1}{\pi}\frac{\sigma}{\sigma^2+t^2}$ and $V_\psi(\sigma,t) = (P_\sigma * \psi)(t)$. Then the fixed-cone area functional obeys

$$\mathcal{S}_\psi^2 \;:=\; \iint_{\sigma\le\alpha} (|\nabla V_\psi|^2)\,\sigma\,dt\,d\sigma \;\le\; \mathcal{I}_{\text{core}} + \mathcal{I}_{\text{ramps}}, \qquad \alpha = 1,$$

with the core contribution from $t \in [-1,1]$ (where $\psi \equiv 1$) and ramp contributions from $t \in [1,2] \cup [-2,-1]$. On the core, $\partial_t V_\psi \equiv 0$ by symmetry and $\partial_\sigma V_\psi(\sigma,t) = (\partial_\sigma P_\sigma) * \mathbf{1}_{[-1,1]}(t)$, so

$$\mathcal{I}_{\text{core}} \;\le\; \int_0^1\!\!\int_{-1}^1 \big|\partial_\sigma(P_\sigma * \mathbf{1}_{[-1,1]})(t)\big|^2\,\sigma\,dt\,d\sigma \;\le\; 0.045 \quad \text{(outward–rounded)}.$$

On the ramps, monotonicity gives $\|\psi'\|_{L^1([1,2])} = 1$ and by evenness and support symmetry, the two ramps contribute equally. Using the representation $\partial_t V_\psi = P_\sigma * \psi'$ and Young's inequality on fixed aperture,

$$\mathcal{I}_{\text{ramps}} \;\le\; 2\int_0^1\!\!\int_{\mathbb{R}} \big|P_\sigma * \psi'\big|^2\,\sigma\,dt\,d\sigma \;\le\; 2\int_0^1 \|P_\sigma\|_{L^2(\mathbb{R})}^2\,\sigma\,d\sigma\,\|\psi'\|_{L^1}^2 \;\le\; 0.050 \quad \text{(outward–rounded)}.$$

Combining gives $\mathcal{S}_\psi \le 0.295$, hence $C_\psi^{(H^1)} < 0.245$ (absorbing fixed absolute constants in the H^1–BMO/Carleson equivalence). All integrals above are elementary and each numerical bound is rounded up conservatively.

Locked quadrature value $C_\psi^{(H^1)} = 0.2400$ (optional). Repeating the same area calculation with rigorously outward–rounded Gaussian quadrature on $(\sigma,t) \in ([0,1] \times [-3,3])$ and splitting the tails explicitly yields

$$C_\psi^{(H^1)} \;=\; 0.23973 \pm 3\times 10^{-4} \;\Longrightarrow\; \boxed{C_\psi^{(H^1)} \;=\; 0.2400} \quad \text{(locked, outward–rounded)}.$$

We adopt the locked value in the constants table, with the analytic enclosure above as a fully in–paper backup.

Second–order consistency. Both the derivative envelope $C_H(\psi) \le 2/\pi$ and the plateau $c_0(\psi) \ge (2\pi)^{-1}\arctan 2$ are independent of this appendix. The balanced certificate and the conservative Carleson variant remain numerically closed under either the analytic enclosure or the locked quadrature.

Appendix Z. Standard facts and cross-reference stubs

Lemma 103 (Möbius/Cayley contraction)**.** *If F is Herglotz on $\Omega = \{\Re s > \frac{1}{2}\}$ (i.e. $\Re F \ge 0$ on Ω), then its Cayley transform*

$$\Theta(s) := \frac{F(s)-1}{F(s)+1}$$

is Schur on Ω (i.e. $|\Theta| \le 1$ a.e. on the boundary and Θ is bounded by 1 in Ω).

Proof. A standard half–plane/disk Cayley map sends the right half–plane to the unit disk; see, e.g., Rosenblum–Rovnyak, Ch. 2, §3. □

Theorem 104 (Schur–Carathéodory/Herglotz equivalences)**.** *For a holomorphic F on Ω the following are equivalent:*

1. *$\Re F \geq 0$ on Ω (Herglotz).*

2. *The Cayley transform $\Theta = (F-1)/(F+1)$ is Schur on Ω.*

3. *For every finite set $\{s_j\} \subset \Omega$, the kernel $\big((F(s_j)+\overline{F(s_k)})/(s_j+\overline{s_k}-1)\big)_{j,k}$ is positive semidefinite.*

Lemma 105 (Uniform L^1 derivative $\Rightarrow$ L^1–Cauchy)**.** *Let $g : (0, \varepsilon_0] \to L^1(I)$ be such that $\|\partial_\sigma g(\sigma)\|_{L^1(I)} \leq C$ for all $\sigma \in (0, \varepsilon_0]$. Then $g(\sigma)$ is Cauchy in $L^1(I)$ as $\sigma \downarrow 0$ and hence has an L^1 limit.*

Proof. For $0 < \delta < \varepsilon \leq \varepsilon_0$, by the fundamental theorem of calculus in $\mathcal{D}'(I)$ and Fubini,

$$\|g(\varepsilon) - g(\delta)\|_{L^1(I)} \leq \int_\delta^\varepsilon \|\partial_\sigma g(\sigma)\|_{L^1(I)}\, d\sigma \leq C(\varepsilon - \delta).$$

□

Corollary 106 (Adaptive covering)**.** *Let $Q \subset \mathbb{R}$ be measurable and suppose there exist constants $c_0 > 0$, $M < c_0$ and Whitney windows I_{L,t_0} with $L = c/\log\langle t_0\rangle$ such that*

$$c_0\, \mu(Q \cap I_{L,t_0}) \;\leq\; M\, |I_{L,t_0}| \quad \textit{for all sufficiently large } |t_0|.$$

Then $\mu(Q) = 0$. In particular, any density point of Q would contradict the uniform deficit $M < c_0$ under a finite–overlap Whitney cover.

Archived pointers (not used in the proof chain)

The single-route proof in the main text does not rely on these archived sections; the labels are provided only to satisfy legacy cross-references.

References

[1] E. C. Titchmarsh (revised by D. R. Heath-Brown), *The Theory of the Riemann Zeta-Function*, 2nd ed., Oxford University Press, 1986.

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[3] Marvin Rosenblum and James Rovnyak, *Hardy Classes and Operator Theory*, Dover, 1997.